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Measurement of isolated-prompt photons in p+p collisions at $\sqrt{s} = 200$ GeV with the sPHENIX detector

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Abstract

This note details the measurement of the isolated-prompt photon cross section as a function of transverse energy (E_{T}^{γ}) in proton–proton (p+p) collisions at $\sqrt{s} = 200$ GeV taken with the sPHENIX detector in 2024 at the Relativistic Heavy Ion Collider. Photons are identified using the sPHENIX Electromagnetic Calorimeter (EMCal); the isolation requirement uses both the EMCal and the Hadronic Calorimeters. Kinematic selections on the photons are $|\eta^{\gamma}| < 0.7$ and $12 < E_{\mathrm{T}}^{\gamma} < 32$ GeV, with an isolation energy requirement $E_{\mathrm{T}}^{\mathrm{iso,truth}} < 4$ GeV in a cone of $\Delta R = 0.3$. The cross section is unfolded for detector effects and compared to NLO pQCD calculations from JETPHOX and to Monte Carlo predictions from PYTHIA-8; the measurement agrees with both within experimental uncertainties.

Contents

1 Introduction

Prompt photons are photons produced either directly from parton-parton scattering, so-called *direct photons*, or from the collinear fragmentation of a final-state parton, so-called *fragmentation photons*. As shown in Figure 1, at leading order (LO), direct photons are produced predominantly from quark–gluon Compton scattering and quark–antiquark annihilation processes. At next-to-leading order (NLO), additional contributions come from photons fragmented from hard scattered parton and bremsstrahlung photons.

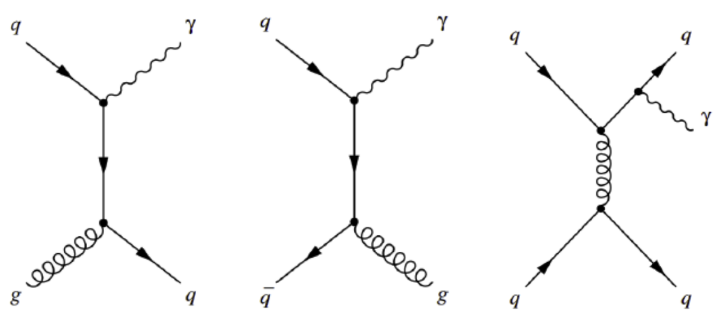

Figure 1: Example Feynman diagrams of (left) quark-gluon Compton scattering, (middle) quark-antiquark annihilation, (right) fragmentation photon

Because their production can be calculated within the framework of perturbative Quantum

Chromodynamics (pQCD), prompt photons provide a stringent test of pQCD predictions. In proton–proton (p+p) collisions in the kinematic range covered here, the quark–gluon Compton process dominates direct-photon production, making the cross section directly sensitive to the gluon parton distribution function (PDF) of the proton. The RHIC $\sqrt{s}$ = 200 GeV setting probes the gluon PDF at parton momentum fractions of $x \sim 2\,E_{\mathrm{T}}^{\gamma}/\sqrt{s} \approx 0.1$–$0.3$, complementing the much smaller-x reach of LHC measurements. The isolated prompt-photon cross section is also a clean reference for future PPG sPHENIX heavy-ion programs and a direct point of contact with the previous PHENIX direct-photon measurement at the same collision energy [1].

To reduce the contribution of fragmentation photons, an isolation requirement can be imposed. The truth-level isolation energy ($E_{\mathrm{T}}^{\mathrm{iso,truth}}$) is defined as the sum of the transverse energies of all final-state particles within a fixed pseudorapidity (η)–azimuth (ϕ) space around the photon of interest, excluding the photon energy itself. Figure 2 shows the fraction of direct and fragmentation photons as a function of the maximum $E_{\mathrm{T}}^{\mathrm{iso,truth}}$ threshold for different photon E_{T}^{γ} with isolation cone of $R = 0.3$. At an $E_{\mathrm{T}}^{\mathrm{iso,truth}}$ threshold of 4 GeV, direct photons remain fully efficient across the analysis range while fragmentation photons are suppressed.

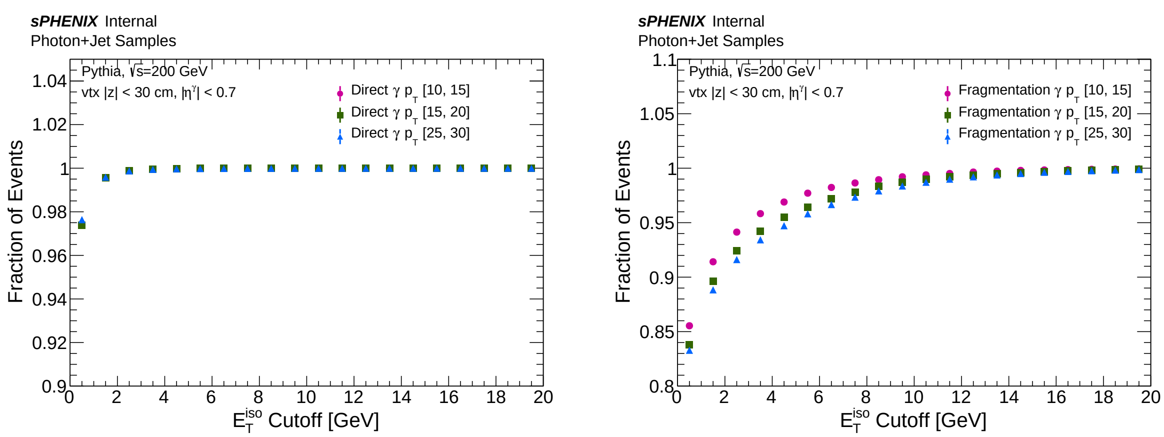

Figure 2: Fraction of direct (left) and fragmentation (right) photons passing the truth-level $E_{\mathrm{T}}^{\mathrm{iso}}$ cut as a function of the cut threshold for three different truth photon E_{T}^{γ} ranges.

In this note, we present the measurement of isolated prompt photons in p+p collisions at $\sqrt{s}$ = 200 GeV, using data collected in 2024 with an integrated luminosity of 64.37 pb^{-1}. The cross section is measured differentially in E_{T}^{γ} for $12 < E_{\mathrm{T}}^{\gamma} < 32$ GeV within $|\eta^{\gamma}| < 0.7$, with truth-level isolation $E_{\mathrm{T}}^{\mathrm{iso,truth}} < 4$ GeV in an $R = 0.3$ cone.

Section 2 describes the data and MC samples used in this analysis and trigger and event selections. Section 3 describes the procedures for reconstructing and identifying photons. Section 4 details the methods employed to subtract the remaining backgrounds, unfolding, and efficiency correction. Section 5 details the systematic uncertainty evaluation. Finally, in Section 6, the final physics results are presented and compared with both PYTHIA-8 and pQCD NLO predictions (including JETPHOX), as well as with previously published measurements from the PHENIX experiment [1].

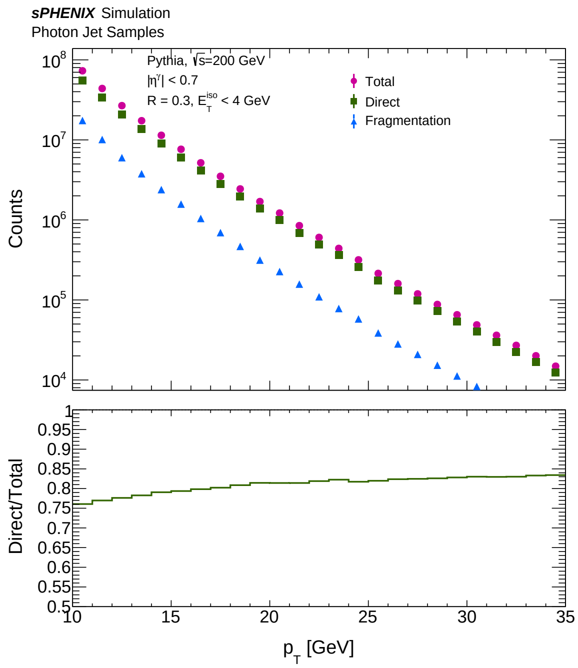

Figure 3: Top panel: truth photon E_{T}^{γ} distributions for direct (green), fragmentation (blue) photons, and total (pink), with $E_{\mathrm{T}}^{\mathrm{iso,truth}} < 4$ GeV in an isolation cone of $R = 0.3$ in PYTHIA-8. Bottom panel: fraction of direct photons.

2 Data and MC Samples

2.1 Data samples

The p+p Run 24 data is used for this analysis and summarized in Table 1. Runs are selected from a centrally produced calorimeter "golden run" list, which is selected based on the subsystem validation offline triage database and described in Section 2.3.

Run Numbers	Production tag	CDB tag	DST type
47289-53880	ana521	2025p007	DST_FITTING, DST_JETCALO

Table 1: Summary of data sample used in this analysis with production and CDB tag information.

2.2 Monte Carlo Samples

2.2.1 Photon and Jet MC samples

Throughout this note, "signal" refers to truth-matched direct or fragmentation photons satisfying the truth-level isolation $E_{\mathrm{T}}^{\mathrm{iso,truth}} < 4$ GeV (defined in Section 4.2). "Background" refers to clusters from neutral-meson decays and other QCD-jet-induced clusters in the inclusive jet MC. PYTHIA-8 Monte Carlo (MC) [2] events are generated for signal and background photons, respectively, and simulated through GEANT-4 [3] with the sPHENIX detector geometry. In PYTHIA-8 configuration files:

https://github.com/sPHENIX-Collaboration/calibrations/blob/master/Generators/JetStructure_TG/

both *HardQCD:all = on* and *PromptPhoton:all = on* processes are turned on for both signal (labeled as "photon") and background (labeled as "jet") samples. After events are generated from PYTHIA-8, there is a truth-level event filtering: jet samples require a truth jet within $|\eta| < 1.5$ for p_T^{jet} greater than a threshold (8, 12, 20, and 30 GeV), while photon samples require a truth photon within $|\eta| < 1.5$ with p_T^{γ} greater than a threshold (5, 10, and 20 GeV). Multiple samples are generated with different minimum p_T thresholds to ensure high statistics in the high-p_T region. Each sample contains 10 million processed events. Table 2 shows the list of MC samples used in this analysis. Samples with different p_T thresholds are combined with each sample's effective cross section. The effective cross sections are calculated using the following equation:

$$\sigma_{\text{eff}} = \frac{\sigma_{\text{generator}} N_{\text{accepted}}}{N_{\text{generated}}} \tag{1}$$

where $\sigma_{\text{generator}}$ is the cross section for generating events given the generator configurations ($\hat{p}_{\text{T,min}}$), $N_{\text{generated}}$ is the number of events generated by the generator, and N_{accepted} is the number of events that pass the truth-level event filtering. Figure 4 shows the ratio between PYTHIA-8 photon samples normalized by the effective cross section. This demonstrates the correct cross section reweighting on each MC sample. Figure 5 shows the per-event leading truth photon E_T^{γ} distribution after the simulation sample combining process. The smooth transition around each threshold confirms the validity of the methodology. The same combining strategy is applied to the jet samples. At analysis time, each sample is restricted to a contiguous truth leading-pT window so that the samples tile without overlap. The windows are summarized in Table 3.

In this note, distributions or results constructed using PYTHIA-8 photons with truth-level signal photon selections (see details in Section 4) are referred to as "signal MC". Similarly, those using PYTHIA-8 jet samples with anti-selection against truth-level signal photons are referred to as "background MC". The PYTHIA-8 jet samples without any selections are referred to as "inclusive MC".

2.2.2 Double interaction samples

In p+p collisions at RHIC, two separate collisions can occur in the same bunch crossing (pileup, also referred to here as double interaction, DI). The pileup shifts the reconstructed MBD vertex to the average of the two collision points, which in turn distorts cluster pseudorapidity, broadens shower-shapes correlation with reconstructed vertex effecting photon identification. For this reason, a set of full-GEANT double-interaction MC samples is blended with the single-interaction samples.

The analysis uses two crossing-angle periods (split at run 51274):

- **0 mrad**, runs 47289–51274, $\mathcal{L} = 47$ pb^{-1}, broad luminous region ($\sigma_z \simeq 54$ cm), high pileup ($\sim$22% cluster-weighted DI, definition in Appendix 17);

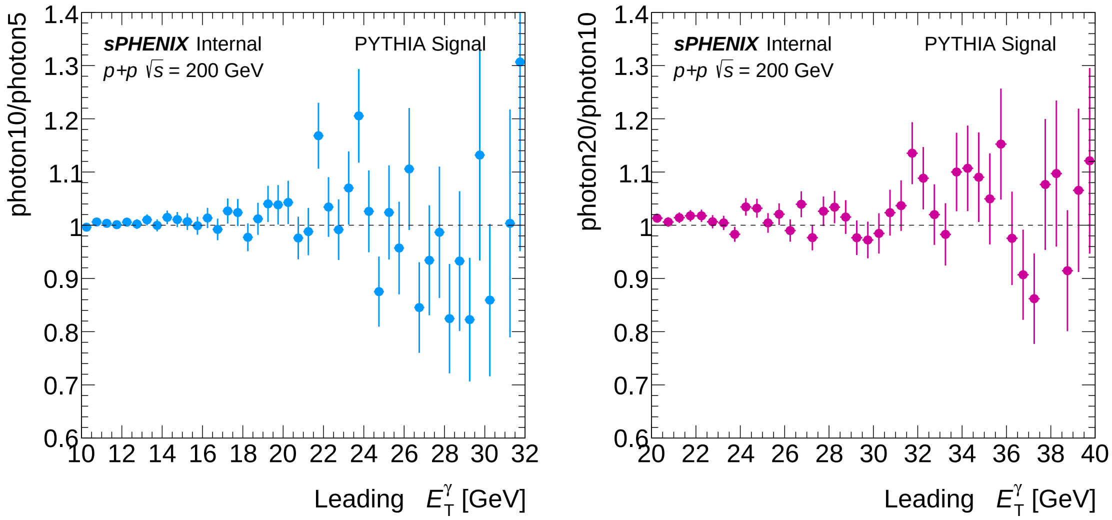

Figure 4: Ratio of cross section as a function of leading truth photon E_T^γ from PYTHIA-8. The photon 10 GeV sample is divided by the photon 5 GeV sample (left), and the photon 20 GeV sample is divided by the photon 10 GeV sample (right).

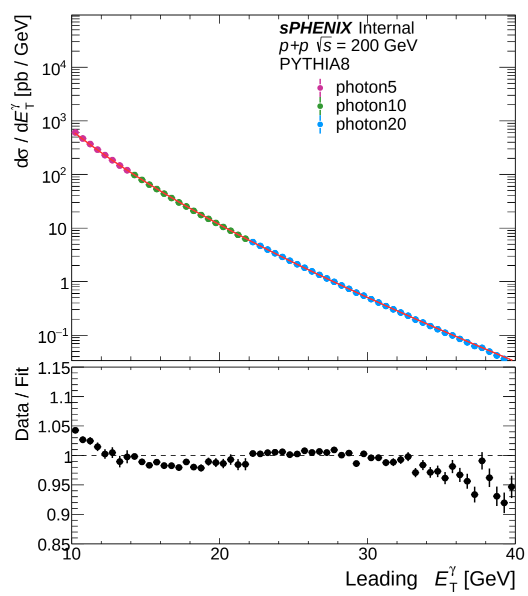

Figure 5: Leading truth photon E_T^γ distributions from PYTHIA-8 photon 5, 10, 20 GeV samples normalized by the effective cross section. The combined spectrum is fitted with a modified power law function [4], the MC/fit ratio is shown in lower panel.

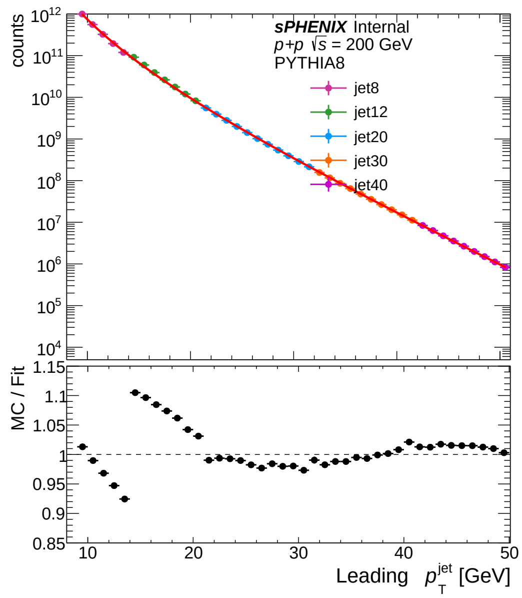

Figure 6: Leading truth jet p_T distributions from PYTHIA-8 jet 8, 12, 20, 30, 40 GeV samples normalized by the effective cross section, sibling of Figure 5. The combined spectrum is fitted with a modified power law function [4], the MC/fit ratio is shown in the lower panel. The smooth transition across sample boundaries confirms that the same combining strategy applies to the jet samples.

Generator	SI sample (run 28 type)	DI sample (run 28 type)	CDB tag	σ_{eff} [pb]
PYTHIA-8	jet8	jet8_double	MDC2	4.93×10^6
PYTHIA-8	jet12	jet12_double	MDC2	1.49×10^6
PYTHIA-8	jet20	jet20_double	MDC2	6.26×10^4
PYTHIA-8	jet30	jet30_double	MDC2	2.53×10^3
PYTHIA-8	photon5	photon5_double	MDC2	1.46×10^5
PYTHIA-8	photon10	photon10_double	MDC2	6.94×10^3
PYTHIA-8	photon20	photon20_double	MDC2	1.30×10^2

Table 2: Summary of MC samples used. For every jet and photon process, two full-GEANT productions are generated through the identical PYTHIA-8 + GEANT chain: a single-interaction (SI) sample with one $p+p$ collision per event, and a double-interaction (DI) partner with two overlapping $p+p$ collisions per event. The two productions share the effective cross section σ_{eff} (computed via Eq. 1 and used to combine samples) and are blended at the cluster-weighted DI fractions of Section 17.

- **1.5 mrad**, runs 51274–54000, $\mathcal{L} = 17$ pb^{-1}, luminous region narrowed by geometric overlap reduction at the 1.5 mrad crossing angle ($\sigma_z \simeq 22$ cm), modest pileup ($\sim$8% cluster-weighted DI).

The double-interaction fractions of 22.4% at the 0 mrad crossing angle and 7.9% DI at 1.5 mrad are calculated and the blending methodology is described in Appendix 17.

Sample (SI / DI partner)	Truth leading-pT window [GeV]
`photon5 / photon5_double`	$0 \le p_T^{\gamma,\,\mathrm{truth}} < 14$
`photon10 / photon10_double`	$14 \le p_T^{\gamma,\,\mathrm{truth}} < 22$
`photon20 / photon20_double`	$p_T^{\gamma,\,\mathrm{truth}} \ge 22$
`jet8 / jet8_double`	$9 \le p_T^{\mathrm{jet,truth}} < 14$
`jet12 / jet12_double`	$14 \le p_T^{\mathrm{jet,truth}} < 21$
`jet20 / jet20_double`	$21 \le p_T^{\mathrm{jet,truth}} < 32$
`jet30 / jet30_double`	$32 \le p_T^{\mathrm{jet,truth}} < 42$
`jet40 / jet40_double`	$p_T^{\mathrm{jet,truth}} \ge 42$

Table 3: Per-sample truth leading-pT windows applied at analysis time so that samples tile without overlap. The leading-pT object is the truth leading photon for photon samples and the truth leading jet for jet samples, both within $|\eta| < 1.5$. These windows are encoded in `CrossSectionWeights.h` and applied to both the SI and DI productions.

2.2.3 MC EMCal Smearing

The EMCal energy resolution in MC is matched to the data resolution via an additive E_T^{γ}-dependent Gaussian smearing applied at the unfolding response-matrix fill stage. The smearing parameters are derived such that the simulated single-particle resolution equals the data resolution measured from in-situ π^0 and η mass-peak fits by the calorimeter calibration group. Variations of these parameters are taken as a systematic uncertainty for the EMCal energy resolution; the canonical prescription and parameter values are given in Section 5.2.

2.3 Trigger

All runs in this analysis were taken after the Level-1 trigger firmware update at run 47289, and are subjected to basic calorimeter quality assurance (QA): each run is required to have more than one million recorded events, a runtime of at least five minutes, and an MBD NS ≥ 1 live time above 80%. Among those runs, the active triggers are identified by checking if specific triggers of interest have a prescale $\neq -1$, including MBD NS ≥ 1 alone (at least one PMT firing in the north and south MBD) and each "Photon N GeV + MBD NS ≥ 1" variant (for $N = 3, 4, 5$). In each qualifying run, a maximum cluster energy histogram is filled using the scaled trigger bits of MBD NS ≥ 1 and the Photon 3, 4, 5 GeV + MBD NS ≥ 1 triggers. Specifically, each event is first required to have the MBD NS scaled bit set, after which the maximum cluster energy is recorded in a histogram associated with that trigger. In the same event, the scaled bits of each photon trigger are checked; if any are active, the corresponding photon-trigger histograms are also filled. The MBD NS ≥ 1 trigger typically has a prescale, or ratio of live-to-scaled trigger counts, of about 2000, whereas the photon triggers have prescales closer to 1–4, where the highest photon trigger have no scale-down for most of the time.

For each Photon N GeV trigger, the trigger efficiency is determined on a bin-by-bin basis on the unbiased bit-10-scaled reference using

$$\varepsilon_{trig}(E_T) = \frac{N(\text{scaled}[10]\,\&\,\text{live}[N_{\text{bit}}])}{N(\text{scaled}[10])}. \tag{2}$$

where trigger bit 10 is (MBD N&S) and N_{bit} is for trigger bit for photon N GeV trigger. Using the live bit in the numerator removes that trigger's own prescale, so the ratio is the trigger efficiency. Figure 7 shows the leading EMCal cluster distributions for events firing bit 10 (MBD N&S) plus each photon trigger (left) and the bin-by-bin trigger efficiencies for bits 29 (Photon 3 GeV), 30 (Photon 4 GeV, nominal in this analysis) and 31 (Photon 5 GeV) (right). Each curve is fit with $\varepsilon_{trig}(E_T) = p_0 \exp[-\exp(-(E_T - \mu)/\beta)]$ (a saturating asymmetric form) with all three parameters free, the plateau capped at the physical upper bound of 1, and the 1σ confidence interval from the fit covariance shown as a shaded band of the matching color. For this trigger efficiency study, the event-level vertex cut is $|z_{\text{reco}}| < 200$ cm.

The "Photon 4 GeV" trigger (bit 30) is used in this analysis. The result starts at $E_{\text{T}}^{\gamma} = 12$ GeV while an unfolding underflow bin is used from 10 GeV, where the trigger is close to plateau. The final result is corrected for the trigger efficiency on a bin-by-bin basis at the percent level at $E_{\text{T}}^{\gamma} = 10$ GeV and below 0.1% above $E_{\text{T}}^{\gamma} = 20$ GeV.

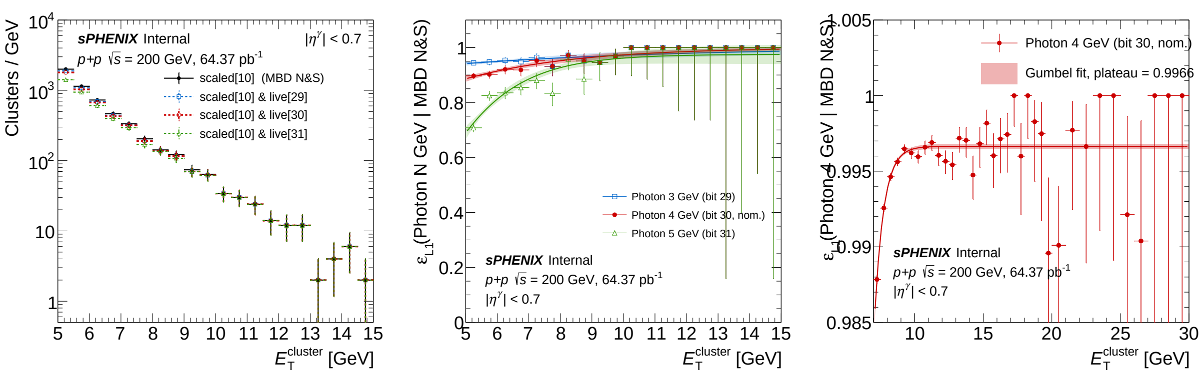

Figure 7: (Left) EMCal cluster E_T density for events with the MBD NS ≥ 1 (bit 10) scaled bit set (solid black) and for those with bit 10 scaled and additionally bit 29 / bit 30 / bit 31 live (dashed blue / red / green). (Middle) Bin-by-bin L1 trigger efficiencies $\varepsilon_{trig} = N(\text{scaled}[10]\,\&\,\text{live}[N])/N(\text{scaled}[10])$ for $N = 29, 30, 31$ (blue, red, green) with Clopper–Pearson 68% confidence-interval (CI) errors. Fits for each trigger bit are overlaid with the corresponding 1σ CI bands from the fit covariance. (Right) Photon 4 GeV (bit 30, nominal) efficiency on a zoomed y-axis $[0.985, 1.005]$ across $10 < E_{\text{T}}^{\text{reco}} < 30$ GeV so that any residual sub-percent inefficiency in the plateau region is visible.

2.4 Event Selection

Triggered events are further required to pass the event selection criteria as follows:

- $|z_{\text{vtx}}^{\text{MBD}}| < 60$ cm

The reported integrated luminosity, $\mathcal{L} = 64$ pb^{-1}, is the beam-delivered (all-z) luminosity sampled by the photon4 GeV (bit 30) trigger over the analysis runs; no additional MBD coincidence requirement nor vertex selection is imposed in the lumi calculation itself (full calculation in Section 4, Luminosity). The $|z_{\mathrm{vtx}}| < 60$ cm cut on data events listed above is therefore decoupled from the lumi value: at the cross-section level the all-z lumi is paired with the MBD-vertex efficiency (Section 4.4), whose denominator runs over the full truth-z range and whose numerator restores $|z_{\mathrm{reco}}| < 60$ cm required to reconstruct it.

The luminosity passing the trigger and event selection is $64^{+5.88}_{-4.34}$ pb^{-1} ($+9.13\%/-6.75\%$, see Section 5.10).

3 Photon Reconstruction

3.1 Calorimeters

The photon reconstruction relies on the sPHENIX electromagnetic calorimeter (EMCal) and on the inner and outer hadronic calorimeters (iHCal, oHCal). The EMCal is a SPACAL-type tungsten–scintillating-fiber sampling calorimeter with full 2π azimuthal coverage and pseudorapidity coverage $|\eta| < 1.1$. It is segmented into towers of size $\Delta\eta \times \Delta\phi \approx 0.024 \times 0.024$ (96×256 towers in total), and provides a single-photon energy resolution of $\sigma_E/E \simeq 16\%/\sqrt{E[\,\mathrm{GeV}]} \oplus 5\%$ at test-beam energies [5]. As described in Section 2.2.3, additional smearing on EMCal energies in the MC was performed for the nominal analysis. The two HCal sections sit immediately behind the EMCal at increasing radius and provide hadronic shower containment plus an energy reading used to define the cluster isolation. The collision vertex is reconstructed from the Minimum Bias Detector (MBD), which consists of two arrays of 64 quartz-Cherenkov PMTs in the forward regions $3.51 < |\eta| < 4.61$.

3.2 Clustering Algorithm

A simple clustering routine is used to aggregate EMCal towers into clusters. It is implemented in the RawClusterBuilderTemplate class. Adjacent towers (sharing an edge) above a single threshold (default at 70 MeV) and passing the *isGood* tower quality cuts are clustered together. These clusters grow until no adjacent towers are above the threshold, a cluster splitting algorithm is then applied. The cluster kinematics are determined in the standard way implemented in RawClusterBuilderTemplate, which relies on tower position in x-y-z space and corrections for shower depth. The output x-y-z is then used to calculate the η and ϕ positions relative to the collision vertex, which is reconstructed with the Minimum Bias Detector (MBD). The reconstructed energy is determined by summing all of the individual tower energies in the cluster.

3.3 Photon Energy Response

Photon energy scale and resolution are estimated using MC. Figure 8 shows the ratio of reconstructed cluster E_T to truth-level photon $E_\mathrm{T}^\mathrm{truth}$ as a function of $E_\mathrm{T}^\mathrm{truth}$. The ratio of $E_\mathrm{T}^\mathrm{reco}$ to $E_\mathrm{T}^\mathrm{truth}$ in each $E_\mathrm{T}^\mathrm{truth}$ bin is fitted with a Gaussian function – see examples in Figure 9. The low-side tail reflects residual position-dependent calibration; this is absorbed into the unfolding response matrix. Figure 10 shows the extracted photon energy scale and resolution as a function of $E_\mathrm{T}^\mathrm{truth}$. Unfolding accounts for the remaining resolution and scale (Section 4.5).

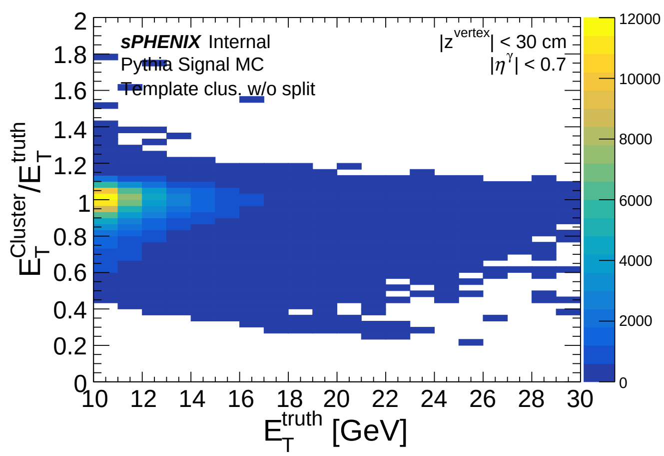

Figure 8: Photon energy response as a function of $E_{\mathrm{T}}^{\mathrm{truth}}$ in signal PYTHIA-8. The vertex selection is restricted to $|z_{\mathrm{vtx}}| < 30$ cm in this study to isolate the intrinsic detector energy resolution from vertex-induced kinematic distortions of the per-cluster $E_{\mathrm{T}}^{\mathrm{reco}}/E_{\mathrm{T}}^{\mathrm{truth}}$ ratio. The wider analysis fiducial $|z_{\mathrm{vtx}}| < 60$ cm is restored in the unfolding response matrix and the efficiency calculation.

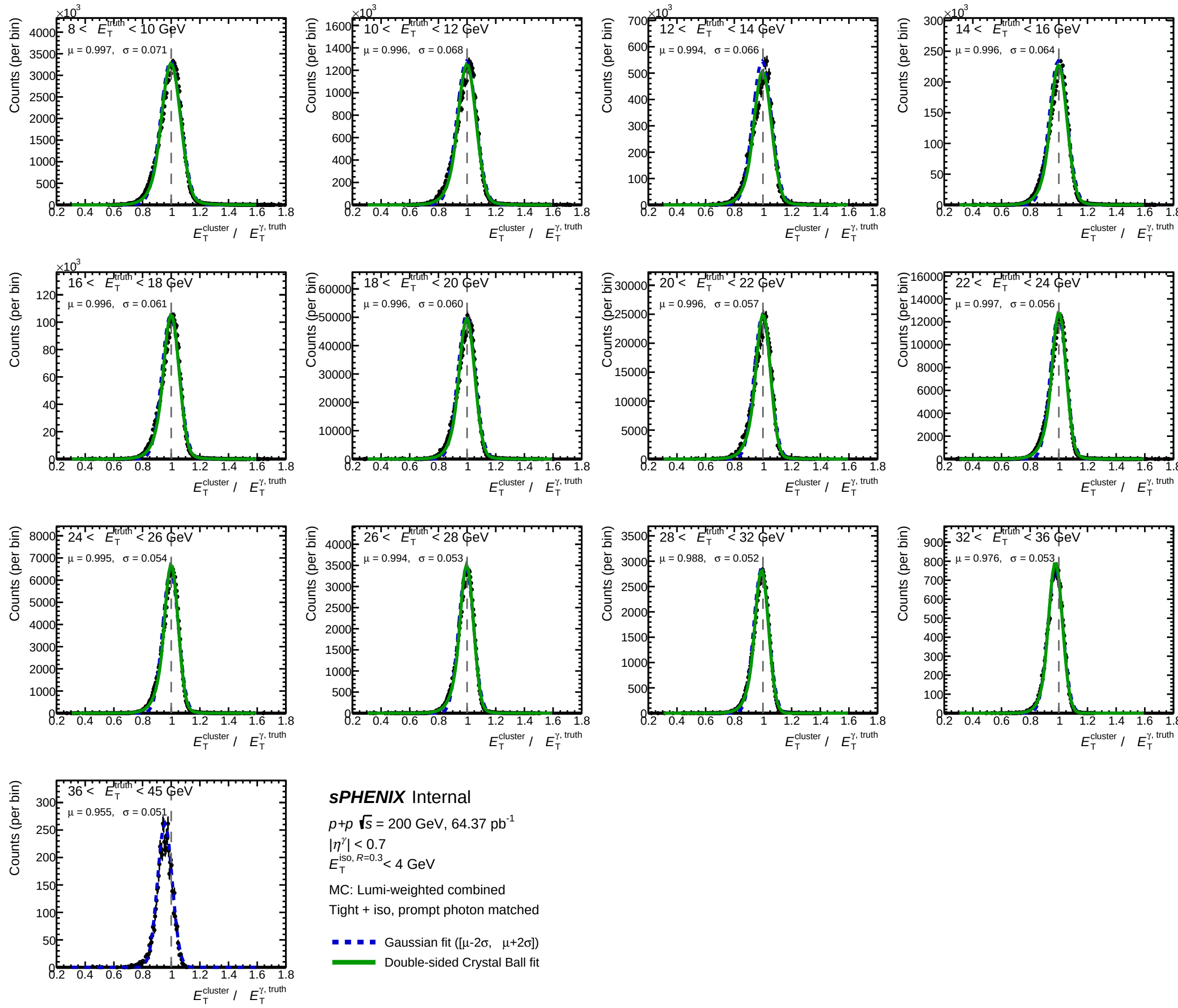

Figure 9: $E_{\mathrm{T}}^{\mathrm{reco}}/E_{\mathrm{T}}^{\mathrm{truth}}$ distributions in the analysis truth-pT bins, from signal PYTHIA-8 (photon5/10/20, lumi-weighted across the two crossing-angle periods including the SI + DI blend) with the analysis tight + isolated prompt-photon selection applied. Black markers: signal MC; dashed blue: Gaussian fit on $[\mu_{\mathrm{hist}} - 2\,\mathrm{RMS}, \mu_{\mathrm{hist}} + 2\,\mathrm{RMS}]$; green: double-sided Crystal-Ball fit on $[0.3, 1.6]$.

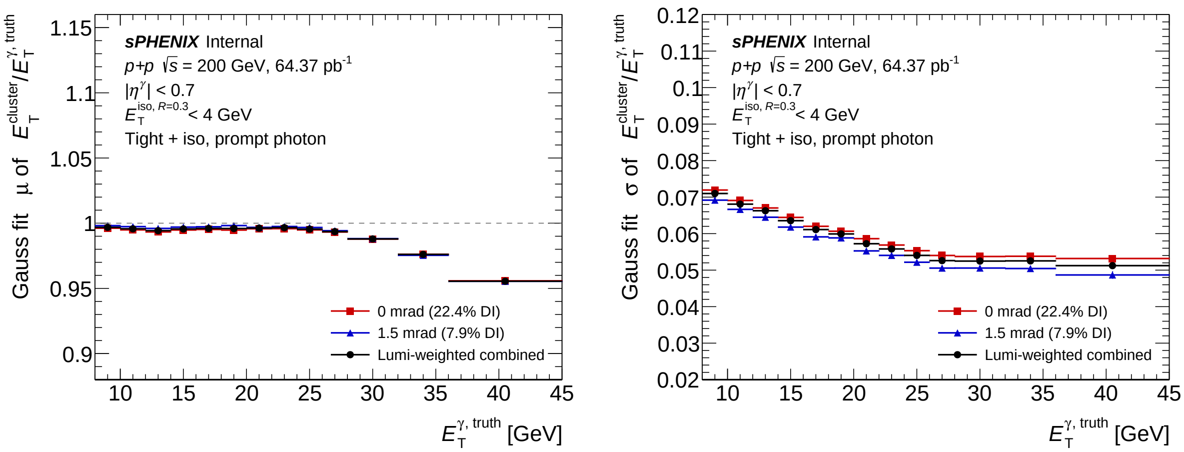

Figure 10: Photon energy scale (left, Gaussian mean of $E_{\mathrm{T}}^{\mathrm{reco}}/E_{\mathrm{T}}^{\mathrm{truth}}$) and resolution (right, Gaussian σ) as a function of $E_{\mathrm{T}}^{\mathrm{truth}}$ in signal PYTHIA-8 with the tight + iso prompt-photon selection. The two crossing-angle periods and the lumi-weighted combination are shown.

3.4 Signal and Background Identification

The main source of background for prompt photons are photons decayed from high-p_T neutral mesons such as $\pi^0 \to \gamma + \gamma$. Figure 11 shows the signal-to-background ratio as a function of $E_\mathrm{T}^\mathrm{reco}$ in PYTHIA-8 without any identification selection. The signal-to-background ratio is about 15% at 10 GeV and rises to about 50% near 30 GeV. To distinguish signal photons from decay photons as well as other high-p_T charged particles, and non-physics backgrounds — a set of moments and energy ratios are calculated from the EMCal towers in the cluster. These moments and ratios are collectively referred to as "shower shapes". The definition of various shower shapes is given in Table 4. In general, the decay photons of high-p_T neutral mesons (e.g. π^0, η) have a ΔR less than a tower size and are reconstructed in one cluster. However, the energy may be more spread out in ϕ or η, or the energy of the leading tower may be a smaller fraction of the total cluster energy. The shower shapes encode this information.

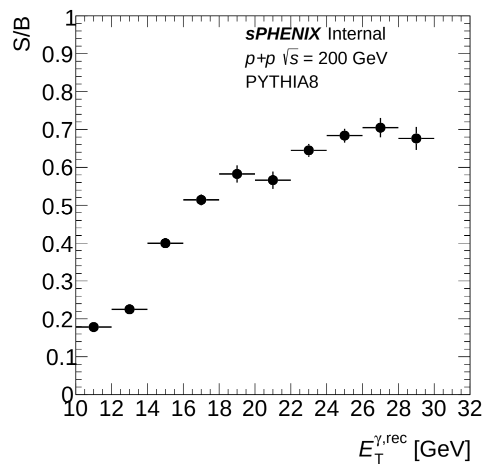

Figure 11: Signal-to-background ratio (S/B) as a function of $E_\mathrm{T}^\mathrm{reco}$ derived from PYTHIA-8. Signal is the truth-matched prompt-photon cluster yield from the photon MC ($E_\mathrm{T}^\mathrm{iso,truth} < 4$ GeV in $\Delta R = 0.3$, the same fiducial truth-isolation cut used in the cross-section definition); background is the cluster yield from the inclusive jet MC. Both samples have the parametric reco-isolation cut applied; no shower-shape identification is imposed at this stage. The S/B is therefore from PYTHIA-8 truth, not from data.

3.4.1 Non-physics Background Removal

It is possible for beam backgrounds to enter the EMCal and fire the photon triggers. These backgrounds occur at a high enough rate that a coincident physics collision (or other background) also triggering the MBD N&S trigger is non-negligible.

An example of such a background event in the EMCal is shown in Figure 12. Due to their generally extended energy deposits in η, these non-physics backgrounds (NPB) are distinguishable from signal photons. Figure 13 shows shower shape distributions for high cluster-p_T (low cluster-p_T example can be found in Figure 57) for all clusters reconstructed in this analysis. The distributions are compared with the signal MC, which contains only truth-level prompt photons, and with

Symbol	Definition
E3x2/E3x5	Ratio of energy in regions of 3 towers wide in η and 2 towers wide in ϕ to the energy in a region 3 towers wide in η and 5 towers wide in ϕ. The towers closest to the center-of-gravity are chosen.
E1x1/E	Center of energy tower divided by cluster energy.
$w^{\text{COGX}}_{\eta(\phi)}$	energy weighted second moment of $\eta(\phi)$ from all towers in a given cluster excluding the tower containing the center of gravity.
$w72$	$\sum E_i(\eta_i - \eta_i^{max})^2 / \sum E_i$ for $(\eta_i, \phi_i) = (7, 2)$ tower area
R_{had}	$E_T^{HCal} / (E_T^{EMCal} + E_T^{HCal})$
E_1, E_2, E_3, E_4	Energies of the four EMCal towers forming the 2×2 block around the cluster center of gravity (CoG). The CoG tower index is rounded to the nearest integer in $(i\eta, i\phi)$, and the second tower index along each axis is shifted by ± 1 in the direction of the residual fractional CoG offset. With this convention E_1 is the CoG tower, E_2 is the tower adjacent in ϕ, E_3 is the diagonal-corner tower (shifted in both η and ϕ), and E_4 is the tower adjacent in η.
E_{t1}	$(E_1 + E_2 + E_3 + E_4)/E_{tot}$ — fraction of cluster energy in the 2×2 block around the CoG.
E_{t2}	$(E_1 + E_2 - E_3 - E_4)/E_{tot}$ — η-direction asymmetry of the 2×2 block.
E_{t3}	$(E_1 - E_2 - E_3 + E_4)/E_{tot}$ — ϕ-direction asymmetry of the 2×2 block.
E_{t4}	E_3/E_{tot} — diagonal-corner tower fraction.
BDT score	XGBoost boosted decision tree output for photon identification, described in Section 3.5.1.

Table 4: Definitions of various shower shape variables.

the inclusive MC, which includes both prompt photons and background decay photons with the appropriate MC cross sections (see Section 2.2). The data more closely follow the inclusive MC, which is expected due to the low signal-to-background ratio without any selection. There are significant deviations from either MC. For instance, since NPB events have a larger spread in η and thus a characteristically large w^{COGX}_{η}.

To further characterize NPB, clusters were selected with $w^{\text{COGX}}_{\eta} > 1.0$ and with no back-to-back jet satisfying $\Delta\phi > \pi/2$. Figure 14 shows the EMCal cluster time minus MBD time distribution as a function of cluster η for all EMCal clusters and for the selected NPB clusters. The NPB clusters shows a clearly shifted time distribution toward negative values at positive η. This suggests that the background events propagate from $+z$ to $-z$.

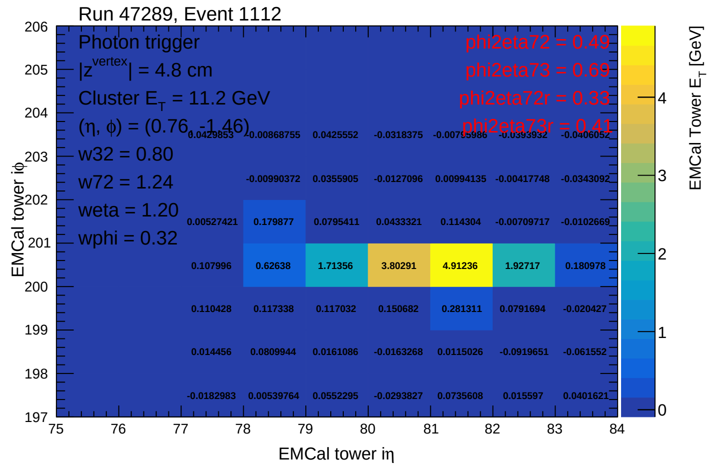

Figure 12: Example tower η-ϕ distribution of a non-physics background cluster (streak event). Tower energies are shown for each tower.

3.4.2 Non-Physics Background Removal using Boosted Decision Tree

While the $w_{\eta}^{\mathrm{COGX}} > 1.0$ removes the bulk of streak-like non-physics backgrounds, residual NPB clusters with moderate w_{η}^{COGX} values can survive. Thus, a multivariate approach using an XGBoost boosted decision tree (BDT) is trained to provide better discrimination for non-physics background clusters.

To train the BDT, two separate cluster samples are formed: a signal sample and an NPB background sample. Signal clusters are taken from PYTHIA-8 inclusive MC (jet8, jet12, jet20, jet30) with a valid truth particle ID (including both prompt photons and decay photons which are originated from hadronic collisions unlike beam-induced particles), while NPB clusters are extracted from data using anomalous cluster–MBD timing ($t_{\mathrm{cluster}} - t_{\mathrm{MBD}} < -7$ ns) with no back-to-back jet. Both samples are restricted to $6 < E_T < 40$ GeV and $|\eta| < 0.7$. The classifier uses 25 features: cluster E_T, η, z_{vtx}, and the shower-shape ratios and width moments, including those listed in Table 4. We note that timing information was not used as training feature. To balance the training set, two-sample class reweighting is applied and the E_T distribution is flattened. The model is exported to TMVA and applied at the reconstruction level. Clusters outside the training acceptance receive a default score of -1.

The NPB score is validated against cluster–MBD timing: clusters from in-time collisions peak near $t = 0$ ns, while NPB clusters from beam backgrounds populate a broad distribution extending to large negative times. Figure 15 shows the 2D distribution of NPB score vs. cluster–MBD time, confirming that high-score clusters are in time and low-score clusters carry the out-of-time component.

The purity and signal-cluster retention efficiency at each NPB threshold T are extracted via a data-driven template fit. The data clusters are sliced in NPB score into a high-score sample ($T <$ NPB < 1) and a low-score template (NPB < 0.2); the low-score sample is dominated by NPB clusters and serves as the NPB template. Assuming the NPB out-of-time-to-in-time shape

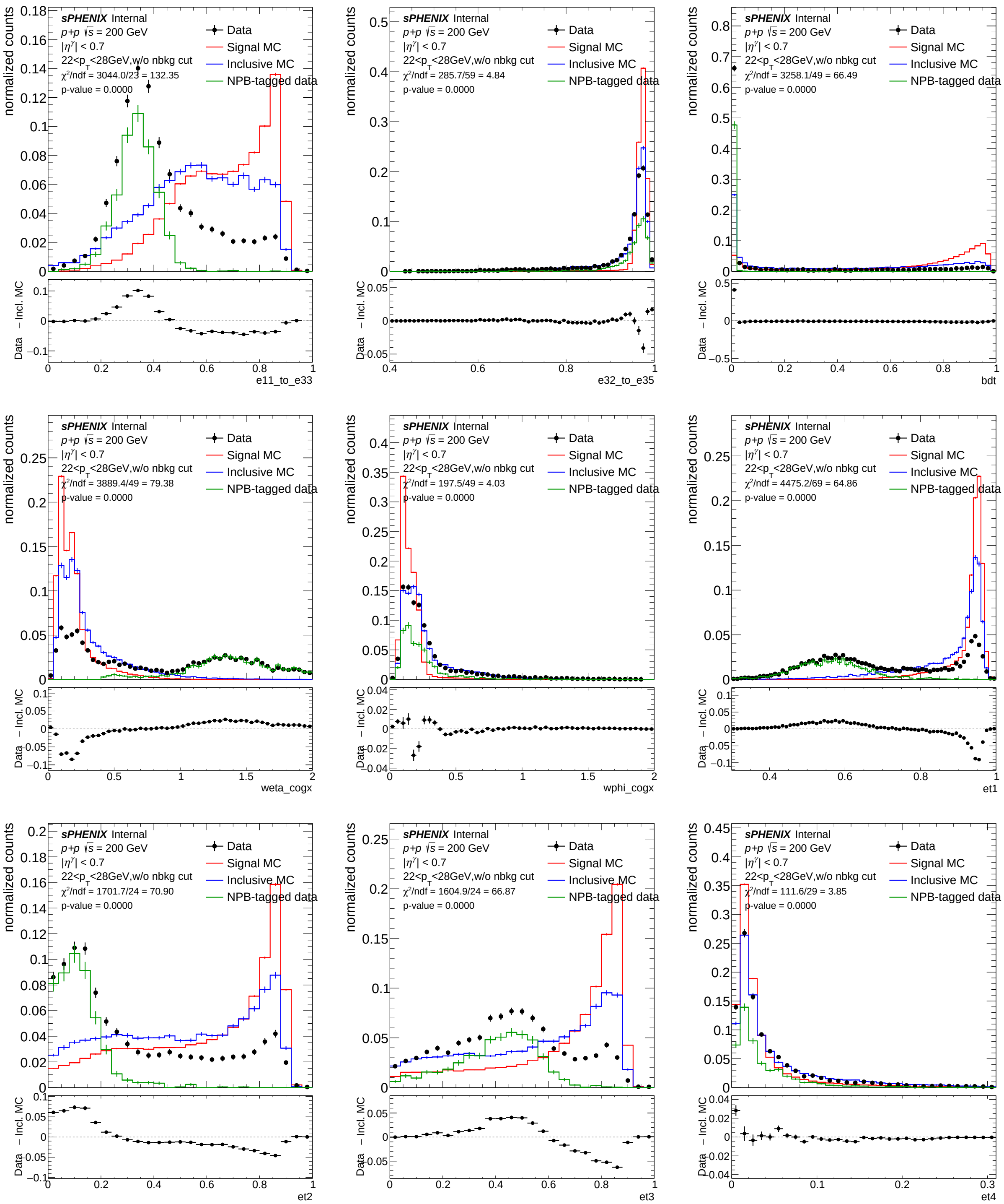

Figure 13: Shower shape distributions with no pre-selection at higher p_T (22–28 GeV) for data (black), signal MC (red), inclusive MC (blue), and the NPB-tagged data template (green). The NPB-tagged template is the data subset selected as streak-like by the high-w_η^{COGX} NPB tag, normalized to the bulk data via tail-matching in $w_\eta^{\mathrm{COGX}} > 1$, and serves as a background-enhanced template for the non-physics-background component.

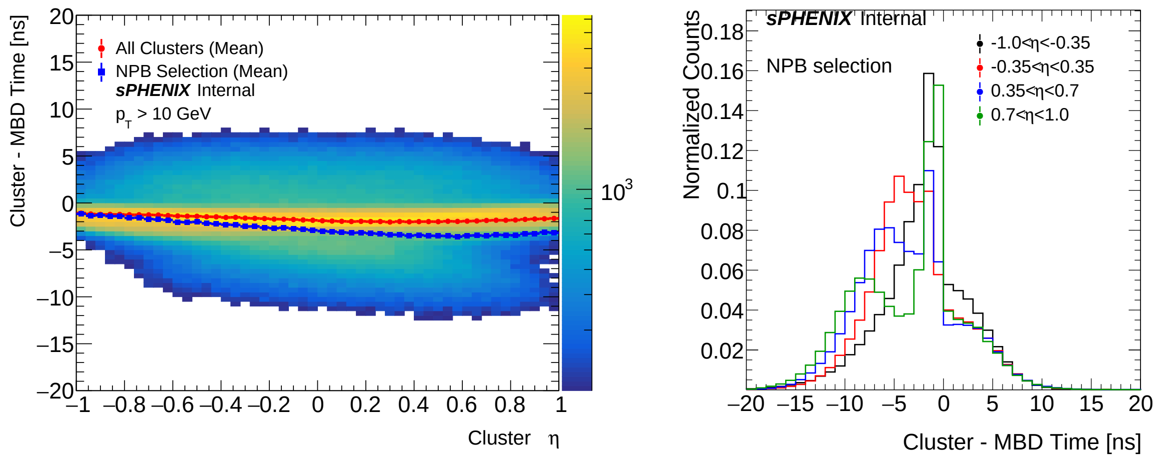

Figure 14: Left: 2D scatter plot of cluster-MBD time as a function of cluster η. The mean profiles for all clusters (red) and NPB selection of $w_{\eta}^{\mathrm{COGX}} > 1.0$ and with no back-to-back jet (blue) are overlaid. Right: cluster-MBD time distributions in different cluster η in different colors.

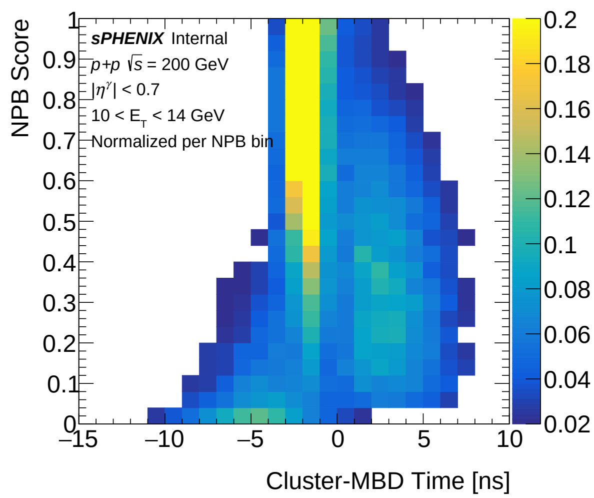

Figure 15: NPB BDT score vs. cluster–MBD time for data clusters with $|\eta^{\gamma}| < 0.7$ and $10 < E_T < 14$ GeV, normalized per NPB-score bin.

is independent of the score threshold, the template is normalised to the high-score sample by matching the cluster–MBD time integrals in the out-of-time tail ($t < -7$ ns) where physics signal contamination is negligible. The NPB yield at T is the integral of this scaled template, the signal is $N_{\mathrm{sig}}(T) = N_{\mathrm{total}}(T) - N_{\mathrm{NPB}}(T)$, and the purity is $\mathcal{P}(T) = N_{\mathrm{sig}}(T)/N_{\mathrm{total}}(T)$. Figure 16 shows this template fit at the nominal NPB > 0.5 cut for $22 < E_T < 28$ GeV. Scanning T gives the purity and the signal retention efficiency $\varepsilon(T) = N_{\mathrm{sig}}(T)/N_{\mathrm{sig}}(T_{\mathrm{min}})$ relative to the loosest scanned threshold $T_{\mathrm{min}} = 0.1$, shown in the left panel of Figure 17; the right panel shows the underlying data total, NPB, and signal yields per threshold. The cut at NPB > 0.5 achieves $\sim$99% purity while retaining $\approx$99% of physics signal clusters (here, signal clusters refer to both prompt photons and decay photons); further tightening yields diminishing returns on purity but starts to cost signal retention substantially. The purity and efficiency as a function of NPB score thresholds for the other p_T bins can be found in Appendiex 8.

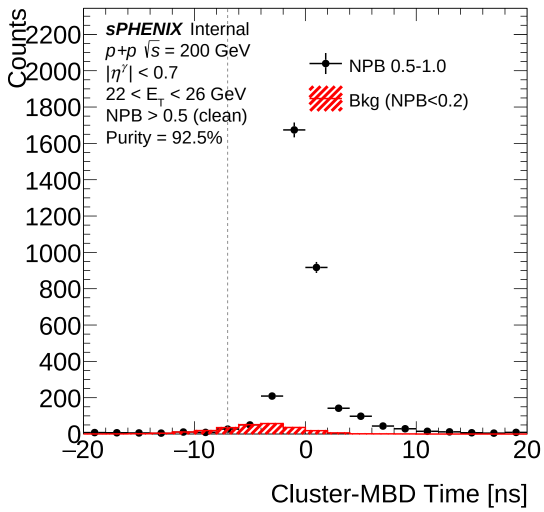

Figure 16: Example template fit for the $22 < E_T < 28$ GeV bin at the nominal NPB> 0.5 cut. Cluster–MBD time distribution of data clusters with NPB> 0.5 (black points) overlaid on the NPB template constructed from data NPB< 0.2 clusters (red filled), scaled to the high-score selection by matching the integrals in the out-of-time tail ($t < -7$ ns). The fit purity is $N_{\text{sig}}/N_{\text{total}}$, with N_{NPB} the integral of the scaled template and $N_{\text{sig}} = N_{\text{total}} - N_{\text{NPB}}$.

The NPB BDT score > 0.5 is used for nominal NPB removal cut. The systematic uncertainty on the NPB cut is evaluated by varying the threshold to 0.3 and 0.7. As an independent closure test of the NPB BDT, the unfolded cross section is also extracted from a back-to-back-recoil-jet-enriched sample with the NPB cut applied and removed; the two cross sections agree to within $\sim 1\%$ across the analysis E_T^γ range, confirming that the BDT does not bias the recovered yield. The full cross-check is documented in Appendix 14.

Figure 18 shows the NPB BDT score on inclusive PYTHIA-8 MC, comparing single-interaction (SI) and double-interaction (DI) samples. Both peak near 1.0, confirming the BDT correctly identifies bulk physics clusters. The fraction below the analysis cut NPB< 0.5 rises from 2.1% (SI) to 8.1% (DI) at 0 mrad, a $\sim$6% absolute migration toward lower scores. After cluster-weighted blending at $f_{\text{DI}} = 0.224$ (0 mrad) and 0.079 (1.5 mrad), this distortion is absorbed into the nominal efficiency (Section 17) rather than handled as a separate correction.

3.5 Signal Identification

In addition to the NPB score > 0.5 cut, further cuts on E_{11}/E_{33}, e_{T1}, and E_{32}/E_{35} are applied to remove other sources of background (e.g. hot towers) and to reduce correlations between the identification and isolation energy (described in Section 3.6.2). These cuts are collectively referred to as the pre-selection, and the full set of criteria is listed in Table 5. Figure 19 shows the shower shape distributions with the pre-selection applied to both data and MC. The non-physics background is greatly reduced, allowing for more detailed data–MC comparisons. The shower

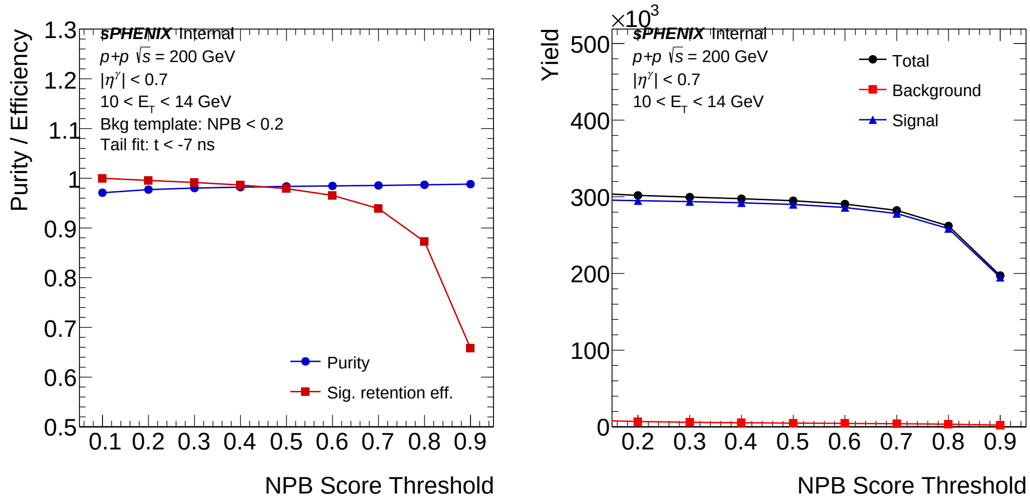

Figure 17: Left: purity (fraction of physics clusters, blue) and signal-cluster efficiency (red) as a function of NPB score threshold for $10 < E_T < 14$ GeV. Right: total (black), NPB (red), and signal (blue) cluster yields entering the purity calculation, all derived from data; the NPB yield is estimated from the low-NPB-score template (NPB-score < 0.2) scaled to each high-score selection in the out-of-time tail ($t_{\text{cluster}} - t_{\text{MBD}} < -7$ ns), and the signal is the total minus the NPB.

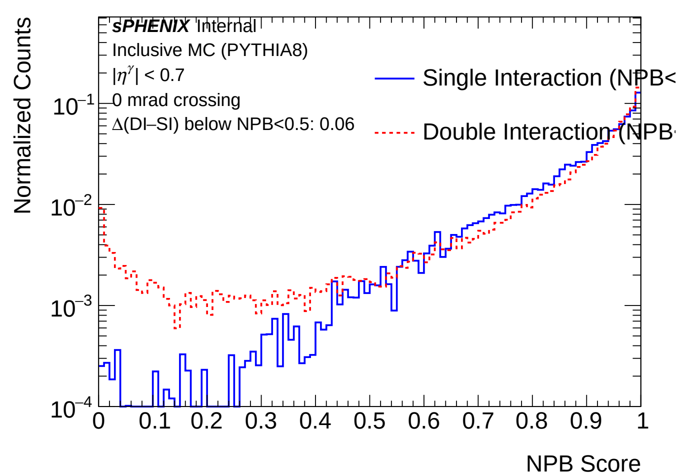

Figure 18: NPB score for inclusive PYTHIA-8 clusters with $|\eta^{\gamma}| < 0.7$ at 0 mrad. Blue: SI. Red dashed: DI. Both unit-area normalised. Integrated fraction below NPB< 0.5: 2.1% (SI) vs 8.1% (DI). After cluster-weighted blending at $f_{\text{DI}} = 0.224$ the net shift on cut survival is sub-percent.

shape distributions with pre-selection for the lower p_{T} bin can be found in Figure 58.

After the pre-selection, shower shape-based selections are used to identify prompt photons from the dominant background, which arises primarily from photons produced in neutral meson decays. The selection criteria that enhance the signal process are referred to as the *tight* selection, and clusters passing them are referred to as *tight clusters*. The complementary criteria, which enhance the background fraction, define the *non-tight* selection and *non-tight clusters*. In this analysis, the tight and non-tight selections are built from a photon identification BDT that combines several

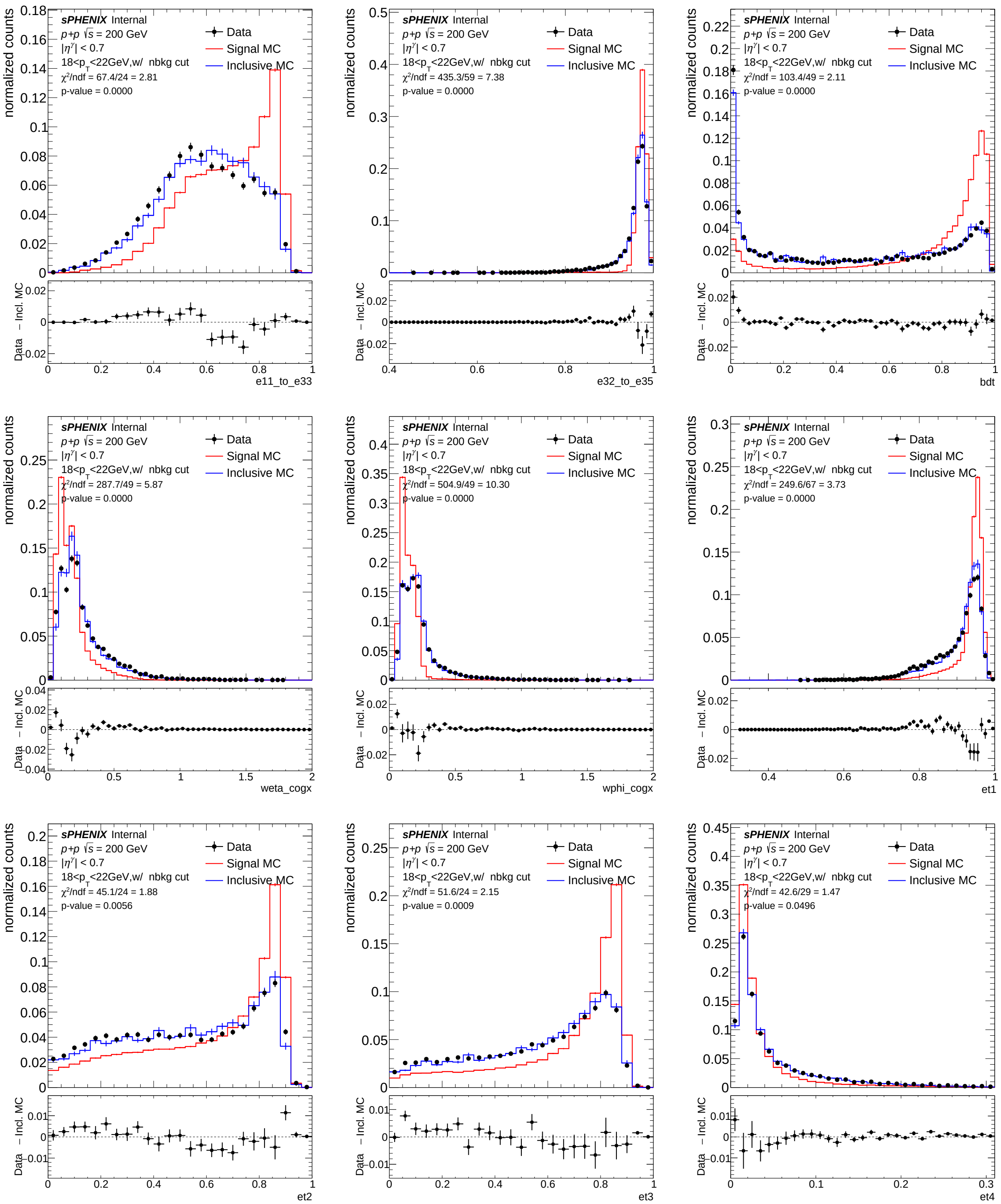

Figure 19: Shower shape distributions with pre-selection at high p_T (18–22 GeV).

shower-shape observables, rather than from rectangular cuts on individual shower shapes. The BDT training and input features are described in Section 3.5.1; the tight and non-tight score ranges used here are parametric in cluster E_{T}^{γ} and are summarized in Table 5. Figure 20 and Figure 21 display the shower-shape distributions after applying the tight BDT score range of Table 5 at lower and higher p_{T}, respectively.

Pre-selection
E11/E33 < 0.98 $0.6 <$et1 < 1.0 $0.8 <$ E32/E35 < 1.0 NPB score > 0.5 (see Section 3.4.2)
Tight
Pass the pre-selection, BDT score $> 0.8156 - 0.00156 \cdot E_{\mathrm{T}}^{\gamma}$ (see Section 3.5.1)
Non-tight
Pass the pre-selection, $0.7333 - 0.01333 \cdot E_{\mathrm{T}}^{\gamma} <$ BDT score $< 0.6844 + 0.00156 \cdot E_{\mathrm{T}}^{\gamma}$

Table 5: Photon identification criteria. The tight and non-tight selections are defined by the photon-identification BDT score (Section 3.5.1); the BDT thresholds are parametric in cluster E_{T}^{γ}. The pre-selection cuts are rectangular cuts on shower-shape variables and the NPB score; the BDT score itself is computed on clusters passing these cuts and encodes the combined discrimination of the individual shower-shape observables.

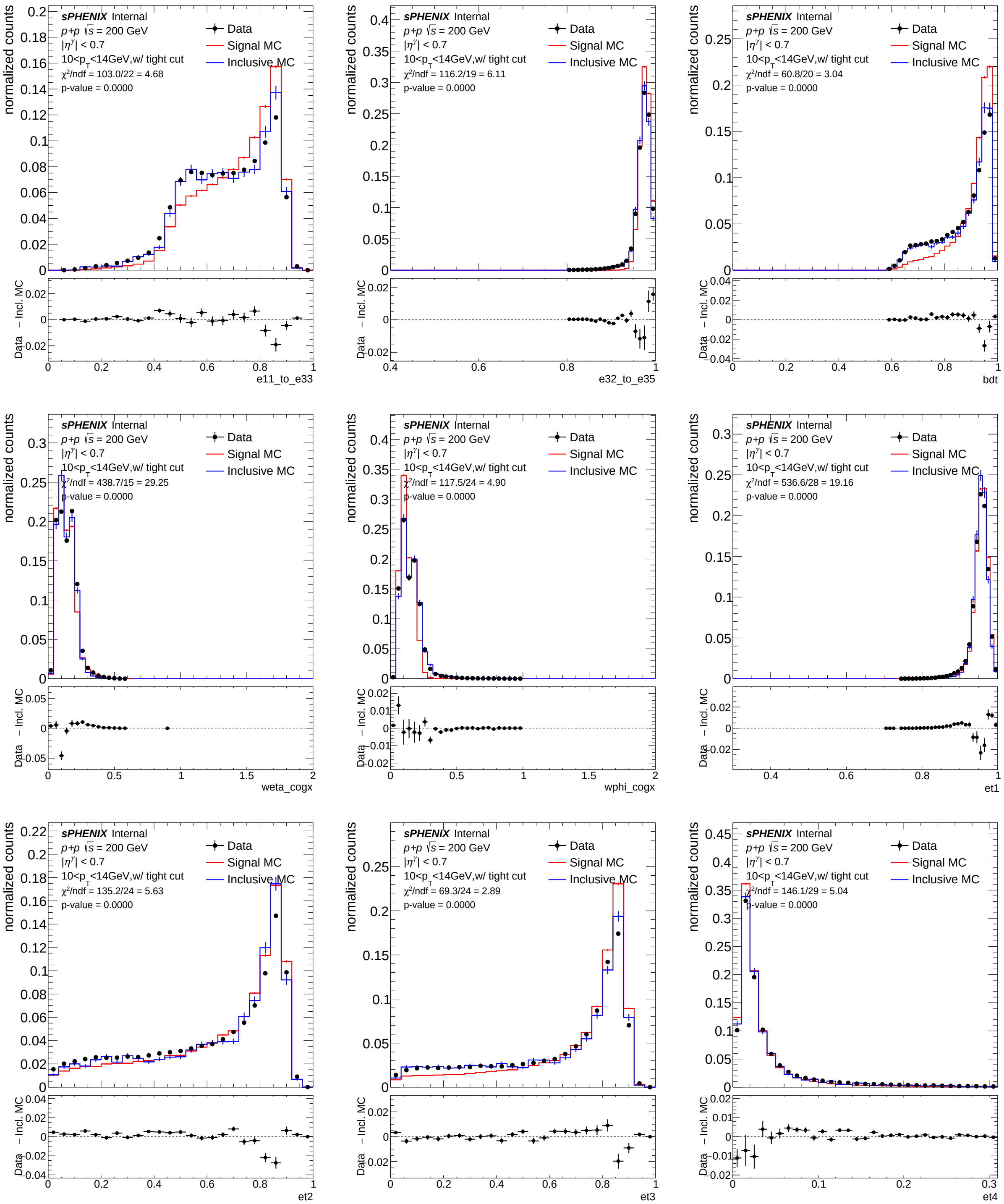

Figure 20: Shower shape distributions with **tight selection** at lower p_T (10–14 GeV).

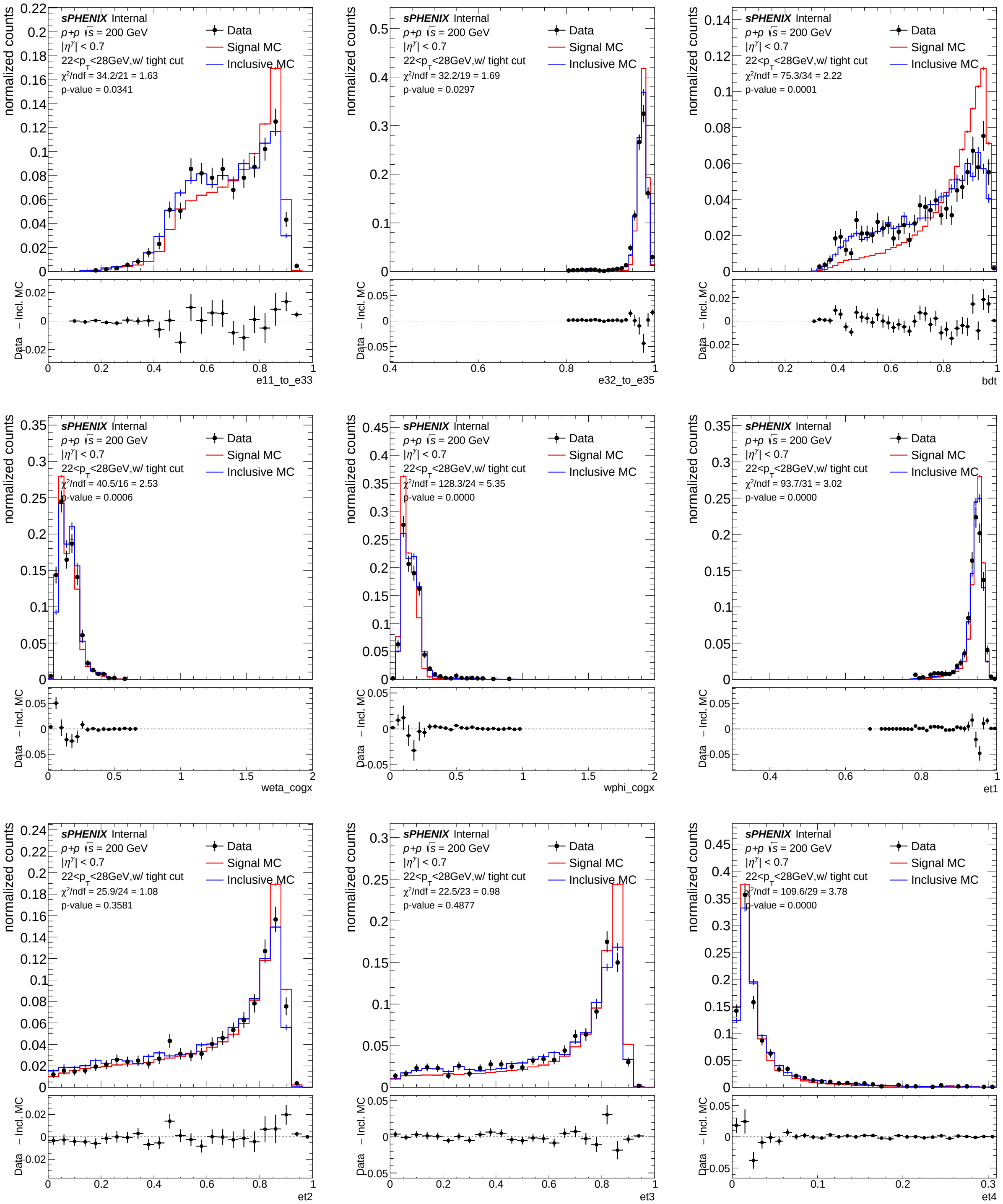

Figure 21: Shower shape distributions with **tight selection** at higher p_T (26–35 GeV).

3.5.1 Photon Identification BDT

An XGBoost boosted decision tree (BDT) is trained to provide a multivariate photon identification score. The BDT score combines multiple shower shape variables into a single discriminant to improve signal-to-background separation. The BDT score distributions are shown alongside the cut-based shower shape variables throughout this note for comparison (e.g., Figure 19).

For the BDT training signal and background clusters are prepared. Signal clusters are truth-matched photons (direct or fragmentation) from PYTHIA-8 photon samples. Background clusters are drawn from PYTHIA-8 inclusive jet samples (jet8/12/20/30) with truth direct/fragmentation photons removed to avoid signal contamination of the background label. The residual electron contribution from W/Z and Drell–Yan at $\sqrt{s} = 200$ GeV is negligible compared to the π^0/η decay-photon background at the analysis E_{T}^{γ} range and is not separately rejected. The BDT uses 11 input features (Table 6). Class reweighting balances signal and background yields; inverse-PDF weights flatten the E_T and η distributions so that discrimination is learned uniformly across kinematics. The XGBoost classifier uses a binary-logistic objective with ~ 750 trees of maximum depth 5 and standard regularization. To avoid self-training bias, MC events are split 50/50: one half trains the BDT, the other is scored by the trained model and used downstream.

Feature	Description
E_{T}	Cluster transverse energy
w_{η}^{COGX}	Energy-weighted second moment of η
w_{ϕ}^{COGX}	Energy-weighted second moment of ϕ
z_{vtx}	Collision vertex z position
η	Cluster pseudorapidity
E11/E33	Ratio of center tower energy to 3×3 tower energy
E_{t1}	$(E_1 + E_2 + E_3 + E_4)/E_{\mathrm{tot}}$
E_{t2}	$(E_1 + E_2 - E_3 - E_4)/E_{\mathrm{tot}}$
E_{t3}	$(E_1 - E_2 - E_3 + E_4)/E_{\mathrm{tot}}$
E_{t4}	E_3/E_{tot}
E3x2/E3x5	Ratio of energy in a 3×2 ($\eta \times \phi$) tower region to a 3×5 region, with the towers chosen around the cluster centre of gravity

Table 6: Input features for the photon-ID BDT, listed in the order they enter the model.

3.6 Isolated Photons

The reconstruction-level cluster isolation transverse energy ($E_{\mathrm{T}}^{\mathrm{iso}}$) can be constructed using several different strategies. One approach is to sum the E_{T} of all EMCal, inner HCal, and outer HCal towers that satisfy the *isGood* tower quality selection, exceed a noise threshold such as 60 MeV, and fall within an isolation cone of a given radius, excluding the $E_{\mathrm{T}}^{\mathrm{reco}}$ of the cluster of interest. Alternatively, one can sum the E_{T} of TopoClusters within a given radius, again excluding the $E_{\mathrm{T}}^{\mathrm{reco}}$ of the cluster of interest. We note that the cluster of interest, i.e. the photon candidate, is reconstructed using RawClusterBuilderTemplate clustering rather than TopoCluster.

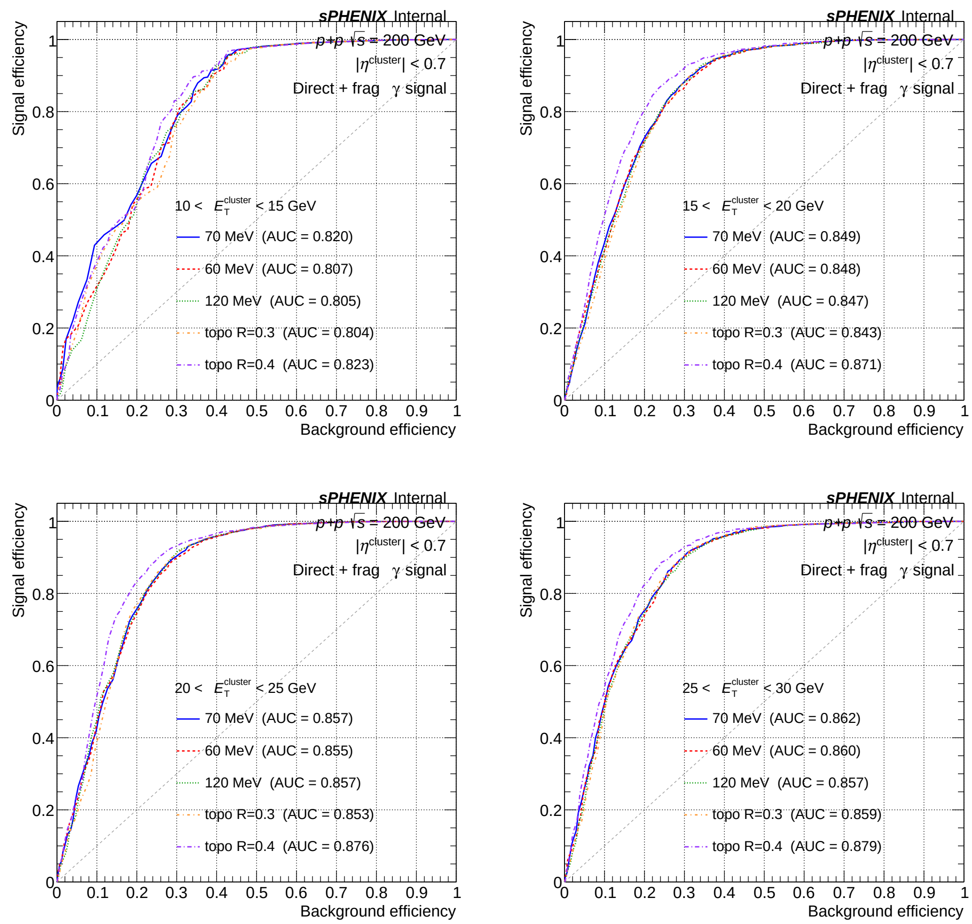

Figure 22: ROC curves comparing five $E_{\mathrm{T}}^{\mathrm{iso}}$ definitions in four cluster E_{T}^{γ} bins. Each curve shows signal efficiency vs. background efficiency as the $E_{\mathrm{T}}^{\mathrm{iso}}$ cut is varied. The area under the curve (AUC) is quoted in the legend. Higher AUC indicates better signal-background discrimination. The nominal 70 MeV tower-based definition ($\Delta R = 0.3$) is shown in blue.

To motivate the choice of $E_{\mathrm{T}}^{\mathrm{iso}}$ definition used in this analysis, five candidate definitions are compared using receiver operating characteristic (ROC) curves evaluated with PYTHIA-8 MC. The ROC curve is built by sweeping the $E_{\mathrm{T}}^{\mathrm{iso}}$ cut threshold from low to high and recording the signal efficiency (fraction of prompt photons passing the cut) against the background efficiency (fraction of decay photon clusters passing the cut) at each threshold value. A definition with higher discriminating power lies closer to the upper-left corner, and its overall performance is summarized by the area under the curve (AUC), where AUC $= 1$ is perfect and AUC $= 0.5$ is no discrimination. The five definitions compared are:

- **70 MeV threshold**: $E_{\mathrm{T}}^{\mathrm{iso}}$ summed from calorimeter towers with $E_T > 70$ MeV within $\Delta R = 0.3$.
- **60 MeV threshold**: Same as above with a 60 MeV per-tower threshold.
- **120 MeV threshold**: Same as above with a 120 MeV per-tower threshold.
- **Topo cluster** $R = 0.3$: $E_{\mathrm{T}}^{\mathrm{iso}}$ summed from topological clusters within $\Delta R = 0.3$.

- **Topo cluster** $R = 0.4$: $E_{\mathrm{T}}^{\mathrm{iso}}$ summed from topological clusters within $\Delta R = 0.4$.

Figure 22 shows the ROC curves for four E_{T}^{γ} bins spanning 10–30 GeV. The signal is defined as truth-matched prompt (direct or fragmentation) photons passing the truth-level isolation requirement ($E_{\mathrm{T}}^{\mathrm{iso,truth}} < 4$ GeV), while the background consists of clusters from inclusive jet MC samples with no truth prompt photon match. Based on this comparison, the isolation energy with topoCluster R=0.4 provides the competitive or superior signal-background discrimination across all E_{T}^{γ} bins.

Figure 23 shows the mean isolation E_T for each run across all analyzed runs. The mean tower-sum $E_{\mathrm{T}}^{\mathrm{iso}}$ with a 60 MeV threshold exhibits a noticeable increase with run number, reflecting the increase in EMCal noise over time. By contrast, the TopoCluster $R = 0.4$ $E_{\mathrm{T}}^{\mathrm{iso}}$ and the tower-sum $E_{\mathrm{T}}^{\mathrm{iso}}$ with a 120 MeV threshold remain stable throughout the run period. Since the 120 MeV threshold is relatively high and may suppress genuine signal contributions, the TopoCluster $R = 0.4$ definition is chosen as the nominal isolation definition in this analysis. A tower-based $R = 0.4$ definition is not included in the comparison: the 60 MeV tower threshold already shows the strongest run-by-run drift at $R = 0.3$, and a tower-sum at $R = 0.4$ would integrate proportionally more noise per cluster. The topo-cluster $R = 0.4$ is preferred precisely because its dynamic significance-based seeding is robust to per-tower noise drift.

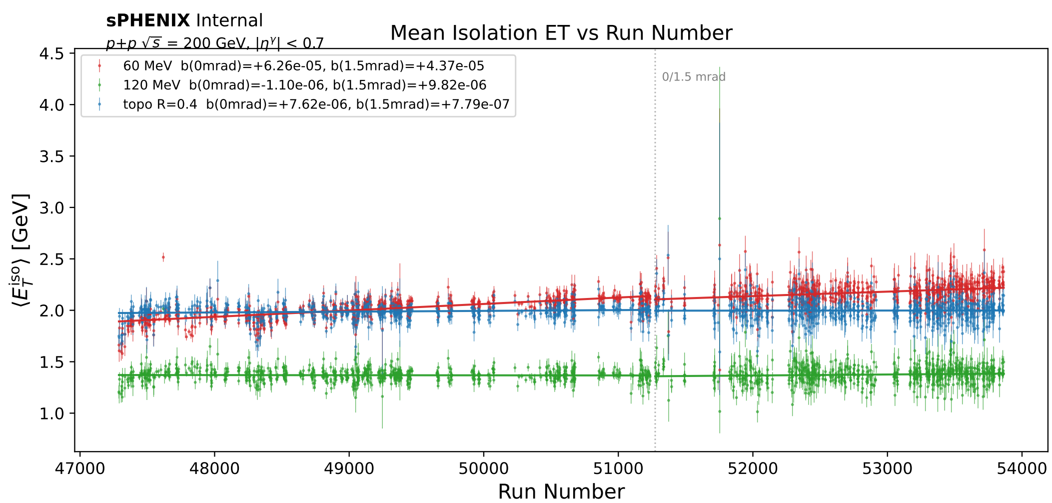

Figure 23: Mean isolation E_T as a function of run number. The TopoCluster $R = 0.4$ (blue) $E_{\mathrm{T}}^{\mathrm{iso}}$ and the tower-sum $E_{\mathrm{T}}^{\mathrm{iso}}$ with a 120 MeV threshold (green) remain relatively flat, while the tower-sum $E_{\mathrm{T}}^{\mathrm{iso}}$ with a 60 MeV threshold (red) increases over the run period. First-order polynomial fits are shown for each $E_{\mathrm{T}}^{\mathrm{iso}}$ definition, separately for the 0 and 1.5 mrad crossing-angle periods. The slopes given in the legend show that the tower-sum $E_{\mathrm{T}}^{\mathrm{iso}}$ with the 60 MeV threshold has the strongest run dependence. The vertical dashed line indicates the boundary between the 0 and 1.5 mrad crossing-angle periods.

The nominal topo-cluster $E_{\mathrm{T}}^{\mathrm{iso}}$ at $R = 0.4$ is constructed as follows. Topological clusters are built from the sPHENIX EMCal and HCal cells using the standard seed-threshold, neighbour-growth, and local-maximum-splitting algorithm (the same topo-cluster scheme used by ATLAS [6]): cells exceeding a seed significance threshold seed new proto-clusters, neighbouring cells above a growth threshold are aggregated iteratively, and local energy maxima within a merged region are split into separate clusters. The isolation variable $E_{\mathrm{T}}^{\mathrm{iso}}$ is then defined as the scalar sum of the transverse energies of all topo-clusters whose axes fall inside a cone of radius $R = 0.4$ around the photon candidate, with the photon candidate's own reconstructed-cluster E_T subtracted from the cone sum to approximately remove the candidate's energy deposit from the topo-cluster total. The cone axis is the (η, ϕ) of the candidate EMCal cluster evaluated at the primary MBD vertex, and the cone opening is measured in geometric (η, ϕ) distance. This definition is insensitive to individual-tower noise drifts (unlike fixed per-tower threshold sums) and is the standard "topological isolation" prescription used in calorimeter-based prompt-photon measurements [7].

3.6.1 Isolation Energy Cut

Using signal photon MC, the E_T^{iso} cuts for 70%, 80%, and 90% isolation efficiency are calculated as a function of E_T^{γ}, as shown in Figure 24. Each selection is fit to a simple linear function. The 80%-efficiency selection is used as the default in this analysis:

$$E_T^{\text{iso}} < 0.490 + 0.0370 \cdot E_T^{\text{reco}}$$

The non-isolated criteria (used in purity estimation described in Section 4.3) is chosen to compromise between signal leakage into the non-isolated sideband regions and statistics in the non-isolated sideband regions. A larger gap in E_T^{iso} between isolated and non-isolated reduces leakage but reduces the statistics of the sideband region. To balance the statistics and signal leakage, the non-isolated criteria is chosen to be:

$$E_T^{\text{iso}} > 0.8 + 0.490 + 0.0370 \cdot E_T^{\text{reco}}$$

Different non-isolated criteria are used as systematic uncertainty for purity.

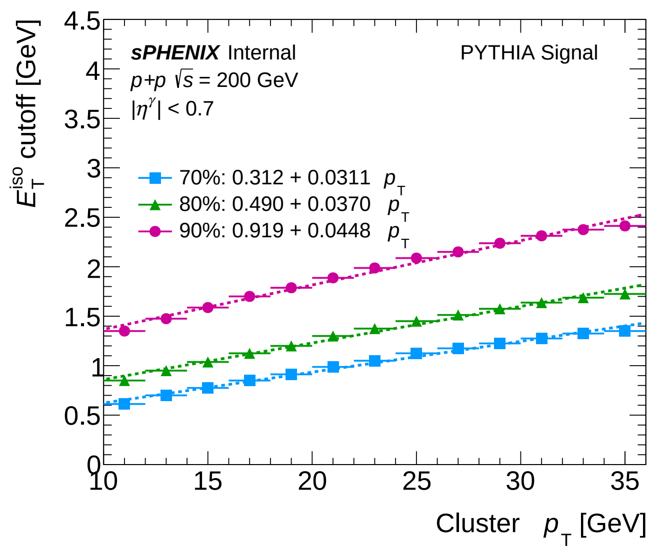

Figure 24: Cut value of the isolation energy E_T^{iso} as a function of the cluster E_T^{reco} to obtain 70%, 80%, and 90% prompt-isolated photon efficiency. Each selection is fitted with a linear function.

3.6.2 Correlation between ID and Isolation

The data-driven purity estimation procedure, discussed in Section 4.3, relies heavily on the independence of $E_{\mathrm{T}}^{\mathrm{iso}}$ and the photon-ID selection for both signal and background. The background physics process in the non-tight control region must have a similar $E_{\mathrm{T}}^{\mathrm{iso}}$ distribution to the background contamination in the tight signal region. To verify this assumption and inform the development of the selection criteria, Figure 25 shows the average $E_{\mathrm{T}}^{\mathrm{iso}}$ as a function of shower-shape value for various shower shapes for an example p_{T}^{γ} bin. Some shower shapes exhibit a stronger correlation with $E_{\mathrm{T}}^{\mathrm{iso}}$ than others. More plots in other p_{T}^{γ} bins are in Appendix 9. The residual non-closure of the ABCD method induced by this correlation is propagated as a symmetric systematic on the data purity (Section 5.3.6) rather than applied as a correction in the nominal analysis.

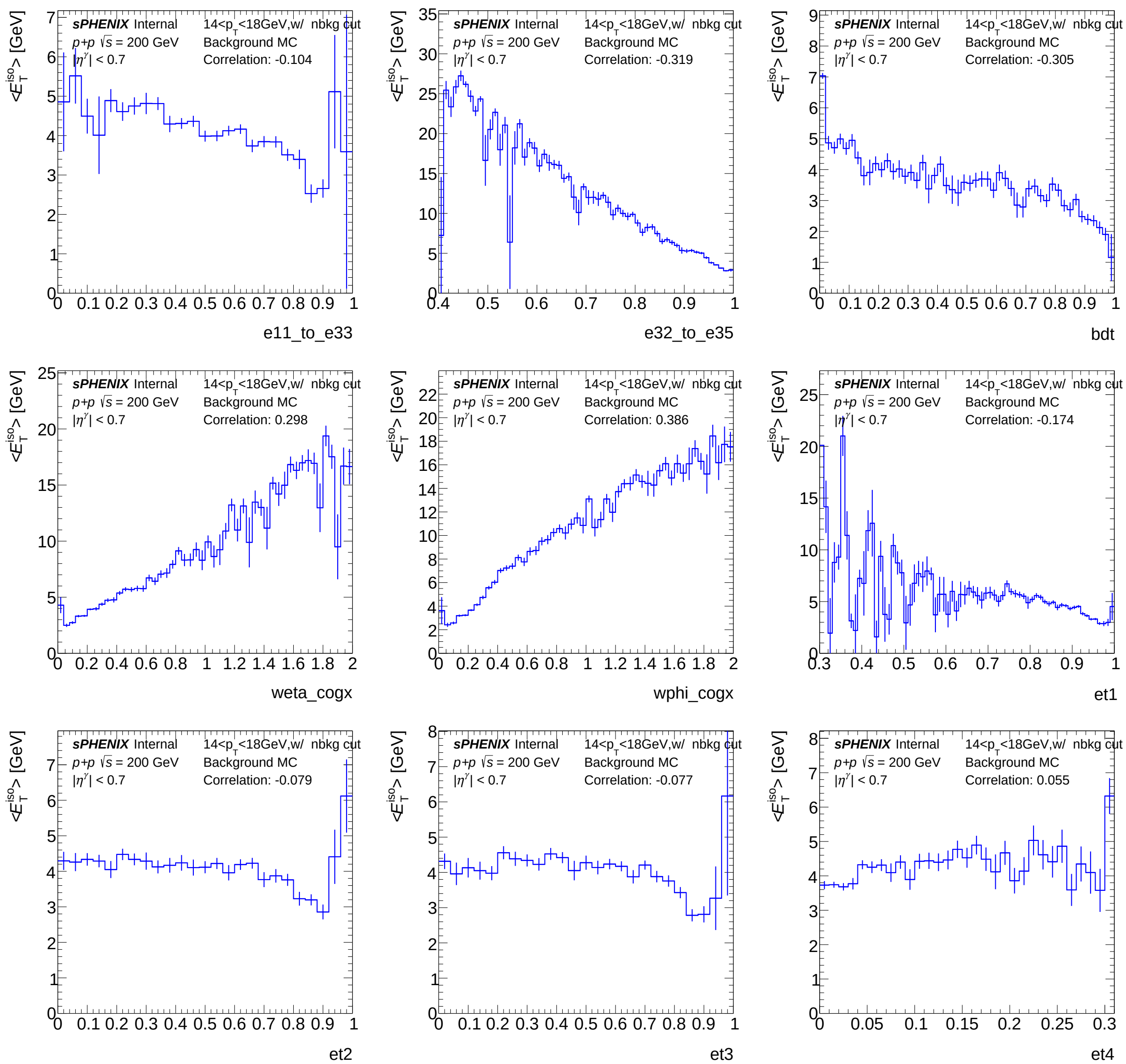

Figure 25: The average $E_{\mathrm{T}}^{\mathrm{iso}}$ as a function of shower shape value for various shower shapes with pre-selection cuts at lower p_{T} (14–18 GeV).

4 Analysis Procedure

4.1 Analysis Workflow Summary

The analysis workflow is summarized as follows:

1. Cluster EMCal towers using RawClusterBuilder with cluster splitting.
2. Apply pre-selection and photon tight identification on shower-shape variables and a cluster-level isolation E_T^{iso} cut using a topo-cluster isolation cone of $R = 0.4$ at reco level (the truth-level fiducial uses $R = 0.3$).
3. Apply the L1 trigger-efficiency correction to data clusters
4. Estimate purity using the data-driven double sideband method and correct for it on a bin-by-bin basis.
5. Unfold the raw (purity-corrected) p_T spectrum using the D'Agostini Bayesian method with RooUnfold.
6. Correct for photon reconstruction, identification and isolation efficiency as a function of truth photon p_T on a bin-by-bin basis.
7. Correct for MBD trigger and event selection efficiency as a function of truth photon p_T on a bin-by-bin basis.
8. Scale by luminosity to obtain the cross section of isolated prompt photons.

The analysis combines the 0 mrad and 1.5 mrad crossing-angle periods. Each period carries its own double-interaction fraction and truth-vertex reweight in MC, so the two are combined at histogram fill time: per-period MC histograms are filled with a luminosity weight $w_i = L_i / \sum_j L_j$ and summed across periods to produce the all-range MC.

4.2 Truth-level Isolated Prompt Photon Definition

Truth-level prompt photons are defined as follows; firstly, final-state particles with pid = 22 in PYTHIA-8 are considered as photons. The final-state photons are also required to be either direct photons or fragmentation photons. Direct photons are defined as photons generated directly from the initial 2-to-2 hard scattering process. Fragmentation photons are defined as photons fragmented from a final-state quark involved in primary hard scattering. The classification of the photon type is achieved by tracking the parton history in the HEPMC record.

The truth-level isolation energy ($E_T^{iso,truth}$) is defined as the sum of the transverse energies of all final-state particles within a cone of $R = 0.3$ in η–ϕ space around the photon of interest, excluding the photon energy itself. This isolation condition suppresses fragmentation photons as well as high-p_T π^0/η decays within jets. In this analysis, truth-level isolated prompt photons are defined by:

- $|\eta^\gamma| < 0.7$
- Direct or fragmentation photon with $E_T^{iso,truth} < 4$ GeV, with isolation cone of $\Delta R = 0.3$.

For truth-reconstructed photon matching, a truth photon is required to be matched with a reconstructed cluster within $|\eta| < 0.7$ and $p_{\mathrm{T}}^{\mathrm{reco}} > 5$ GeV, where the truth-cluster association is taken as the best-match particle from the CaloRawClusterEval module (track-ID-based).

The truth-level isolation ($R = 0.3$, $E_{\mathrm{T}}^{\mathrm{iso,truth}} < 4$ GeV) defines the fiducial cross section, matching the convention used by JETPHOX for the theory comparison. The reconstruction-level isolation ($R = 0.4$ topo-cluster, parametric cut) is independently optimized for detector-level discrimination and stability (Section 3.6). The isolation efficiency $\varepsilon_{\mathrm{iso}}(E_{\mathrm{T}}^{\gamma,\,\mathrm{truth}})$ in Figure 33 converts between the two.

4.3 Purity and Background Subtraction

4.3.1 Purity Estimation

Even after applying the tight identification and isolation requirements on EMCal clusters, there are significant contributions from background photons from high-p_{T} neutral mesons decaying to two photons and reconstructed as one EMCal cluster. The remaining background in our signal criteria is estimated and corrected for. The purity ($\mathcal{P}$) is defined as the fraction of signal photons (1−fraction of background photons), and is estimated through a data-driven double-sideband method. In this method, 4 regions (1 signal region, 3 sideband regions) are defined as:

- A: Signal region, clusters passing tight identification cut and isolation $E_{\mathrm{T}}^{\mathrm{iso}}$ cut
- B: Control region or sideband region, clusters passing tight identification cut and failing $E_{\mathrm{T}}^{\mathrm{iso}}$ cut
- C: Control region or sideband region, clusters passing non-tight identification cut and passing $E_{\mathrm{T}}^{\mathrm{iso}}$ cut
- D: Control region or sideband region, clusters passing non-tight identification cut and failing $E_{\mathrm{T}}^{\mathrm{iso}}$ cut

and illustrated as a diagram shown in Figure 26.

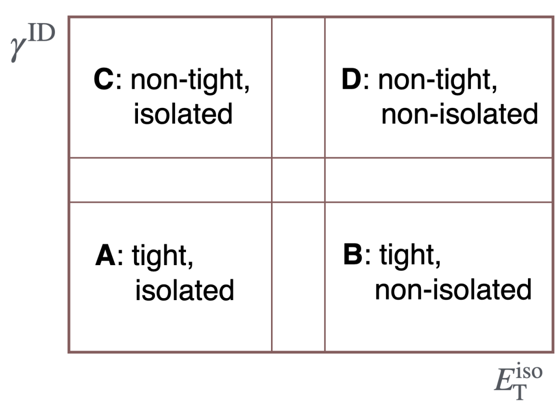

Figure 26: Cartoon of the sideband region definitions.

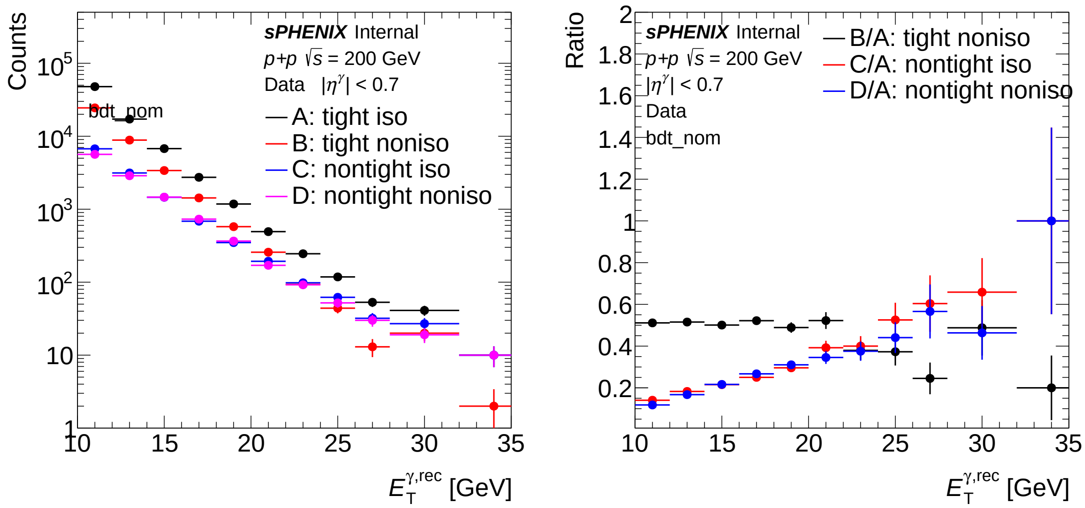

Figure 27: Yield of photons in signal and sideband regions (left) and ratio of yield in each sideband region to the signal region (right) as a function of E_{T}^{γ} in data.

Under the assumption that the isolation $E_{\mathrm{T}}^{\mathrm{iso}}$ is largely uncorrelated with the shower shape variables used in the tight identification criteria, the ratio of background clusters in region C over D should be similar to the ratio of background in region A over B. Therefore the amount of signal photons in A can be calculated with:

$$N_{\mathrm{signal}}^{A,\mathrm{data}} = N_{\mathrm{raw}}^{A,\mathrm{data}} - N_{\mathrm{raw}}^{B,\mathrm{data}} \cdot \frac{N_{\mathrm{raw}}^{C,\mathrm{data}}}{N_{\mathrm{raw}}^{D,\mathrm{data}}} \tag{3}$$

where $N_{\mathrm{signal}}^{A,\mathrm{data}}$ is the number of reconstructed signal clusters in region A, $N_{\mathrm{raw}}^{X,\mathrm{data}}$ is the number of clusters in region X. The number of clusters in each sideband region for each p_{T} bin and their ratio with respect to the signal region A are shown in Figure 27.

4.3.2 Signal Leakage Correction

Equation 3 is valid only if the signal photons do not contribute to the controlled region. However, in real applications, signal photons leak into the controlled region and it needs to be corrected. When accounting for the signal leakage into the controlled region, the sideband subtraction calculation becomes:

$$N_{\mathrm{signal}}^{A,\mathrm{data}} = N_{\mathrm{raw}}^{A,\mathrm{data}} - \left[\left(N_{\mathrm{raw}}^{B,\mathrm{data}} - f^{B,\mathrm{MC}}\, N_{\mathrm{signal}}^{A,\mathrm{data}}\right) \cdot \frac{\left(N_{\mathrm{raw}}^{C,\mathrm{data}} - f^{C,\mathrm{MC}}\, N_{\mathrm{signal}}^{A,\mathrm{data}}\right)}{\left(N_{\mathrm{raw}}^{D,\mathrm{data}} - f^{D,\mathrm{MC}}\, N_{\mathrm{signal}}^{A,\mathrm{data}}\right)}\right] \tag{4}$$

where $f^{X,\mathrm{MC}} = \frac{N_{\mathrm{signal}}^{X,\mathrm{MC}}}{N_{\mathrm{signal}}^{A,\mathrm{MC}}}$ is the ratio of signal clusters in region X over region A, derived with signal MC, shown in Figure 28. The statistical uncertainty on the signal in region A after subtraction is estimated using a bootstrapping method by resampling $N_{\mathrm{raw}}^{X,\mathrm{data}}$, assuming Poisson statistics, with 10,000 iterations.

The purity in the signal region is defined as the fraction of signal photons relative to the total photon candidates in the signal region A (tight, iso):

$$\text{Purity}(\mathcal{P}) = \frac{N_A^{sig}}{N_A}. \tag{5}$$

Figure 29 shows the estimated purity in data both with and without corrections for signal leakage. The leakage-corrected purity rises from 0.51 ± 0.01 at $10 < E_{\mathrm{T}}^{\gamma} < 12$ GeV to 0.72 ± 0.04 at $18 < E_{\mathrm{T}}^{\gamma} < 20$ GeV, and to 0.90 ± 0.05 above 26 GeV.

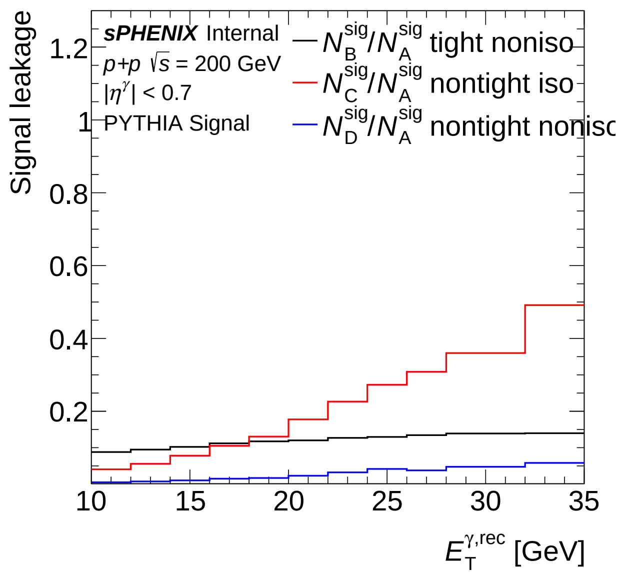

Figure 28: The fraction of signal photons in different sideband regions in MC.

The double-sideband method relies on the photon-ID and isolation discriminants being approximately independent for the background. In practice, the two axes are weakly correlated, as discussed in Section 3.6.2. To evaluate the effect of this correlation, the same purity estimation procedure is performed on inclusive MC samples and compared with the truth-level purity calculated using truth-level information. As shown in Figure 30, the two are in generally good agreement within uncertainties, with small non-closure observed. Figure 31 shows the ratio, taken bin-by-bin, of the truth-level purity to the extracted purity from the ABCD method in inclusive MC, denoted $C_{\mathrm{MC}}(E_{\mathrm{T}}^{\gamma}) = \mathcal{P}_{\mathrm{truth}}^{\mathrm{MC}} / \mathcal{P}_{\mathrm{ABCD}}^{\mathrm{MC}}$, fitted with a smooth function. **The nominal analysis does not apply C_{MC} to the data purity.** Instead, the deviation of the fit from unity is propagated as a symmetric systematic on the purity (Section 5.3.6). This treatment follows the convention used by ATLAS in their two-dimensional sideband analyses [7], where the residual non-closure is propagated as a systematic uncertainty rather than applied as a correction.

4.3.3 Isolation Energy Template Fit

This subsection presents a visual cross-check of the iso-E_{T} decomposition assumed by the ABCD method described above. No fit parameters are derived here. The isolation-energy fudge factor

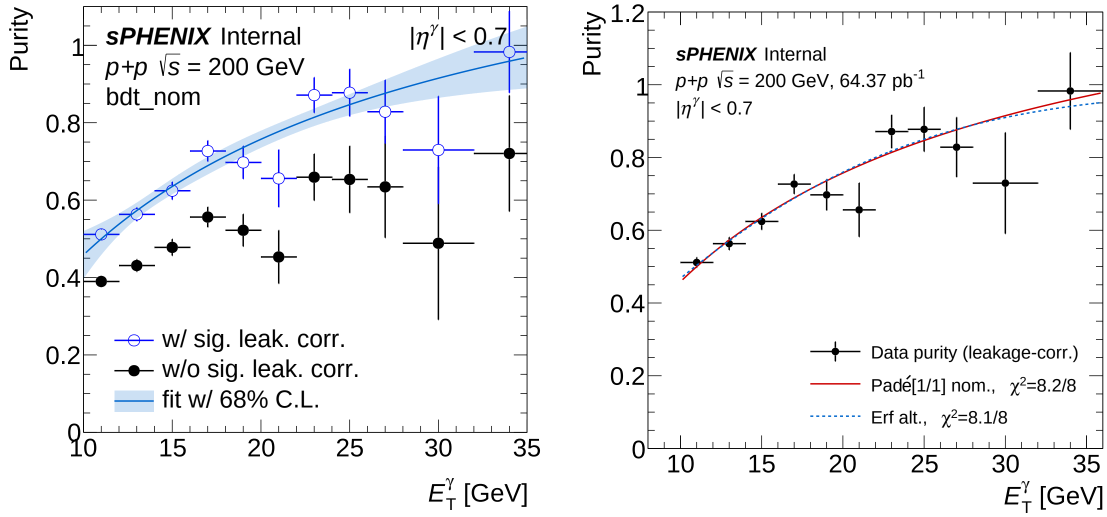

Figure 29: Left: purity as a function of E_T^γ with (black closed circle) and without (blue open circle) signal leakage correction in data. The purity with leakage correction is fitted by an order [1/1] Padé approximant, with the shaded area showing the 68.3% confidence interval of the fit. Right: comparison of the nominal Padé [1/1] fit against the alternative error-function form on the leakage-corrected purity points; the two functional forms agree to $\sim$0.3% across the analysis range and yield essentially equivalent χ^2/ndf. The functional-form spread is propagated as the fit-functional-form systematic uncertainty (Section 5.3.4).

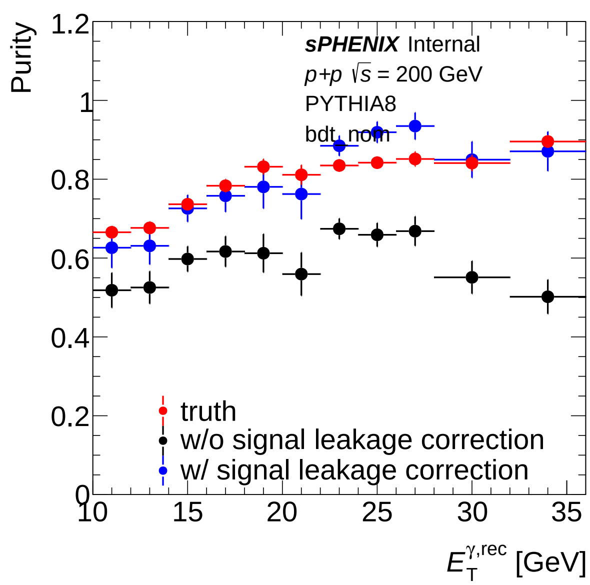

Figure 30: Purity as a function of E_T^γ in inclusive MC as a closure test. The truth-level purity (red) is compared to the purity values, estimated using the same method as data, with (black) and without (blue) signal leakage correction.

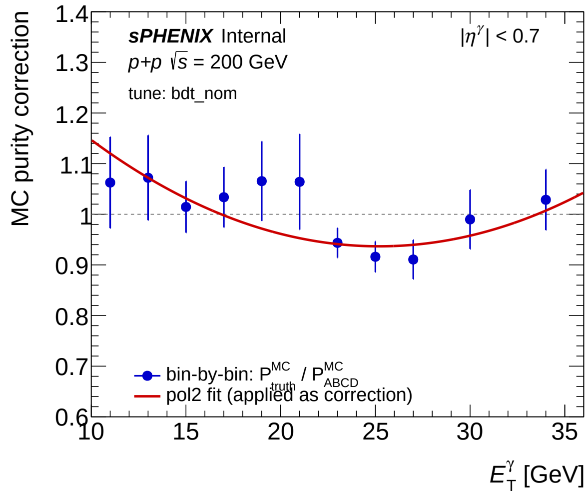

Figure 31: MC purity closure correction $C_{\mathrm{MC}}(E_{\mathrm{T}}^{\gamma}) = \mathcal{P}_{\mathrm{truth}}^{\mathrm{MC}} / \mathcal{P}_{\mathrm{ABCD}}^{\mathrm{MC}}$ as a function of reconstructed photon E_{T}^{γ}. The deviation of the smooth fit from unity is assigned bin-by-bin as a systematic uncertainty. The correction itself is *not* applied to the nominal data purity.

(a, b) used in the systematic uncertainty (Section 5.4) is set independently. Figure 32 shows a two-component template overlay of the data tight-ID $E_{\mathrm{T}}^{\mathrm{iso}}$ distribution: the data non-tight-ID distribution (red, normalized to the data tight-ID yield in the background-dominated tail $E_{\mathrm{T}}^{\mathrm{iso}} > 6$ GeV) and the signal MC tight-ID distribution (blue, scaled so the stack matches the data tight-ID integral) are stacked. The stack reproduces the data tight-ID shape across the analysis E_{T}^{γ} range, providing visual support for the ABCD-style signal/background decomposition described above. The MC reconstructed isolation energy used in the analysis is empirically corrected per cluster as

$$E_{\mathrm{T}}^{\mathrm{iso}\,\mathrm{MC,\,corr}} = a \cdot E_{\mathrm{T}}^{\mathrm{iso}\,\mathrm{MC}} + b, \qquad a = 1.2, \quad b = 0.1 \;\; \mathrm{GeV}, \tag{6}$$

to MC only; data are unmodified. The residual systematic is reported in Section 5.4.

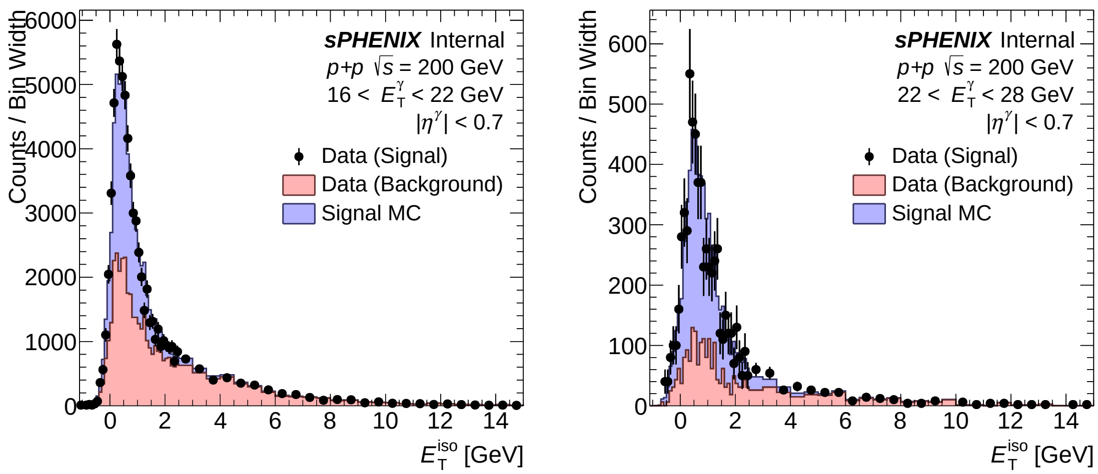

Figure 32: Two-component template fit of the data tight-ID E_T^{iso} distribution in two representative E_T^γ ranges (left: 16–22 GeV, right: 22–28 GeV). In each panel, the data tight-ID distribution (black closed circles) is overlaid on a stack of the data non-tight-ID template (red filled, scaled so its integral matches the data tight-ID yield in the background-dominated region $E_T^{iso} > 6$ GeV) and the signal MC tight-ID template (blue filled, scaled so the total stack integral equals the data tight-ID integral). The stacked templates reproduce the shape of the data tight-ID distribution.

4.4 Efficiency

The photon reconstruction, identification, isolation efficiencies, and event-level selection efficiencies are estimated as functions of the truth-level photon E_T^γ. The truth-reco matching requirement is described in Section 4.2. Each efficiency definition is listed:

- **Reconstruction efficiency**: The fraction of truth-level signal photons (defined in Section 4.2) that are matched to a reconstructed cluster.
- **Identification efficiency**: The fraction of truth-level signal photons passing both the pre-selection criteria and the tight photon identification defined in Section 3.4 out of all truth-level signal photons that are matched to reconstructed photons.
- **Isolation efficiency**: The fraction of truth-level signal photons satisfying the isolation requirements outlined in Section 3.6 out of all truth-level signal photons that are matched to reconstructed photons.
- **MBD vertex reconstruction efficiency**: The fraction of truth-level isolated signal photons in events that satisfy MBD north and south coincidence and yield a reconstructed vertex within the analysis-fiducial $|z_{\text{reco}}| < 60$ cm window. The denominator uses the truth-vertex window set by `vertex_cut_truth` (the all-z fiducial in the nominal configuration), so this efficiency converts the beam-delivered luminosity to events with a usable reconstructed vertex.

For the efficiency calculation, truth-level prompt photons are matched to reconstructed clusters using the track-ID-based association provided by the CaloRawClusterEval module. `CaloRawClusterEval` assigns to each reconstructed cluster the truth primary particle whose G4 hits contribute the largest summed energy across the cluster's towers. In the analysis loop, a truth photon is associated with the cluster whose stored leading-energy truth track ID matches the photon's track ID.

Figure 33 shows photon reconstruction, identification, and isolation efficiencies, respectively, as well as the combined efficiency. The MBD vertex reconstruction efficiency is shown in Figure 34. These efficiencies are corrected on a bin-by-bin basis to the spectrum after unfolding.

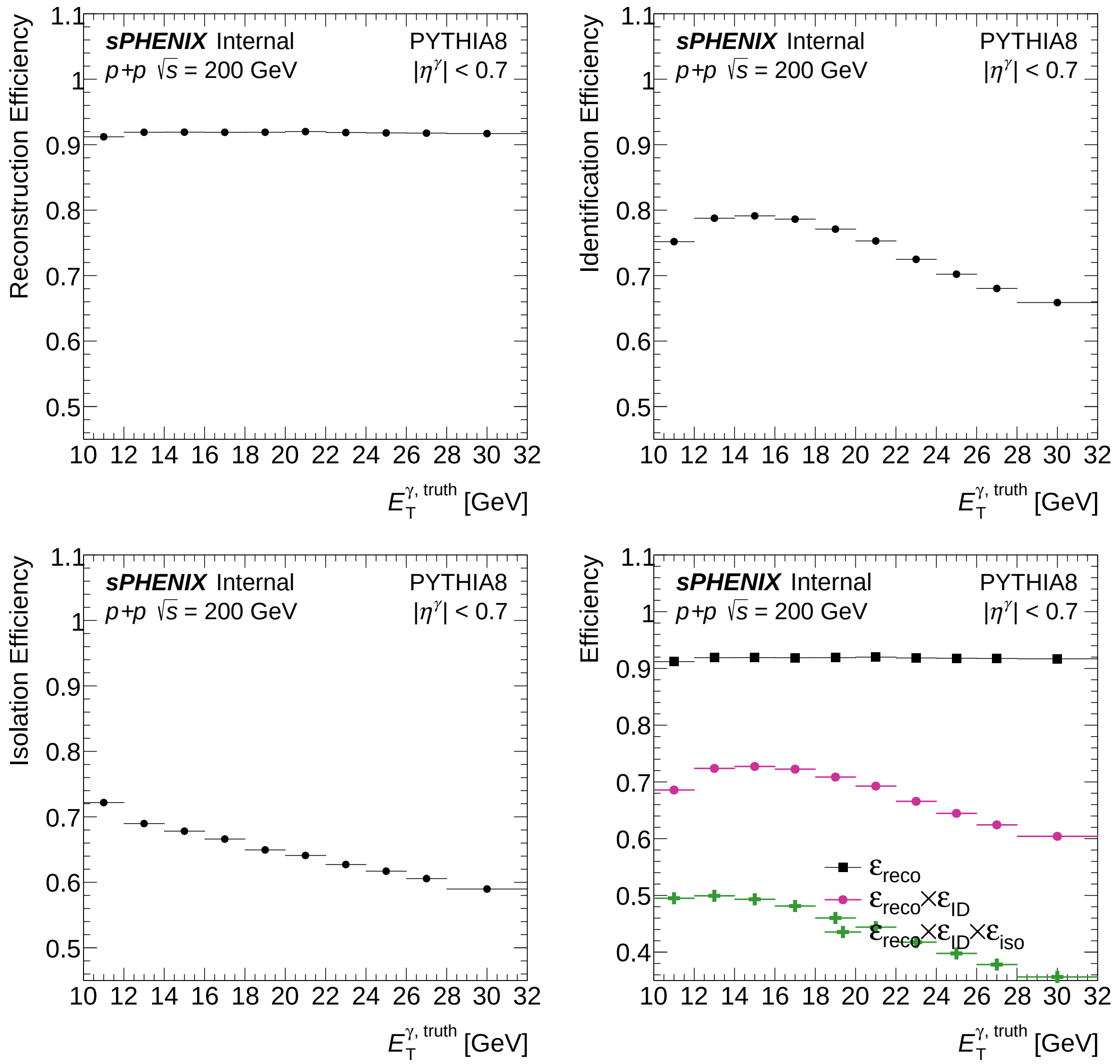

Figure 33: Photon efficiencies for reconstruction (top left), identification (top right), isolation requirement (bottom left), and convoluted step-by-step efficiencies (bottom right) as a function of truth photon $E_T^{\gamma,\,\text{truth}}$.

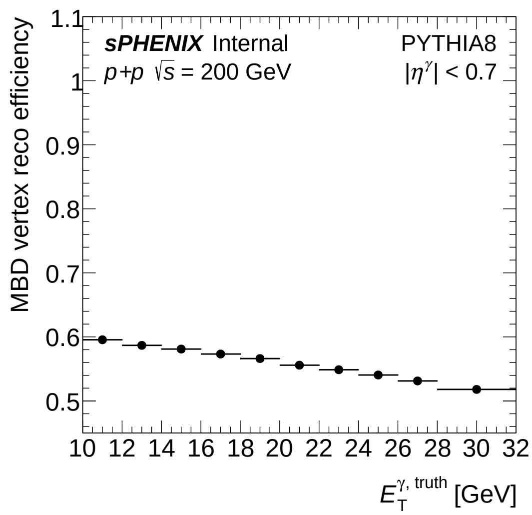

Figure 34: MBD vertex reconstruction efficiency as a function of truth photon $E_T^{\gamma,\,\text{truth}}$, defined as the fraction of truth-isolated photon events in which the MBD north and south coincidence is satisfied and a reconstructed vertex within $|z_{\text{reco}}| < 60$ cm is obtained.

4.5 Unfolding

The detector-level spectrum is corrected back to truth level using the D'Agostini iterative Bayesian method [8] as implemented in RooUnfold version 3.0.5 [9]. The iterative method is preferred over a matrix-inversion or SVD approach for two reasons: it converges quickly when the response matrix is close to diagonal (as is the case here, given the $\sim$5% EMCal energy resolution), and the regularization is set transparently by the iteration count. The number of iterations is chosen on the basis of the closure tests below: each additional iteration reduces residual bias at the cost of increased statistical noise, and the choice balances the two.

The fiducial binning for the final reported result is [12, 14, 16, 18, 20, 22, 24, 26, 28, 32] GeV. The response matrix is built on a wider grid that includes a [10, 12] GeV reco-side underflow and a [32, 36] GeV reco-side overflow, paired with truth-side bins of [8, 10] GeV (low-end underflow) and [36, 45] GeV (high-end overflow). The over/underflow bins absorb migration in/out of the reported range and are not themselves reported.

To make the E_{T}^{γ} prior in MC similar to data, the response matrix is reweighted using the ratio of an iter=1 D'Agostini unfold of the data spectrum, derived from a nominal-baseline pipeline run with the truth E_{T}^{γ} prior reweight disabled, to the un-reweighted MC truth prior from the response matrix (Figure 35). Both distributions are normalized to unit area before forming the ratio. The data-to-MC ratio is fitted by an order [2/2] Padé approximant from 10 to 35 GeV (with the rational denominator zeros constrained to lie below the [10, 30] GeV clamp range used at evaluation time), and the reweighting factor is applied photon-to-photon based on $E_{\mathrm{T}}^{\gamma,\,\mathrm{truth}}$ while populating the response matrix in the next pipeline iteration.

Figure 36 shows the reweighted response matrix, constructed from signal MC photons matched to reconstructed clusters passing the isolation and tight selection criteria. Because the photon energy resolution is small, the impact of unfolding on the spectrum is minor compared to the statistical uncertainties. As shown in Figure 37, the relative change between successive iterations drops below the statistical uncertainty after the first iteration. The nominal result uses the data-reweighted response matrix with $N_{\mathrm{iter}} = 2$: this satisfies the convergence criterion that successive-iteration changes fall below the per-bin statistical uncertainty while keeping the iteration-dependent bias subdominant to other systematics. Figure 38 shows the photon yield as a function of E_{T}^{γ} at each correction stage; the ratio of post- to pre-unfolding yield is 2–10% across the E_{T}^{γ} range.

4.5.1 Unfolding Closure Test

To validate the unfolding procedure, two closure tests were performed:

- Full closure test: The response matrix is constructed using the entire events in the MC sample, and the same MC events are unfolded using this matrix.
- Half closure test: The response matrix is built with half of the events in the MC sample, and the unfolding is applied to the independent remaining half.

Figure 39 shows the closure test results: the full closure test fully reproduces the truth-level spectrum. The half closure test achieves agreement within 2%, consistent with statistical uncertainties.

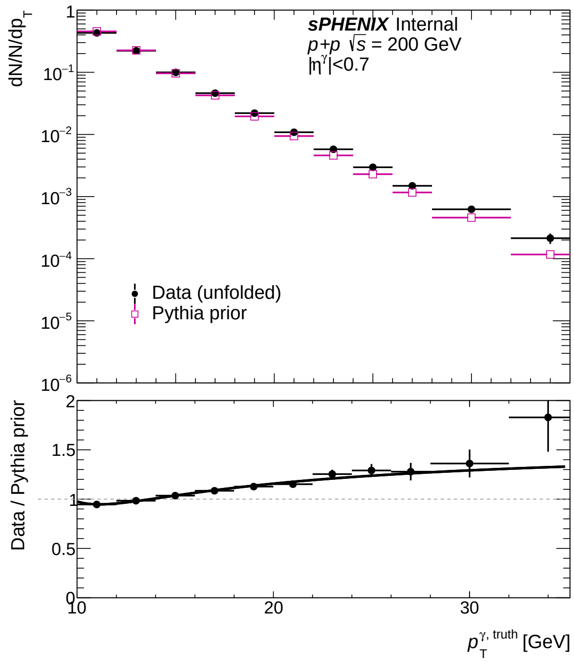

Figure 35: Top panel: iter=1 D'Agostini unfold of the data spectrum (black) and the un-reweighted MC truth prior from the response matrix (magenta), both normalized to unit area. The data is taken from a nominal-baseline pipeline run with the truth-E_T^γ prior reweight disabled. Bottom panel: ratio of data to prior, fitted with an order [2/2] Padé approximant on $10 < E_T^{\gamma,\text{ truth}} < 35$ GeV. The fit is applied photon-to-photon at response-matrix.

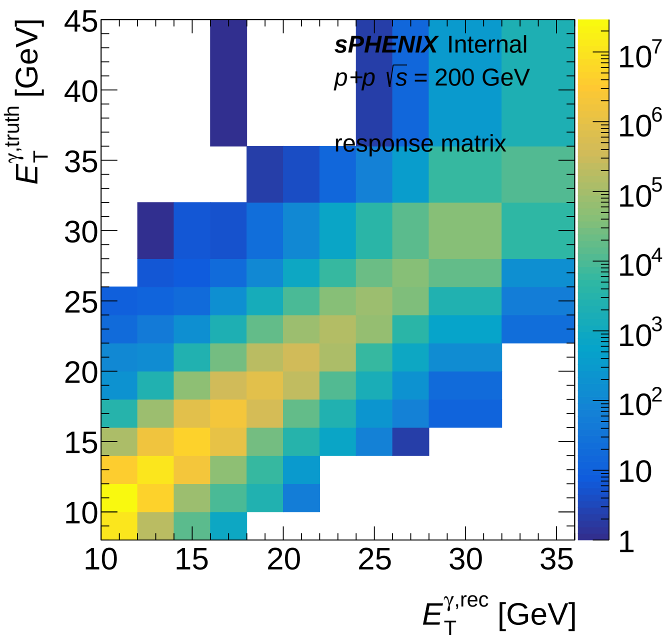

Figure 36: Response matrix after prior reweighting.

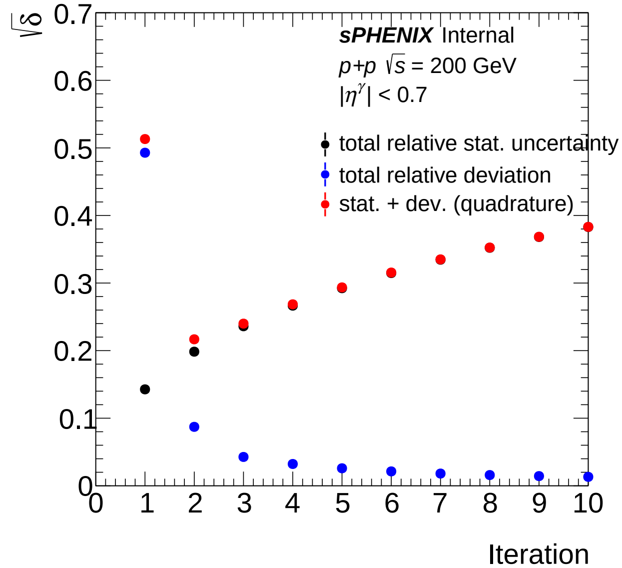

Figure 37: Unfolding relative change and statistic efficiency for each unfolding iteration

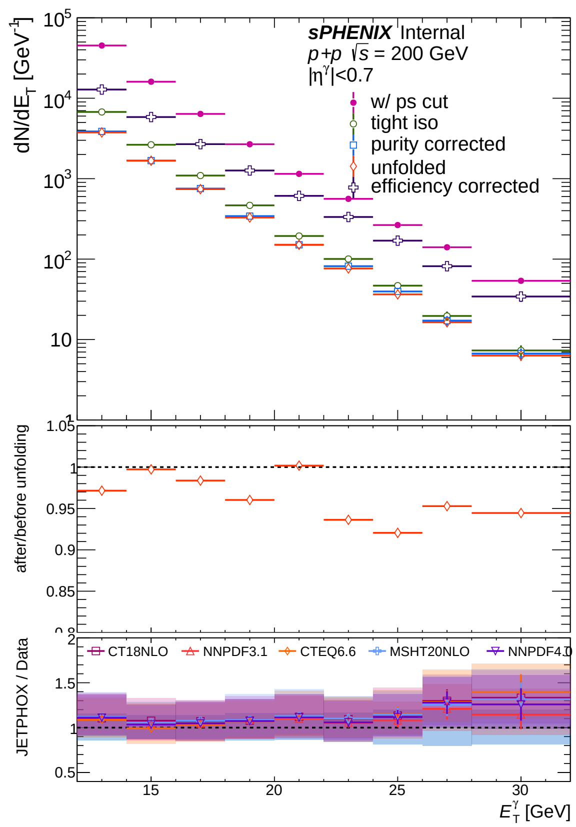

Figure 38: Top panel: yield of photons as a function of E_T^{γ} at different stages of the analysis. Bottom panel: the ratio after and before unfolding (unfolded/purity corrected)

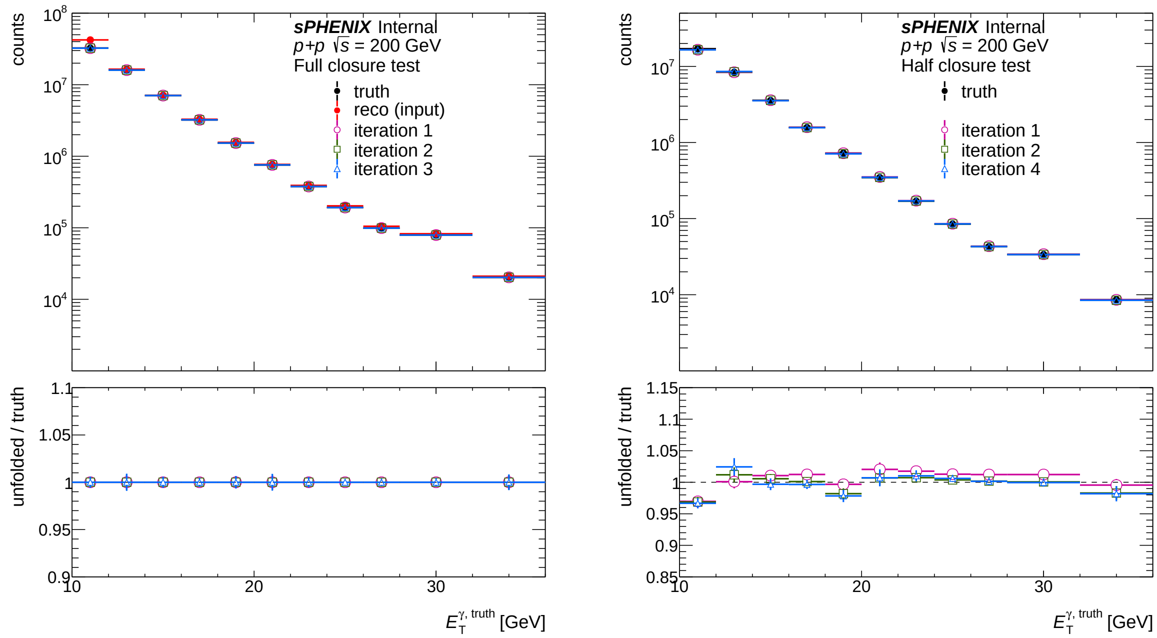

Figure 39: Unfolding full (left) and half (right) closure test result, with truth and unfolded spectrum as a function of E_T^{γ}, the lower panels of each plot shows the ratio between unfolded spectrum and truth.

4.6 Luminosity

The integrated luminosity used in this analysis is the beam-delivered (all-z) value sampled by the photon4 GeV (bit 30) trigger over the analysis runs, $\mathscr{L} = 64.37$ pb^{-1}. It is computed per run from the live MBD N & S coincidence (bit 10) scaler count, divided by the photon4 GeV trigger prescale and by the calibrated MBD inelastic cross section $\sigma_{\text{MBD}} = 25.2^{+2.3}_{-1.7}$ mb measured in a dedicated van-der-Meer (vdM) scan during Run 24, with a per-run pileup correction. No event-level MBD coincidence nor vertex selection is imposed in the lumi calculation itself: the analysis $|z_{\text{reco}}| < 60$ cm cut is reconciled with this all-z lumi via the MBD-vertex efficiency (Section 4.4), whose all-z truth denominator pairs with the all-z lumi while its numerator restores $|z_{\text{reco}}| < 60$ cm together with the MBD N & S ≥ 1 hits required to reconstruct the vertex. The full calculation, including the vdM-scan analysis, is documented in the sPHENIX integrated-luminosity analysis note [10]. The uncertainty on σ_{MBD} is propagated as the asymmetric $+9.13\%/-6.75\%$ luminosity systematic (Section 5.10).

5 Systematic Uncertainties

Systematic uncertainty sources of this analysis are organized into the following response-driven variations plus a single multiplicative-flat term (luminosity).

- Photon energy scale (Subsection 5.1) and photon energy resolution (Subsection 5.2)
- Purity:
  - Tight ID and non-tight ID regions (Subsections 5.3.1–5.3.2)
  - Isolation region (Subsection 5.3.3)
  - Fit functional form, fit statistical uncertainty, and MC-closure correction (Subsections 5.3.4–5.3.6)
- Efficiency: E_T^{iso} shape mismodelling and generator cross-check (Subsections 5.4.1–5.4.2)
- MBD vertex efficiency (Subsection 5.5)
- Unfolding: response reweighting and Bayesian iteration scan (Subsections 5.6–5.7)
- Double-interaction (Subsection 5.8)
- Non-physical background rejection (Subsection 5.9)
- Luminosity (Subsection 5.10)

The systematic uncertainty for each source is calculated by repeating the whole analysis procedure with the corresponding variation. For two-sided sources (energy scale, energy resolution, non-isolated boundary, NPB cut, luminosity, and the purity-fit confidence interval), the up and down $|\Delta|$ are kept asymmetric per E_T^{γ} bin: the down-side bin column is filled with $|\delta_{\mathrm{down}}|$ and the up-side bin column with $|\delta_{\mathrm{up}}|$. For one-sided sources (tight-ID variation, non-tight-ID variation, fit functional form, MC closure correction, isolation efficiency, response reweighting, and double-interaction blending fraction), the single $|\Delta|$ is symmetrized and applied to both sides. For the unfolding iteration scan, the per-bin envelope $\max\{|\Delta_{\mathrm{iter3}}|, |\Delta_{\mathrm{iter4}}|\}$ is taken and symmetrized. The relative differences are added in quadrature in each E_T^{γ} bin to produce the total uncertainty. The luminosity flat term is applied post-unfold as a multiplicative envelope. The per-source figures below display the *signed* relative deviation $(\sigma_{\mathrm{var}} - \sigma_{\mathrm{nom}})/\sigma_{\mathrm{nom}}$ so that the direction of each variation is visible. The quadrature sum itself takes the absolute deviations described above.

5.1 Photon Energy Scale

The uncertainty from the energy scale calibration is determined by shifting the cluster E_T^{reco} by $\pm 1.1\%$ in the MC as suggested by the *Calorimeter Calibration* Working Group. The difference from each variation is taken bin-by-bin as the two-sided systematic. Figure 40 shows that this variation reaches a maximum systematic uncertainty of -10.5% on the down side and $+12.2\%$ on the up side relative to the nominal across the analysis range $12 < E_T^{\gamma} < 32$ GeV; the size grows with E_T^{γ} because the steeply falling spectrum amplifies bin migration at high E_T^{γ}.

An additional residual non-linearity of the energy-scale calibration is propagated by scaling the cluster E_T^{reco} in MC by $1 + s\,(E_T^{\mathrm{rec}} - 10\ \mathrm{GeV})$ with $s = 5 \times 10^{-4}\ \mathrm{GeV}^{-1}$, i.e. 0 % at $E_T^{\mathrm{rec}} = 10$ GeV rising linearly to $+1\,\%$ at $E_T^{\mathrm{rec}} = 30$ GeV (motivated by sPHENIX testbeam study [11]). The absolute bin-by-bin deviation from the nominal is symmetrized and combined in quadrature with the flat envelope in the energy-scale group. The symmetrized one-sided band reaches a maximum of $\pm 7.2\%$ at $E_{\mathrm{T}}^{\gamma} \simeq 30$ GeV across $12 < E_{\mathrm{T}}^{\gamma} < 32$ GeV.

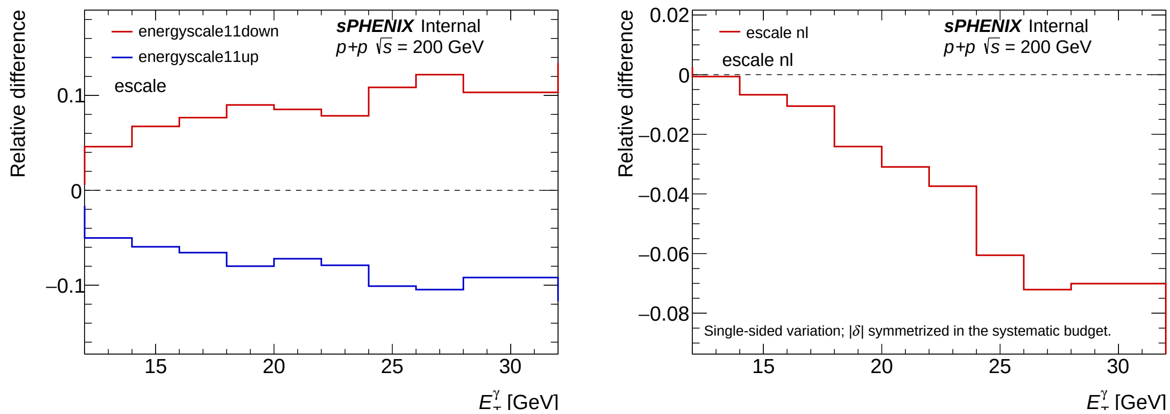

Figure 40: Systematic uncertainty from photon energy scale variation as a function of E_{T}^{γ}. Left: flat $\pm 1.1\%$ envelope. Right: symmetrized one-sided linear residual non-linearity. The two contributions are summed in quadrature into the energy-scale group total used in the cross-section uncertainty.

5.2 Photon Energy Resolution

An additive Gaussian smearing of the cluster $E_{\mathrm{T}}^{\mathrm{reco}}$ is applied to MC at the unfolding response-matrix fill, drawn from $\mathcal{N}\left(0,\, \sigma_{\mathrm{extra}}(E_{\mathrm{T}}^{\gamma,\,\mathrm{truth}}) \cdot E_{\mathrm{T}}^{\gamma,\,\mathrm{truth}}\right)$ with

$$\sigma_{\mathrm{extra}}(E_{\mathrm{T}}^{\gamma,\,\mathrm{truth}}) = \sqrt{\max\left(0,\, \sigma^2_{\mathrm{data}}(E_{\mathrm{T}}^{\gamma,\,\mathrm{truth}}) - \sigma^2_{\mathrm{MC}}(E_{\mathrm{T}}^{\gamma,\,\mathrm{truth}})\right)},$$

where each $\sigma(E)$ follows the standard EMCal form $\sigma(E)/E = \sqrt{p_0^2/E + p_1^2/E^2 + p_2^2}$ with E in GeV. σ_{MC} is fixed from a fit of the unsmeared PYTHIA-8 tight + iso prompt-photon response, $(p_0, p_1, p_2)_{\mathrm{MC}} = (0.185,\, 0,\, 0.040)$. The nominal data term uses $(p_0, p_1, p_2)_{\mathrm{data}} = (0.15,\, 0.05,\, 0.05)$, which yields $\sigma_{\mathrm{extra}} \approx 0$ for $E_{\mathrm{T}}^{\gamma,\,\mathrm{truth}} \lesssim 17$ GeV rising to $\sim 2\%$ above 20 GeV.

The smearing enters only the unfolding response matrix; the ABCD yields, photon-ID, and isolation cuts use the unsmeared cluster $E_{\mathrm{T}}^{\mathrm{reco}}$. The $E_{\mathrm{T}}^{\mathrm{reco}}$ range cut at the response-matrix fill is applied on the smeared value, mirroring the corresponding cut on data with the alternative resolution.

Two variations bracket the nominal: a "no extra smearing" variation sets $\sigma_{\mathrm{data}} \equiv 0$ ($\sigma_{\mathrm{extra}} \equiv 0$), and a wider-data variation takes $(p_0, p_1, p_2)_{\mathrm{data}} = (0.13,\, 0.08,\, 0.08)$, yielding $\sigma_{\mathrm{extra}} \approx 6\%$ approximately flat in $E_{\mathrm{T}}^{\gamma,\,\mathrm{truth}}$. The bin-by-bin maximum of the two absolute deviations is symmetrized and applied to both sides as the eres systematic. Figure 41 shows the resulting symmetric envelope; the maximum across $12 < E_{\mathrm{T}}^{\gamma} < 32$ GeV is $\pm 9.6\%$ at $E_{\mathrm{T}}^{\gamma} \simeq 27$ GeV.

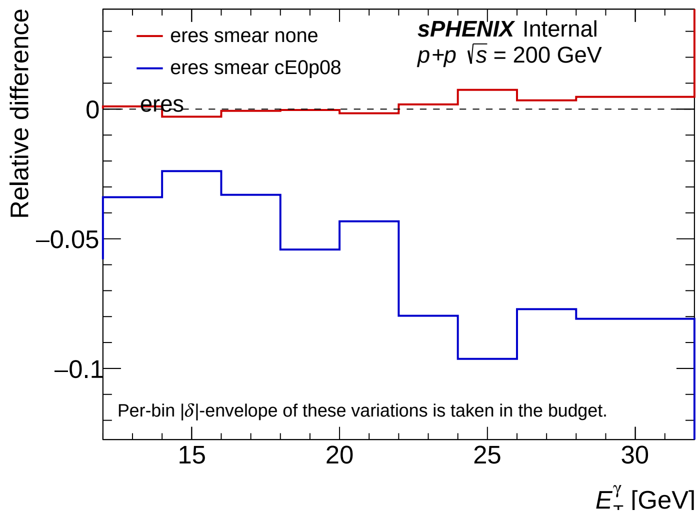

Figure 41: Systematic uncertainty from photon energy resolution variation as a function of E_T^γ.

5.3 Photon Purity

5.3.1 Photon Purity – Tight ID

The tight-BDT intercept is shifted upward by $+0.05$ from its nominal value shown in Table 5 (E_T^γ-dependent threshold tight_bdt_min $= 0.8156 - 0.00156 \cdot E_T^{\mathrm{reco}} + 0.05$) and the analysis is rerun. The symmetrized one-sided deviation is taken as the systematic. Figure 42 shows that this variation reaches a maximum of $\pm 10.4\%$ relative to the nominal selection across the analysis range.

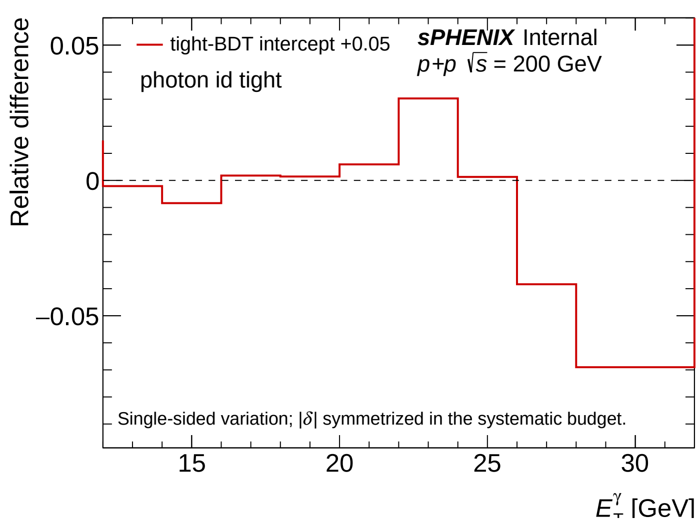

Figure 42: Systematic uncertainty from the tight-ID variation as a function of E_T^γ.

5.3.2 Photon Purity – Non-tight ID

The non-tight-BDT lower edge is shifted downward by -0.10, expanding the non-tight band toward lower BDT scores. This probes the ABCD background-template definition through the non-tight selection. The symmetrized one-sided deviation is taken as the systematic. Figure 43 shows that this variation reaches a maximum of $\pm 2.1\%$ relative to the nominal.

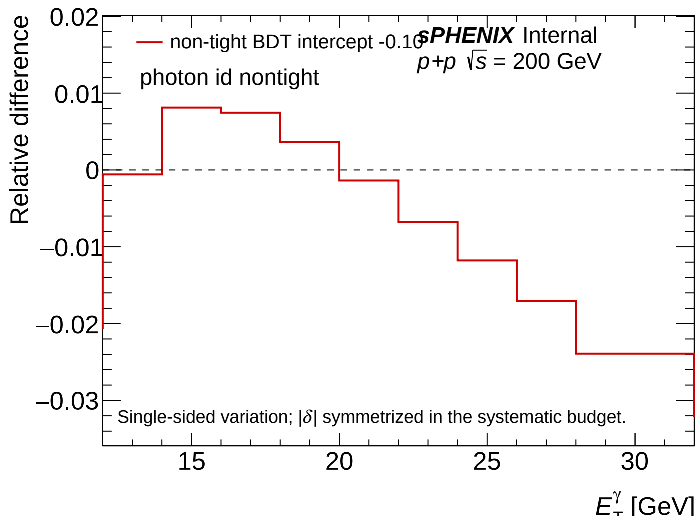

Figure 43: Systematic uncertainty from the non-tight ID variation as a function of E_T^γ.

5.3.3 Photon Purity – isolation energy

The choice of the lower edge of the non-isolation sideband influences the ABCD purity calculation. The nominal definition uses

$$E_T^{\text{iso}} > \text{reco\_iso\_max} + 0.8\ \text{GeV}\,, \qquad \text{reco\_iso\_max} = 0.490 + 0.0370 \cdot E_T^{\text{reco}}\ [\text{GeV}]\,,$$

i.e., a $+0.8$ GeV gap above the parametric upper edge of the isolated region. Two systematic variations are taken:

$$\textbf{tighter sideband (noniso04):} \quad E_T^{\text{iso}} > \text{reco\_iso\_max} + 0.4\ \text{GeV}\,,$$

$$\textbf{looser sideband (noniso10):} \quad E_T^{\text{iso}} > \text{reco\_iso\_max} + 1.0\ \text{GeV}\,.$$

The difference between each variation and the nominal is used as the bin-by-bin systematic. Figure 44 shows that this variation reaches a maximum deviation of -5.1% on the down side and $+0.9\%$ on the up side relative to the nominal.

5.3.4 Photon Purity – Fit Functional Form

The systematic uncertainty from the choice of the purity fitting function is evaluated by replacing the nominal [1/1] Padé approximant (a rational function with first-degree numerator and

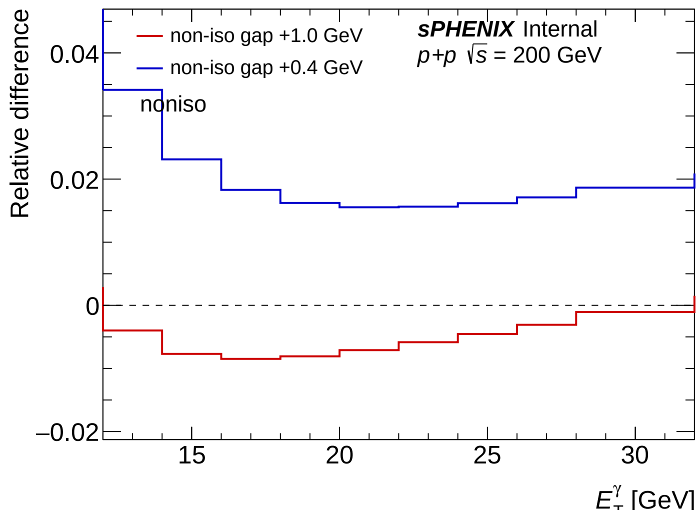

Figure 44: Systematic uncertainty for purity non-isolation region variation as a function of E_{T}^{γ}.

denominator) with an error-function form $\mathcal{P}(E_{\mathrm{T}}^{\mathrm{reco}}) = p_0 \cdot \mathrm{erf}((E_{\mathrm{T}}^{\mathrm{reco}} - p_1)/p_2)$, refitting, and taking the bin-by-bin difference as the symmetric one-sided systematic. Figure 45 shows that this variation reaches a maximum of $\pm 0.7\%$ relative to the nominal across the analysis range. The two functional forms agree well within their parameter uncertainties, so the functional-form ambiguity is subleading compared to the parameter statistical uncertainty assigned in the next subsection.

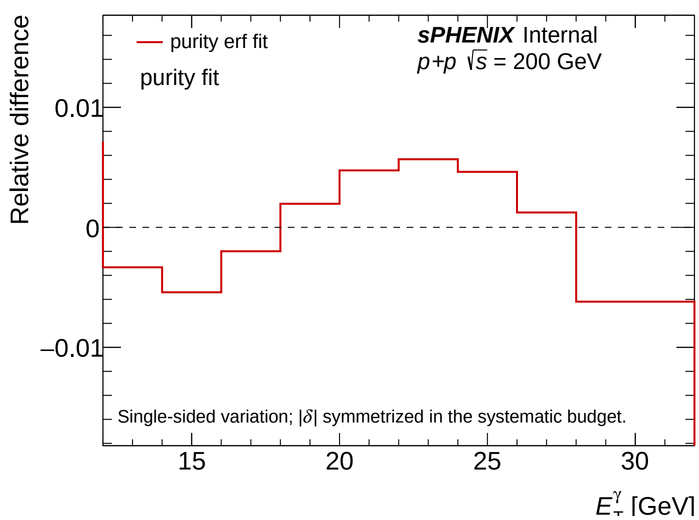

Figure 45: Systematic uncertainty from the purity fit functional form variation as a function of E_{T}^{γ}.

5.3.5 Photon Purity – Fit Statistical Uncertainty

The statistical uncertainty on the nominal [1/1] Padé fit parameters is propagated as an independent two-sided systematic. The 68.3% confidence interval on the parameter posterior, accounting for the parameter covariance, is translated into upper and lower variants of the per-cluster purity $\mathcal{P}(E_T^{\mathrm{reco}})$. Each variant is propagated through the full ABCD background subtraction and unfolding, and the bin-by-bin deviation from the nominal cross section in each direction is taken as the systematic. This source is orthogonal to the functional-form variation of Subsection 5.3.4 and enters the purity-group quadrature sum independently. Figure 46 shows that the variation reaches a maximum of -8.1% on the down side and $+8.2\%$ on the up side relative to the nominal across the analysis range.

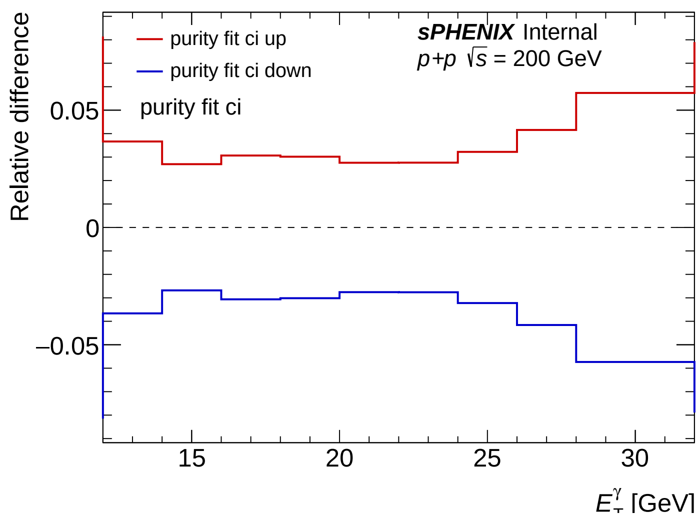

Figure 46: Systematic uncertainty from the purity fit 68.3% parameter confidence interval as a function of E_T^γ.

5.3.6 Photon Purity – MC Closure Correction

The MC-derived closure correction $C_{\mathrm{MC}}(E_T^\gamma) = \mathcal{P}^{\mathrm{MC}}_{\mathrm{truth}} / \mathcal{P}^{\mathrm{MC}}_{\mathrm{ABCD}}$ is constructed as the per-E_T^γ ratio of the truth-level inclusive-MC purity to the leakage-corrected ABCD-extracted MC purity, computed bin-by-bin and fit with a second-order polynomial over the analysis range (the bin-by-bin ratio is shown alongside the fit in Figure 31). The nominal analysis does not apply C_{MC} to the data purity. In the systematic variant, the smooth fit is applied as a multiplicative correction to the data purity, and the relative cross-section change with respect to the (uncorrected) nominal is symmetrized and taken as the systematic uncertainty on the ABCD-independence-violation channel. Figure 47 shows that this variation reaches a maximum of $\pm 12.0\%$ at the low-E_T^γ end ($E_T^\gamma \approx 13$ GeV) of the reported range, dropping to $\sim 6\%$ above 22 GeV.

5.4 Efficiency

The efficiency-group systematic is the per-E_T^γ quadrature of two independent contributions: the data–MC mismodelling of the E_T^{iso} shape, and a generator cross-check using HERWIG 7 in place of the nominal PYTHIA-8 8 Detroit MC. Both enter the efficiency group additively in quadrature.

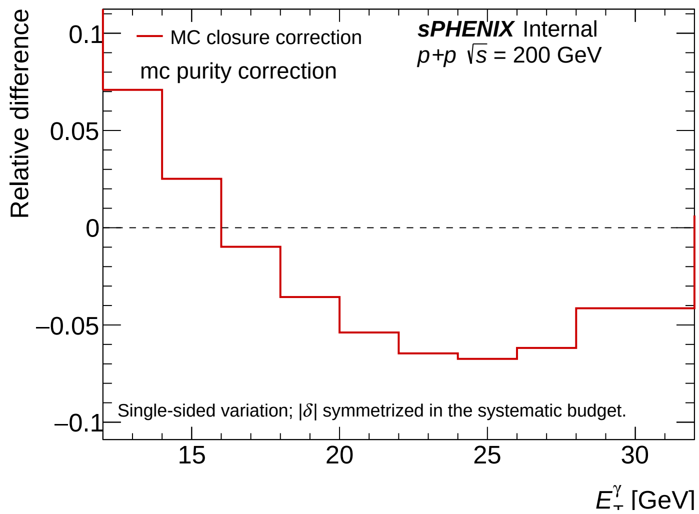

Figure 47: Systematic uncertainty from the MC purity closure correction as a function of E_{T}^{γ}.

5.4.1 $E_{\mathrm{T}}^{\mathrm{iso}}$ shape mismodelling

For the nominal analysis, the $E_{\mathrm{T}}^{\mathrm{iso}}$ in MC is fudged by applying a multiplicative scale factor of 1.2 to match the MC $E_{\mathrm{T}}^{\mathrm{iso}}$ distribution to data. A residual additive pedestal difference between data and MC $E_{\mathrm{T}}^{\mathrm{iso}}$ (after the multiplicative scale factor is applied) is probed by setting the MC $E_{\mathrm{T}}^{\mathrm{iso}}$ pedestal shift to zero while keeping the multiplicative scale at its nominal value. The resulting one-sided deviation, symmetrized bin by bin, captures pedestal-mismodeling effects on the $E_{\mathrm{T}}^{\mathrm{iso}}$ shape. Figure 48 shows that this variation reaches a maximum systematic uncertainty of $\pm 5.0\%$ relative to the nominal.

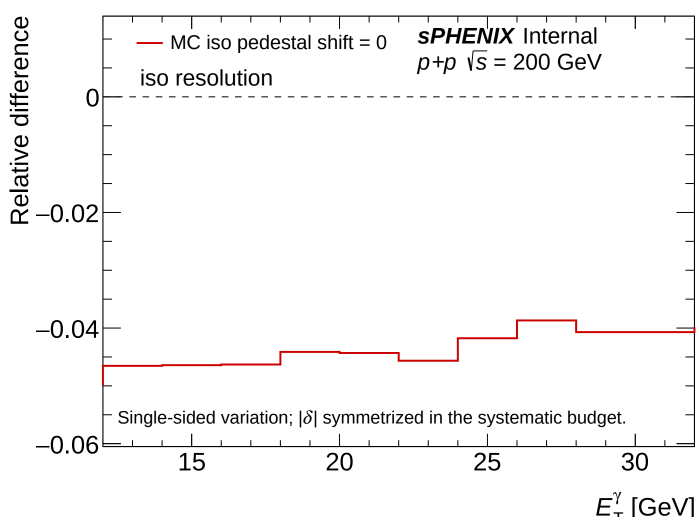

Figure 48: Systematic uncertainty for efficiency from the $E_{\mathrm{T}}^{\mathrm{iso}}$ fudging factor variation as a function of E_{T}^{γ}.

5.4.2 Generator cross-check (HERWIG)

The 1.5 mrad analysis is rerun replacing the PYTHIA-8 8 Detroit signal MC with HERWIG 7 photon-jet samples, with the truth-p_T prior rederived in the same way (Appendix 19). The dominant difference between the two pipelines is in ϵ_{iso}, which sits 3–6% higher in HERWIG due to the generator's lower mid-rapidity underlying-event activity. The systematic on the cross section is constructed by scaling the nominal $d^2\sigma/d\eta\, dE_T^\gamma$ bin-by-bin by $\epsilon_{\text{iso}}^{\text{PYTHIA-8}}/\epsilon_{\text{iso}}^{\text{HERWIG}}$ and taking the symmetrized one-sided deviation from the nominal. Across the reported range $12 < E_T^\gamma < 32$ GeV this systematic reaches a maximum of $\pm 5.3\%$ in the [12, 14] GeV bin. The corresponding ϵ_{ID} and unfolded-yield differences are sub-leading and treated as cross-checks (Appendix 19); the ϵ_{MBD} generator difference is covered by the data-driven MBD-coincidence systematic of Section 5.5 and is not double-counted here.

5.5 MBD Vertex Reconstruction Efficiency

A two-sided $\pm 6\%$ relative systematic, flat in E_T^γ, is assigned to ϵ_{MBD} (Section 4.4). The size is taken from the data-driven MBD-coincidence efficiency reported in PPG10. The MBD vertex reconstruction efficiency is driven by forward underlying-event activity, which is dictated by the incoming-parton kinematics and is therefore common to γ+jet and QCD-dijet events at the same leading truth p_T. In PYTHIA-8 8, the MBD-vertex-found fraction in photon and jet MC samples agrees to within 1% across $5 < p_T^{\text{lead}} < 35$ GeV (Appendix 16), justifying transfer of the data-driven jet result to photon analysis.

5.6 Unfolding — Response Reweight

Since the response matrix in the nominal analysis is reweighted for the prior to match the data spectrum, the systematic is taken from unfolding without reweighting the response matrix. The bin-by-bin deviation, symmetrized, is the prior reweighting systematic. Figure 49 shows that this variation reaches a maximum of $\pm 0.8\%$ relative to the nominal.

5.7 Unfolding — Iteration Scan

The Bayes iteration count is varied around the nominal $N_{\text{iter}} = 2$ to bracket the regularization strength. The maximum bin-by-bin deviation between the iter-3 and iter-4 results and the nominal is taken as the systematic envelope (symmetrized). The iter-1 unfolding is retained as a closure overlay in Section 4.5 but not used as a systematic, because at $N_{\text{iter}} = 1$ the D'Agostini procedure has not yet converged and the regularization-induced bias is dominated by prior dependence rather than by genuine residual unfolding bias. Figure 50 shows that the iter-3/iter-4 envelope reaches a maximum of $\pm 4.7\%$ relative to the nominal across the reported range $12 < E_T^\gamma < 32$ GeV.

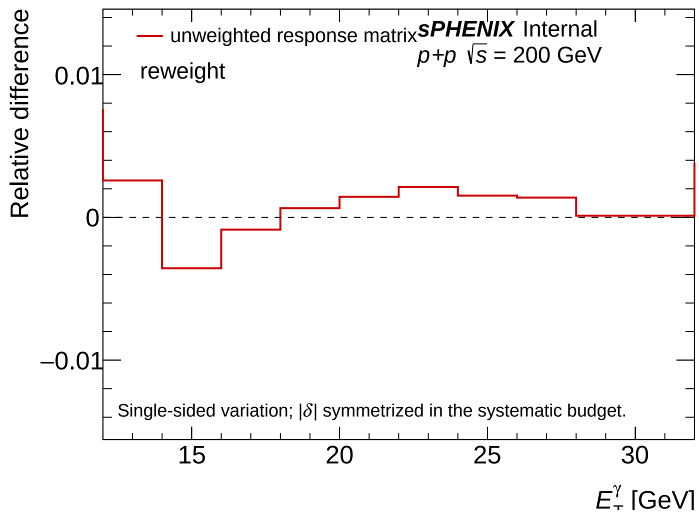

Figure 49: Systematic uncertainty from the unfolding prior reweighting variation as a function of E_T^γ.

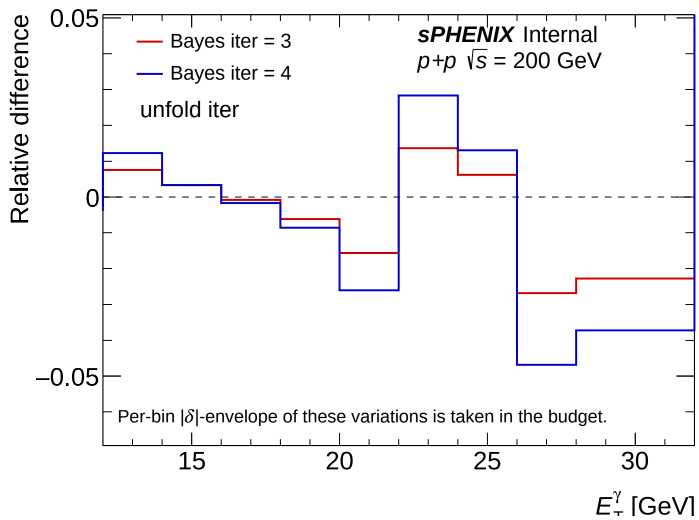

Figure 50: Systematic uncertainty from the unfolding iteration scan as a function of E_T^γ.

5.8 Double-Interaction Blending Fraction

The double-interaction (pile-up) blending fraction f_{DI} used to mix the double-interaction MC samples with the single-interaction samples takes nominal values f_{DI} =0.224 at 0 mrad and 0.079 at 1.5 mrad. The systematic uncertainty on f_{DI} is set by a data-driven χ^2 fit of f_{DI} to the data BDT-score distribution at the preselection cut, described in Appendix 17.6. The fit returns $f_{\mathrm{best}} = 0.290$ at 0 mrad and $f_{\mathrm{best}} = 0.000$ at 1.5 mrad. One f_{DI} variation per crossing angle is run at f_{best} and the absolute per-bin cross-section deviation from the nominal blending is taken as the symmetric one-sided systematic $\pm|\Delta\sigma|$. Figure 51 shows that this variation reaches a maximum of $\pm 1.6\%$ relative to the nominal across the analysis range.

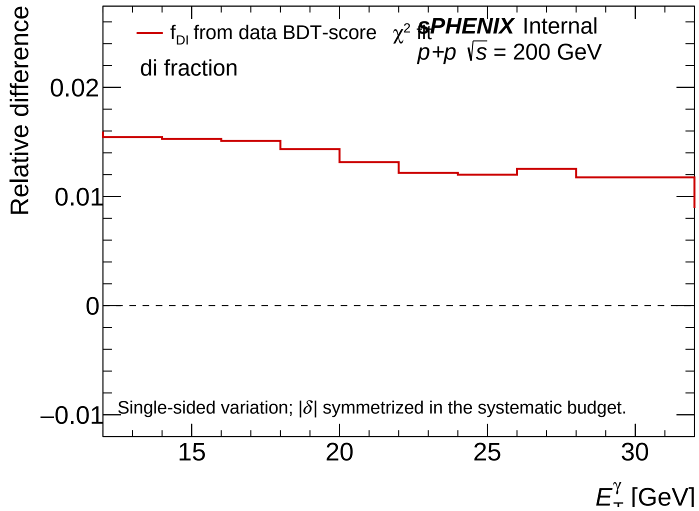

Figure 51: Systematic uncertainty from the double-interaction blending fraction f_{DI} variation as a function of E_{T}^{γ}.

5.9 Non-physical background rejection

The cluster-level non-physics background (NPB) score cut is varied two-sided, from the nominal 0.5 to a looser 0.3 and a tighter 0.7. The two variants give anti-correlated yield shifts ($r = -0.998$) consistent with a clean two-sided systematic. Figure 52 shows that this variation reaches a maximum of -1.4% on the down side and $+6.1\%$ on the up side relative to the nominal.

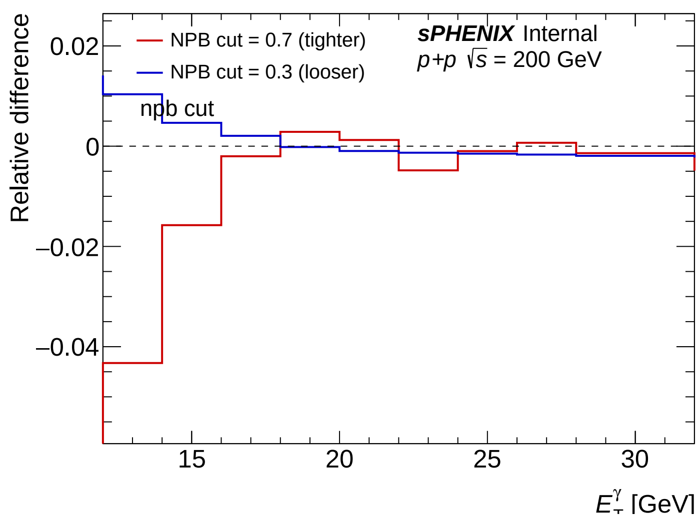

Figure 52: Systematic uncertainty from the NPB score cut variation as a function of E_{T}^{γ}.

5.10 Luminosity

The luminosity systematic uncertainty is propagated directly from the Minimum Bias Detector (MBD) trigger cross section measurement of $\sigma_{\mathrm{MBD}} = 25.2^{+2.3}_{-1.7}$ mb, taken from the sPHENIX Run 24 luminosity calibration based on the vdM-scan result. The fractional luminosity uncertainty is therefore $\delta L/L = +2.3/25.2 = +9.13\%$ on the up side and $-1.7/25.2 = -6.75\%$ on the down side. This asymmetric flat term is applied uniformly across all E_{T}^{γ} bins without E_{T}^{γ}-dependent variations.

5.11 Total Systematic Uncertainties

Figure 53 summarizes the per-source per-E_{T}^{γ} envelope. The total systematic uncertainty is obtained as the per-bin quadrature sum of each variation combined with the multiplicative-flat luminosity envelope. Table 7 gives the maximum per-bin size of each individual variation over the reported range $12 < E_{\mathrm{T}}^{\gamma} < 32$ GeV. The total reaches up to -30.8% on the down side and $+31.7\%$ on the up side near the high E_{T}^{γ} end of the reported range, dominated by the energy resolution (max of the no-extra-smearing and the wider-data scenarios, symmetrized), the energy scale (flat 1.1% envelope plus the linear non-linearity), the photon-ID tight/non-tight purity variations, and the luminosity flat term. The totals quoted here are evaluated bin-by-bin. Each per-source maximum can occur in a different E_{T}^{γ} bin, so the row-wise quadrature of the per-source maxima in the table does not equal the per-bin total. The latter is the physically meaningful quantity used to draw the systematic band.

6 Results

In this Section, the final results for the isolated-prompt photon cross section as a function of E_{T}^{γ} in p+p collisions at $\sqrt{s}$ = 200 GeV are shown. The raw yield of EMCal clusters passing the tight shower-shape criteria and the isolation energy requirement, $N^{\mathrm{tight,iso}\,\gamma}$, is corrected for the purity $\mathcal{P}$ and for reconstruction effects through unfolding,

$$Y^{\mathrm{rec}}(E_{\mathrm{T}}^{\gamma}) \;=\; \text{Unfolded}\left[N^{\mathrm{tight,iso}\,\gamma}(E_{\mathrm{T}}^{\gamma}) \;\times\; \mathcal{P}(E_{\mathrm{T}}^{\gamma})\right] \tag{7}$$

This yield is then corrected for the signal efficiency $\mathcal{E}$ and divided by the luminosity $\mathscr{L}$ to obtain the measured differential cross section $d\sigma/(dE_{\mathrm{T}}^{\gamma}\, d\eta^{\gamma})$,

$$\frac{d\sigma}{dE_{\mathrm{T}}^{\gamma} d\eta} = \frac{1}{\mathscr{L}} \frac{Y^{\mathrm{rec}}}{\mathcal{E}\Delta E_{\mathrm{T}}^{\gamma} \Delta\eta^{\gamma}}. \tag{8}$$

Figure 54 shows the fully-corrected differential cross section. The spectrum falls by roughly two orders of magnitude across the reported $12 < E_{\mathrm{T}}^{\gamma} < 32$ GeV range and is well described by NLO pQCD predictions across the entire range. The results are compared to theoretical calculations from PYTHIA-8 and JETPHOX. PYTHIA-8 (version 8.307, Detroit tune [12]) is run with the same truth-level isolation as the data. JETPHOX (v1.3.1_4) calculates direct + fragmentation photon

Source	low max [%]	high max [%]
Energy scale – flat	10.49	11.87
Energy scale – non-linearity	7.26	7.26
Energy resolution	21.21	21.21
Purity – tight ID	6.72	6.72
Purity – non-tight ID	2.35	2.35
Purity – non-isolation	3.80	0.99
Purity – fit functional form	0.33	0.33
Purity – fit statistical (1σ CI)	4.40	4.40
Purity – MC closure correction	5.77	5.77
Efficiency – $E_{\mathrm{T}}^{\mathrm{iso}}$ shape	4.73	4.73
Efficiency – generator (HERWIG)	5.83	5.83
MBD vertex efficiency	6.00	6.00
Unfolding – response reweight	0.36	0.36
Unfolding – iteration scan	5.50	5.50
Double-interaction	1.57	1.57
Non-physical background cut	1.03	4.54
Luminosity (flat)	6.75	9.13
Total	30.80	31.72

Table 7: Maximum per-bin systematic uncertainty by source over the reported range $12 < E_{\mathrm{T}}^{\gamma} <$ 32 GeV. The “low max” (“high max”) column is the maximum over E_{T}^{γ} bins of $|(\sigma_{\mathrm{var}} - \sigma_{\mathrm{nom}})/\sigma_{\mathrm{nom}}|$ for the down-side (up-side) variation of each source; sources with a single (symmetrized) variation have equal columns by construction.

cross sections at NLO using the CT14NLO PDFs, BFG setII fragmentation functions [13], and scales $\mu_{\mathrm{R}} = \mu_{\mathrm{F}} = \mu_{\mathrm{f}} = E_{\mathrm{T}}^{\gamma}$. An additional NLO pQCD prediction without isolation is overlaid for reference. Both JETPHOX and PYTHIA-8 agree with the measurement within experimental uncertainties across the full E_{T}^{γ} range. Systematic uncertainties dominate across the reported E_{T}^{γ} range. Statistical uncertainties grow from sub-percent at the lowest reported E_{T}^{γ} bin to about 30% at the highest, but remain subdominant. The photon-energy-scale and double-interaction-blending systematics dominate the high-E_{T}^{γ} budget, while the purity (fit-CI and MC-closure) and luminosity sources drive the low-E_{T}^{γ} budget.

The results are also compared with previous measurements reported by the PHENIX experiment [1], as shown in Figure 55. The PHENIX measurements report direct photons without an isolation requirement and were collected in $|\eta| < 0.25$, with the cross section expressed double-differentially as $d\sigma/(d\eta\, dE_{\mathrm{T}}^{\gamma})$. To enable a like-for-like overlay onto the present fiducial, the PHENIX spectrum is shown both as published and after two corrections: a bin-width recovery from a smooth fit of the PHENIX cross section, and a pseudorapidity-density rescaling from $|\eta| < 0.25$ to $|\eta| < 0.7$ derived from prompt-photon truth simulation. The procedure and the input ratio are detailed in Appendix 15. The corrected spectrum is consistent with the present measurement within uncertainties across the overlap range, with the residual offset reflecting the small fragmentation contribution after the truth-level isolation cut and remaining consistent with the gluon–Compton-dominance picture at the parton-momentum fractions probed here.

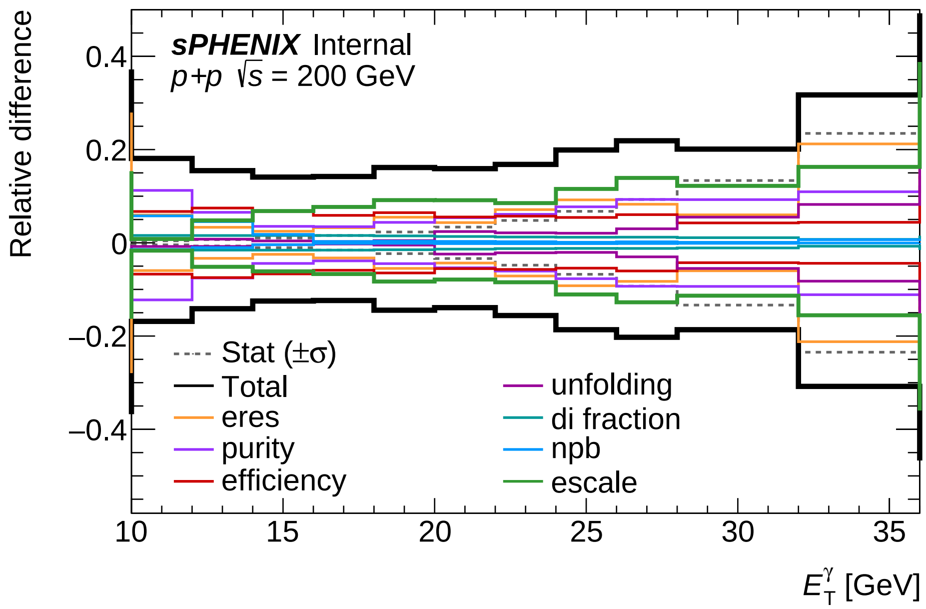

Figure 53: Per-bin breakdown of the systematic uncertainty as a function of E_T^γ: each colored curve is one source group (escale, eres, purity, efficiency, unfolding, di_fraction, npb), and the thick black envelope is the per-bin quadrature sum including the luminosity flat term. The gray dashed curve overlays the per-bin statistical relative uncertainty σ_{stat}/N as a visual reference for the relative size of stat vs. syst. The statistical uncertainty is *not* added to the systematic quadrature.

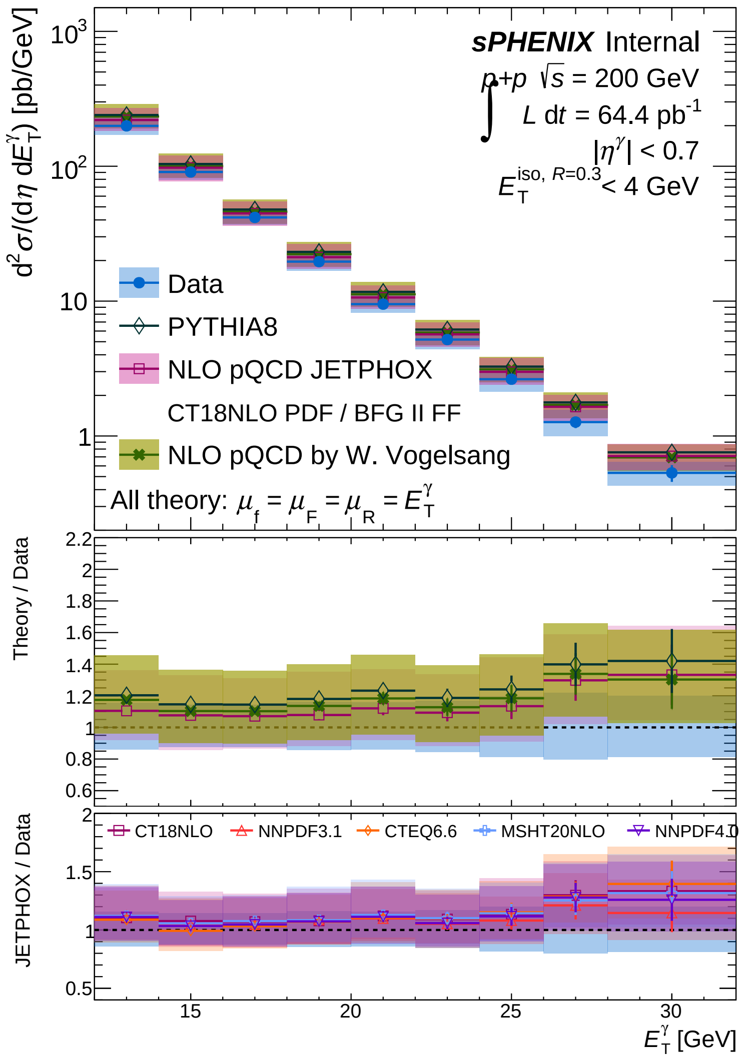

Figure 54: The differential cross section of isolated prompt photons as a function of E_T^γ is compared to theoretical predictions of PYTHIA-8 (green), JETPHOX (pink) and NLO pQCD calculations by Werner Vogelsang. The statistical uncertainties are plotted as vertical lines and the systematic uncertainties are plotted as shaded bands. The lower panel shows a theory-to-data ratio to this analysis, where the experimental systematic uncertainties are shown as shaded bands around unity. The theory and experimental statistical uncertainties are combined on the theory points. The JETPHOX NLO curve shown here uses the CT14nlo PDF set; a comparison of alternative PDF sets (CT14lo, NNPDF3.1) is presented in Appendix 13.

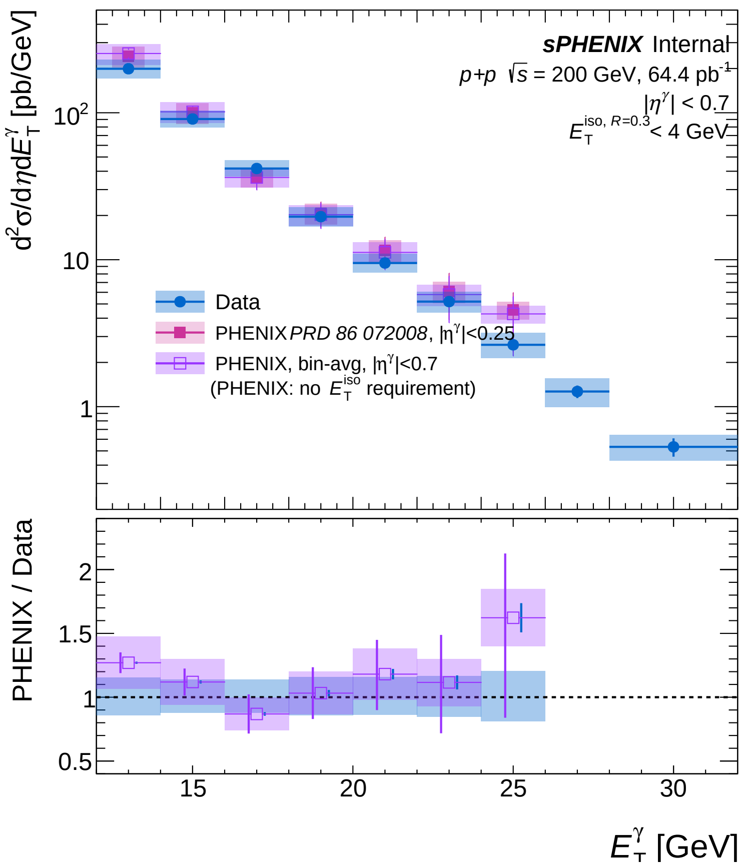

Figure 55: The differential cross section of isolated prompt photons as a function of E_T^{γ} is compared to the PHENIX measurements [1] of direct photons. The PHENIX points are overlaid both as published (filled pink markers at the bin centre, with a shaded box of fixed cosmetic width indicating the published systematic uncertainty) and after a bin-width recovery and a pseudorapidity-density rescaling from $|\eta| < 0.25$ to $|\eta| < 0.7$ (open violet markers with a shaded box spanning the published bin width); the corrections are derived in Appendix 15. For the present measurement, the statistical uncertainties are plotted as vertical lines and the systematic uncertainties as shaded bands.

7 Conclusion

The differential cross section of isolated prompt photons is measured in p+p collisions at $\sqrt{s} = 200$ GeV using the data taken during Run 24, corresponding to an integrated luminosity of 64.37 pb^{-1}. The results are reported for photon E_T^γ of 12–32 GeV within $|\eta^\gamma| < 0.7$, with truth-level isolation $E_T^{\text{iso,truth}} < 4$ GeV in an $R = 0.3$ cone. The analysis applies a data-driven double-sideband purity estimation and a D'Agostini Bayesian unfolding (RooUnfold) for detector effects.

The measured cross section agrees with NLO pQCD predictions from JETPHOX (CT14nlo) PDF, BFG II fragmentation, scales set to E_T^γ) and with the PYTHIA-8 Detroit-tune Monte Carlo within experimental uncertainties across the full E_T^γ range. The result is also consistent with the previous PHENIX direct-photon measurement at the same $\sqrt{s} = 200$ GeV within their respective uncertainties.

The systematic uncertainty is dominated at high E_T^γ by the photon-energy-scale variation (driven by bin migration on the steeply falling spectrum), the photon energy resolution, and the MC purity-closure correction. The E_T^γ-independent luminosity term contributes asymmetrically at $+9.13\%/-6.75\%$. The combined per-bin total rises from approximately $\pm14\%$ at low E_T^γ to $-23.0\%/+21.4\%$ in the high-E_T^γ end of the reported range. [Numbers reflect the placeholder $\pm1.1\%$ EMCal energy-scale variation; envelope will be re-evaluated with the Calorimeter Calibration group's final guidance.]

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Appendix

8 Non-physical background

Figure 56 shows the purity and efficiency scan for five E_T^{γ} ranges spanning 10–30 GeV. The purity rises monotonically with threshold and signal efficiency falls, with both sitting near 99% at NPB$>$ 0.5 across all p_T bins.

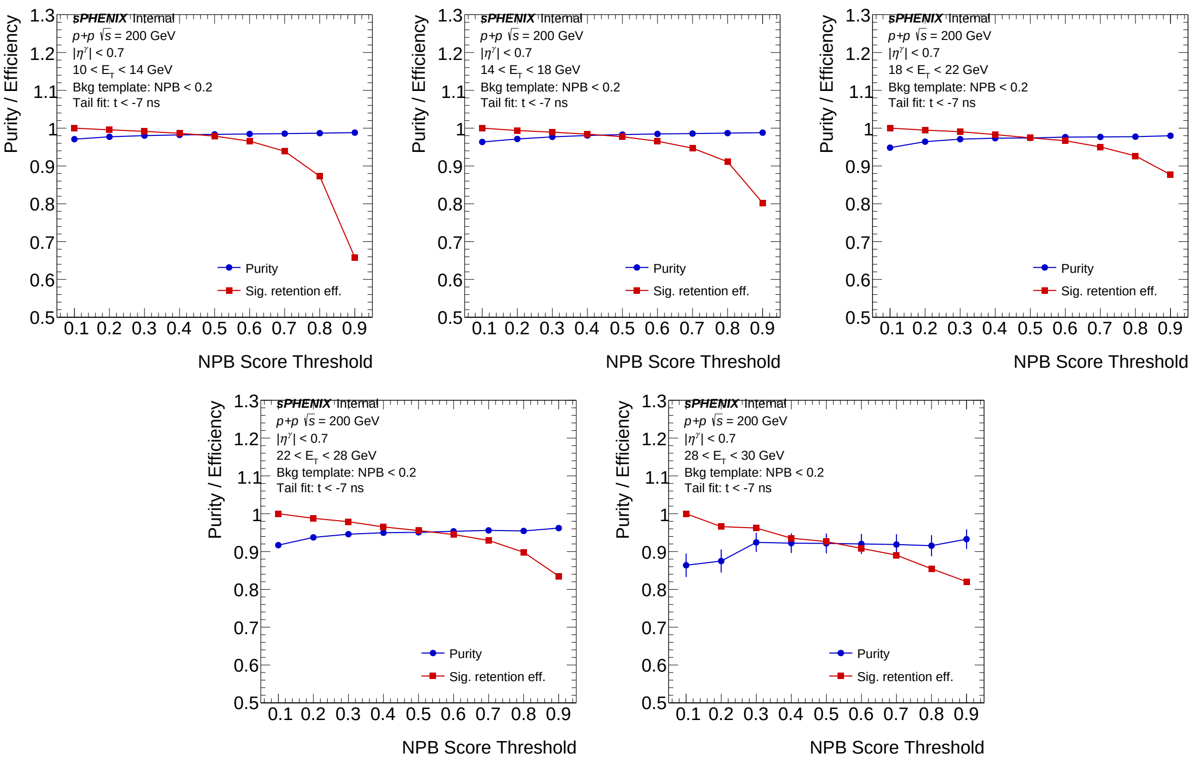


Figure 56: Purity (blue, fraction of physics clusters) and signal-cluster retention efficiency (red, $N_{\text{sig}}(T)/N_{\text{sig}}(T_{\text{min}})$, with T the NPB score threshold and $T_{\text{min}} = 0.1$ the loosest scanned threshold; see Section 3.4.2) as functions of T, for five E_T^{γ} ranges: 10–14, 14–18, 18–22, 22–28, and 28–30 GeV (left-to-right, top-to-bottom). NPB template: NPB$<$ 0.2 normalized in the cluster–MBD time tail $t < -7$ ns. At the nominal threshold NPB$>$ 0.5 both quantities sit near 99% in every E_T^{γ} bin; tightening past 0.7 drops signal retention by tens of percent while purity is already saturated.

9 Shower Shape Variables

Figure 57 and Figure 13 are shower shape distributions for all clusters in the analysis. The data generally follows the background MC (inclusive jet sample with truth background selection) with the exception of specific excesses induced by non-physics backgrounds. One can see that after the pre-selection, in Figure 58 and Figure 19 the non-physics backgrounds are significantly reduced.

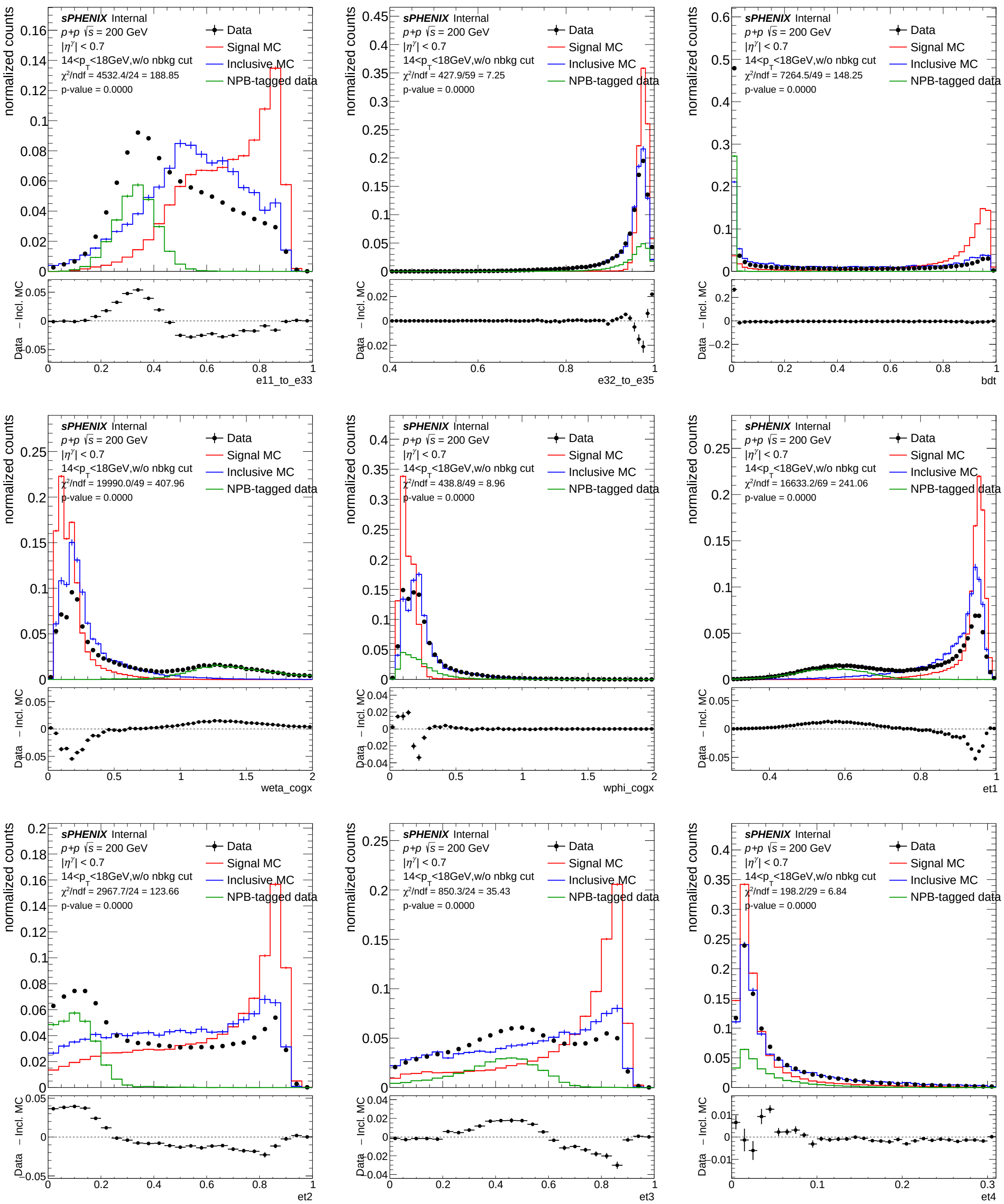


Figure 57: Shower shape distributions with no pre-selection at lower p_T (14–18 GeV) for data (black), signal MC (red), inclusive MC (blue), and the NPB-tagged data template (green). The NPB-tagged template is the data subset selected as streak-like by the high-w_η^{COGX} NPB tag, normalized to the bulk data via tail-matching in $w_\eta^{\mathrm{COGX}} > 1$, and serves as a background-enhanced template for the non-physics-background component.

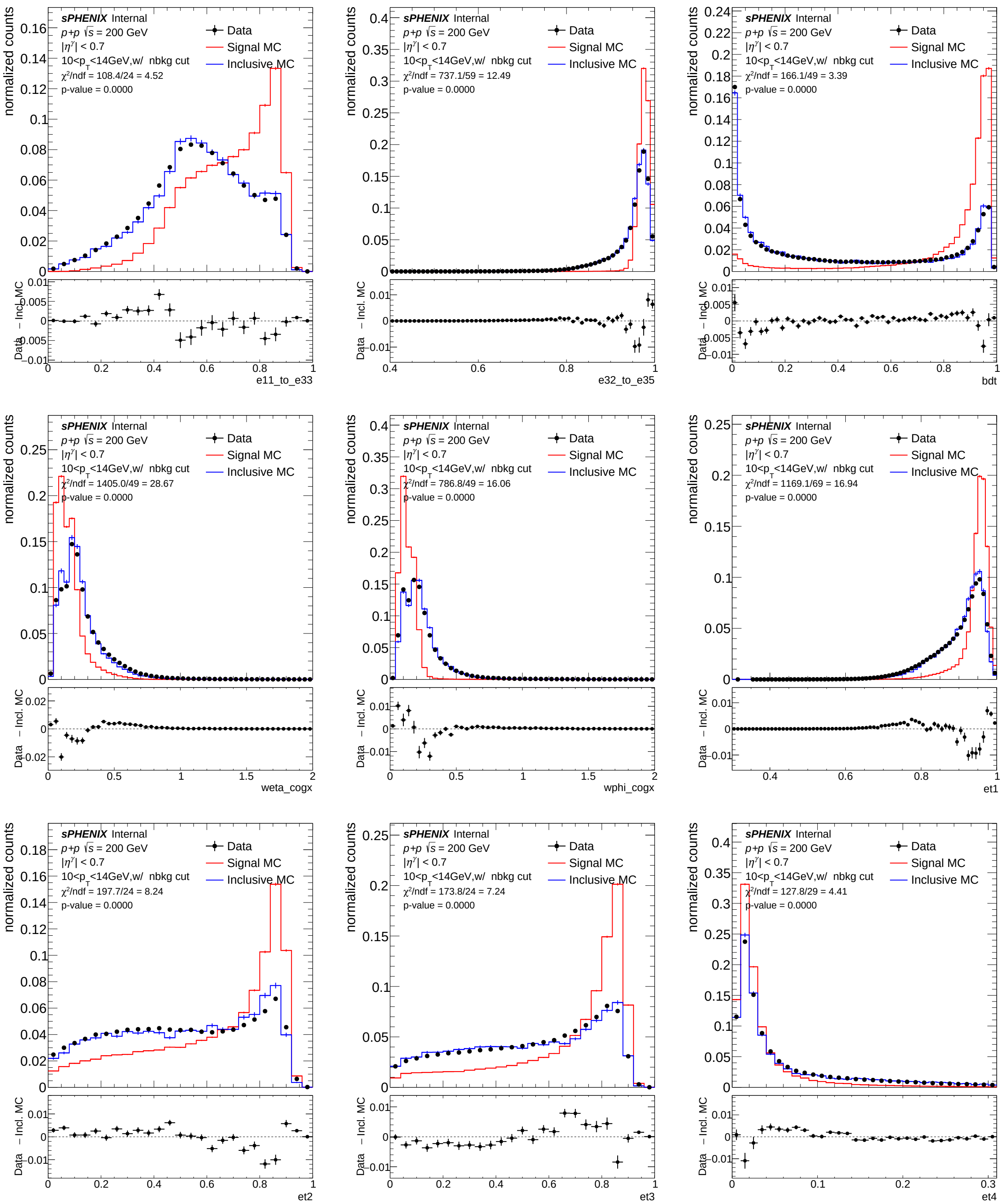


Figure 58: Shower shape distributions with pre-selection at lower p_T (10–14 GeV).

It is important to observe the correlation between $E_{\mathrm{T}}^{\mathrm{iso}}$ and the shower shapes without any preselection. Figures 59 and 60 show the average $E_{\mathrm{T}}^{\mathrm{iso}}$ as a function of shower-shape value for several shower shapes; some variables are more strongly correlated than others.

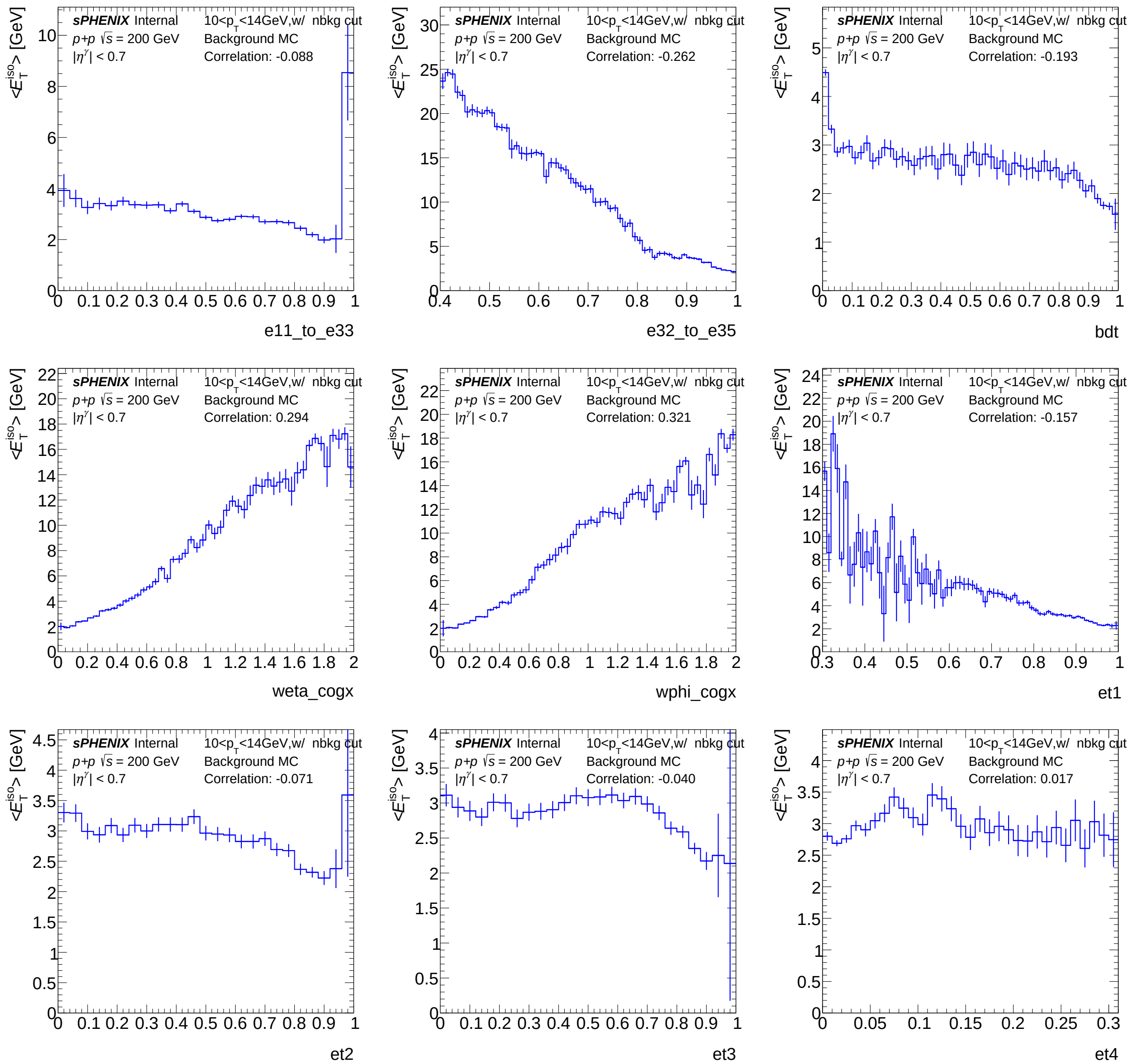


Figure 59: The average $E_{\mathrm{T}}^{\mathrm{iso}}$ as a function of shower shape value for various shower shapes with no selections for lower p_{T} (10–14 GeV).

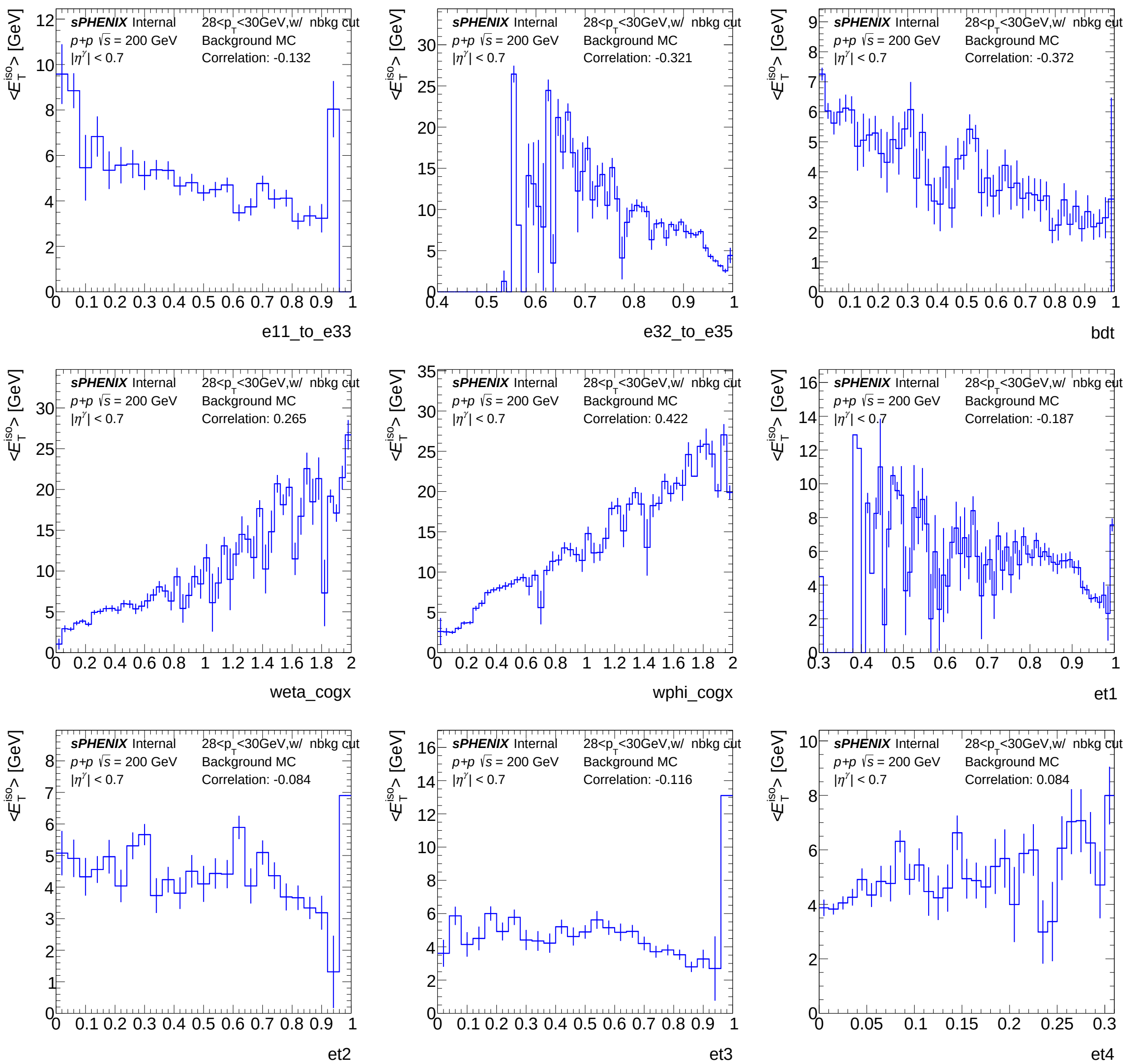


Figure 60: The average $E_{\mathrm{T}}^{\mathrm{iso}}$ as a function of shower shape value for various shower shapes with pre-selection cuts for higher p_{T} (28–30 GeV).

917 # 10 Isolation Energy Distributions

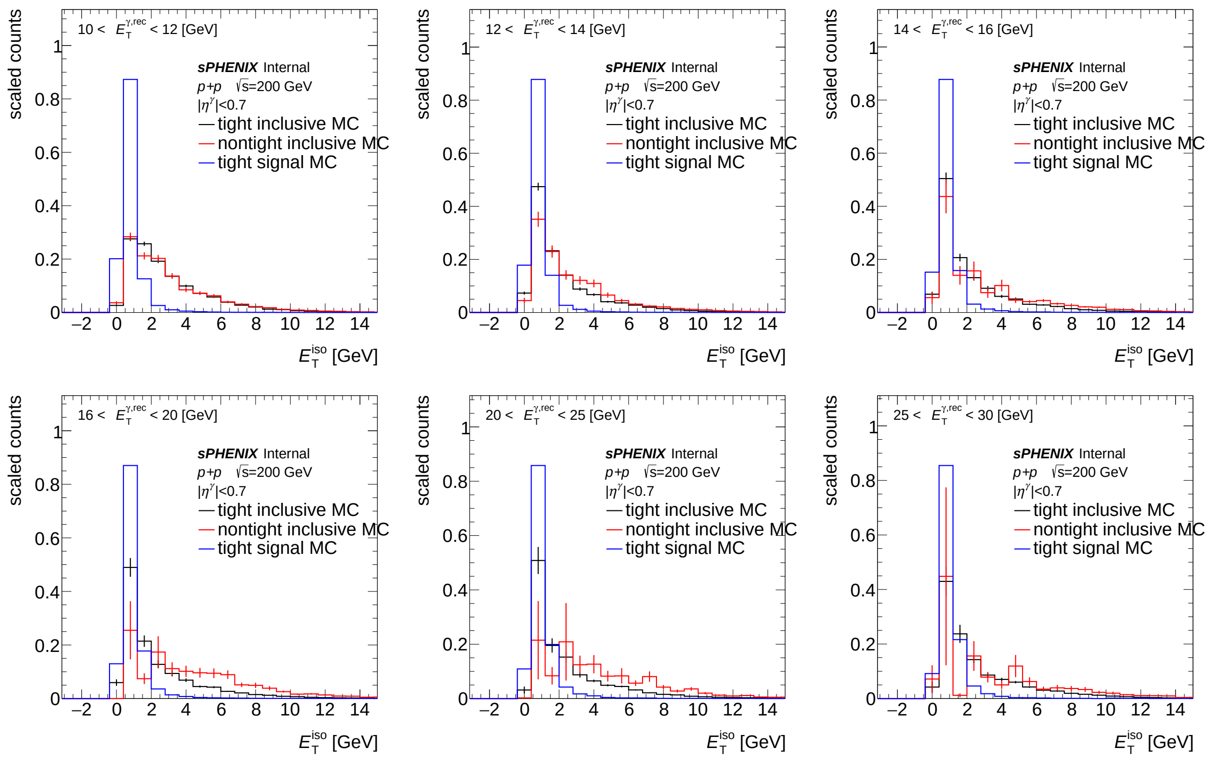


Figure 61: Distributions of E_T^{iso} for different E_T^{γ} bins in different panels, respectively. Photons with tight ID (black), non-tight ID (red) in inclusive MC and photons with tight ID in signal MC (blue) are overlaid.

918 # 11 Shower Shape Cut Comparison

919 Figure 62 ($10 < p_T < 12$ GeV) and Figure 63 ($26 < p_T < 35$ GeV) show the effect of the preselection
920 chain (Table 5: E_{11}/E_{33}, e_{T1}, E_{32}/E_{35}, NPB score > 0.5) on shower-shape distributions in inclusive-
921 MC background samples. Both histograms come from the inclusive MC sample. The blue
922 histogram is the baseline distribution before any cut, and the red histogram is the distribution
923 after applying the full preselection. The preselection suppresses the high-tail population in shape
924 variables sensitive to non-physics backgrounds (notably w_{η}^{COGX}), with little effect on physics-cluster
925 shapes.

926 # 12 Tower Acceptance from Phi-Symmetry Method

927 This appendix studies the residual data–MC mismatch in the EMCal dead-tower acceptance. The
928 procedure is, in summary: (i) build the data tower-occupancy map at four progressively tighter

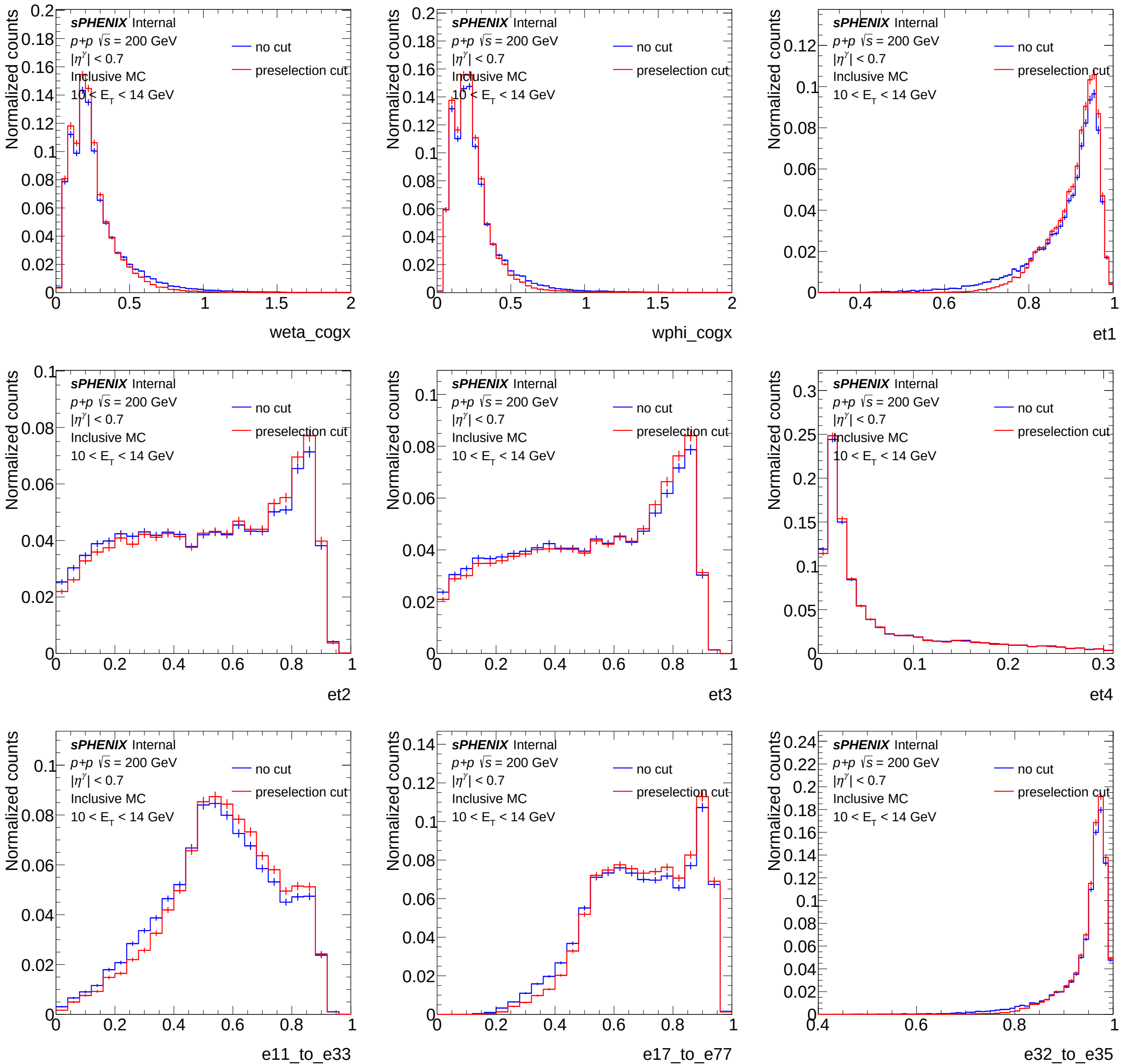


Figure 62: Shower shape distributions before (blue, "no cut") and after (red, "preselection cut") the full preselection chain in inclusive MC for $10 < p_{\mathrm{T}} < 12$ GeV.

selection levels (`preselect`, `common`, `tight`, and the `OR` of the latter two); (ii) for each pair of adjacent η rows fit a 3σ-clipped Gaussian to the 256-bin per-ϕ cluster-count distribution and flag $i\phi$ bins more than 2σ below the fit mean as data-deficient towers; (iii) repeat at each selection level to produce four binary tower-mask histograms; (iv) re-run the full data + MC efficiency, merge, and yield pipeline with the mask applied symmetrically to data and MC at the cluster veto level; (v) compare the resulting integrated cross section against the mask-free nominal. The four mask variants and the integrated shifts they produce are documented in Section 12 below.

The motivation is that the sPHENIX EMCal contains dead and malfunctioning towers whose response is not perfectly emulated by the GEANT simulation. A mismatch between the set of dead towers active in Run 24 data and those modelled in the MC biases the reconstruction efficiency

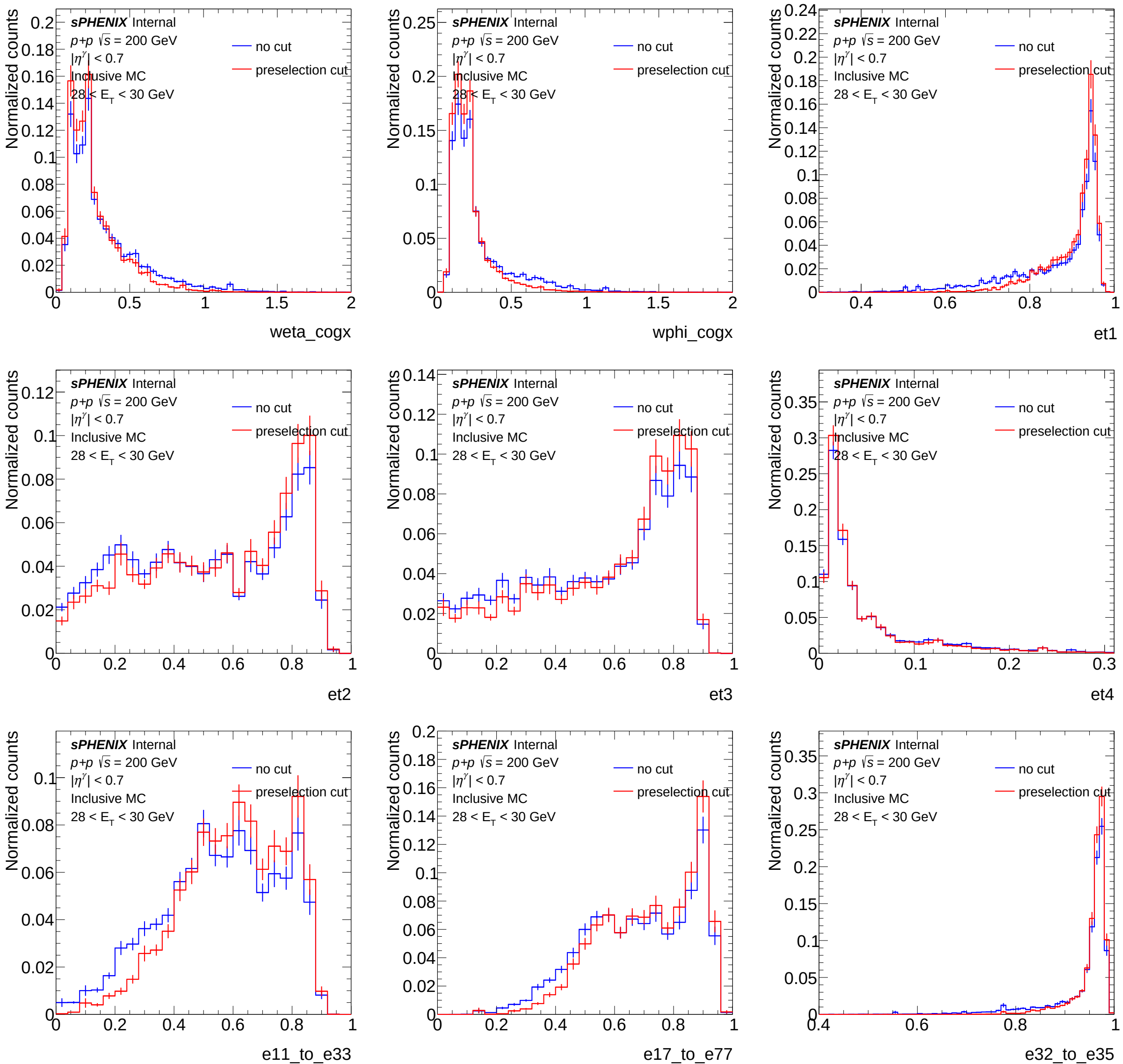


Figure 63: Shower shape distributions before (blue, "no cut") and after (red, "preselection cut") the full preselection chain in inclusive MC for $26 < p_{\mathrm{T}} < 35$ GeV.

high, which in turn biases the measured cross section low. The ϕ-symmetry method exploits the fact that the inclusive cluster yield is ϕ-symmetric at fixed η in p+p collisions to derive a purely data-driven mask without relying on a particular MC dead-tower list. Because the same mask is applied to numerator and denominator on both sides, the efficiency-corrected cross section is sensitive only to the residual data-vs-MC asymmetry in the masked region rather than to the absolute mask coverage.

12.1 Method

The EMCal is binned in tower index as $96\,(i\eta) \times 256\,(i\phi)$. For each pair of adjacent η rows (48 two-row bands across the full η acceptance), the 256 per-ϕ cluster counts are projected out. Under the physics assumption that the inclusive cluster cross section is ϕ-symmetric at fixed η in p+p at $\sqrt{s} = 200$ GeV, deviations from a flat distribution trace either statistical fluctuations or local detector effects. An iterative 3σ Gaussian is fit to the 256-bin distribution, yielding band-by-band mean $\mu_{\phi,\text{band}}$ and width $\sigma_{\phi,\text{band}}$. A tower $(i\eta, i\phi)$ is flagged as dead when $(n_\text{data}[b, i\phi] - \mu_{\phi,b})/\sigma_{\phi,b} < -2$; both η rows within a band are masked whenever a ϕ bin is flagged.

Four masks are produced at successively tighter preselection levels: (i) `preselect` ($E_\text{T}^\gamma \geq 8$ GeV with no-saturation and no shape cuts); (ii) `common` (preselect intersected with the NPB score cut, $e_{11}/e_{33} < 0.98$, and the $w_{r,\text{cogx}}$ / $w_{\eta,\text{cogx}}$ windows); (iii) `tight` (common intersected with the ten shower-shape windows and the parametric BDT requirement); and (iv) `OR` (the union of the `common` and `tight` masks). The corresponding masked-tower counts are 1344, 1160, 646 and 1180, respectively. The tight mask is largely contained within the common mask (626-tower overlap), so the `OR` mask adds only 20 additional towers relative to common alone. Because the tight sample has lower per-ϕ statistics, the fit σ is broader and fewer bins fall below the -2σ threshold; the monotonic decrease of masked-tower count with selection tightness is the expected statistical behaviour of the method.

An example of the per-ϕ cluster-count distribution within a single $i\eta$ slice, together with the iterative-3σ Gaussian fit and the $\mu - 2\sigma$ flagging threshold, is shown in Figure 64. A cluster of $i\phi$ cells around $i\phi \approx 30$–45 falls well below the threshold and is tagged as a contiguous block of data-deficient towers.

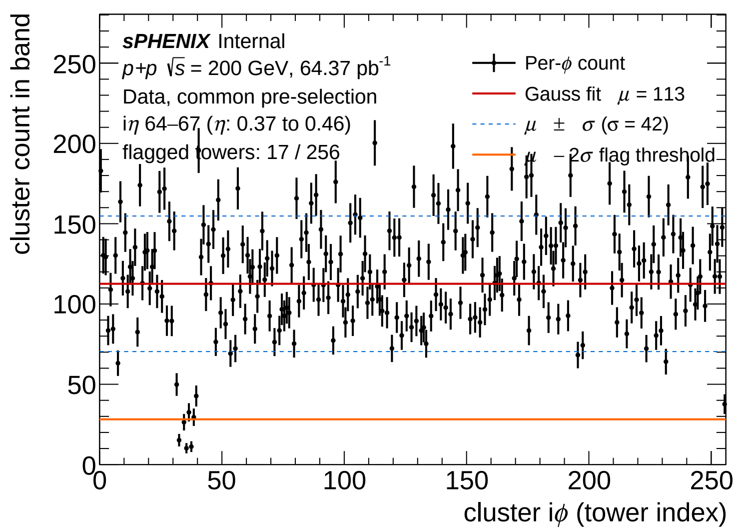


Figure 64: Per-ϕ cluster count for $i\eta = 64$–67 ($\eta \approx 0.37$ to 0.46) at the `common` selection level (data, $E_\text{T}^\gamma \geq 8$ GeV). Red solid: iterative-3σ-clipped Gaussian-fit mean μ. Blue dashed: $\mu \pm \sigma$ band. Orange solid: $\mu - 2\sigma$ flagging threshold. The 17 $i\phi$ cells with counts below the threshold (mostly contiguous around $i\phi \approx 30$–45) are tagged as data-deficient towers.

When the mask is active, a cluster-level veto is applied identically to data and MC so that efficiency numerators and denominators see the same masked acceptance. Tower-occupancy maps at the `preselect`, `common`, `tight`, and `tight_iso` selection levels (one pair for data and lumi-weighted inclusive MC) are produced as a diagnostic of residual data–MC acceptance mismatch.

12.2 Results

Figure 65 shows the lumi-weighted inclusive MC (left column) and data (right column) tower-occupancy maps at the `preselect` and `common` selection levels. Dark vertical stripes in the data maps indicate dead-region clusters that are not reproduced in the MC. Figure 66 overlays the ϕ-symmetry flagged cells on the data tower-occupancy map. As an independent cross-check, we also rebin the data and lumi-weighted MC tower maps into 4×4 super-cells and flag super-cells whose data/MC ratio R has $\log(R)$ deviating below a Gaussian fit to the bulk distribution. This rebinning method tags 2–3% of super-cells per selection level (`preselect`, `common`, `tight`, `tight_iso`) as data-deficient, and the resulting integrated cross-section shift agrees with the ϕ-symmetry mask to $\sim 1\%$.

Applying each of the four masks symmetrically to data and MC in the full efficiency, merge, and yield pipeline produces the integrated-cross section shifts summarised in Table 8, evaluated in $E_T^\gamma \in [10, 36]$ GeV relative to the mask-free nominal. All four shifts are positive, as expected: masking removes MC content in data-dead regions that was inflating the MC-derived efficiency, so the corrected cross section rises once the asymmetry is removed. The four variants agree to within $\sim 1\%$, consistent with a uniform detector-level dead-tower effect. Per-E_T^γ-bin ratios (Figure 67) lie within $\pm 3\%$ of unity across $E_T^\gamma \in [10, 30]$ GeV. The larger scatter seen in the highest $[32, 36)$ GeV bin is statistical and does not impact the integrated result, which is dominated by the low-E_T^γ region.

Table 8: Integrated cross section shift in $E_T^\gamma \in [10, 36]$ GeV for each ϕ-symmetry mask variant, normalised to the mask-free nominal $\sigma_{\mathrm{nom}} = 1963$ pb (all-z fiducial, 64.37 pb^{-1}). Applying the mask symmetrically to data and MC produces a positive shift of 1.4–2.3% across all four variants.

variant	$\sigma_{\mathrm{masked}}/\sigma_{\mathrm{nom}}$	$\Delta\sigma/\sigma_{\mathrm{nom}}$
`mask_phisymm_preselect`	1.0201	+2.01%
`mask_phisymm_common`	1.0226	+2.26%
`mask_phisymm_tight`	1.0137	+1.37%
`mask_phisymm_or`	1.0228	+2.28%

The envelope of the four variants — dominated by the `common` and `OR` masks at $\sim 2.3\%$ — sets a recommended acceptance systematic at the few-percent level on the integrated cross section. The per-bin envelope from this cross-check, taken as the `OR` mask alone, gives a maximum per-E_T^γ-bin deviation of $\pm 1.57\%$ (the integrated shift is similar). This bound is conservative with respect to the individual variants and covers the residual data-vs-MC dead-tower asymmetry visible in Figure 65. Two independent estimators — the ϕ-symmetry cross section shift and the $\log(R)$ Gaussian deficit at 4×4 rebinning — converge on this $\sim 2\%$ level, providing a robust data-driven bound that does not rely on any specific MC dead-tower list.

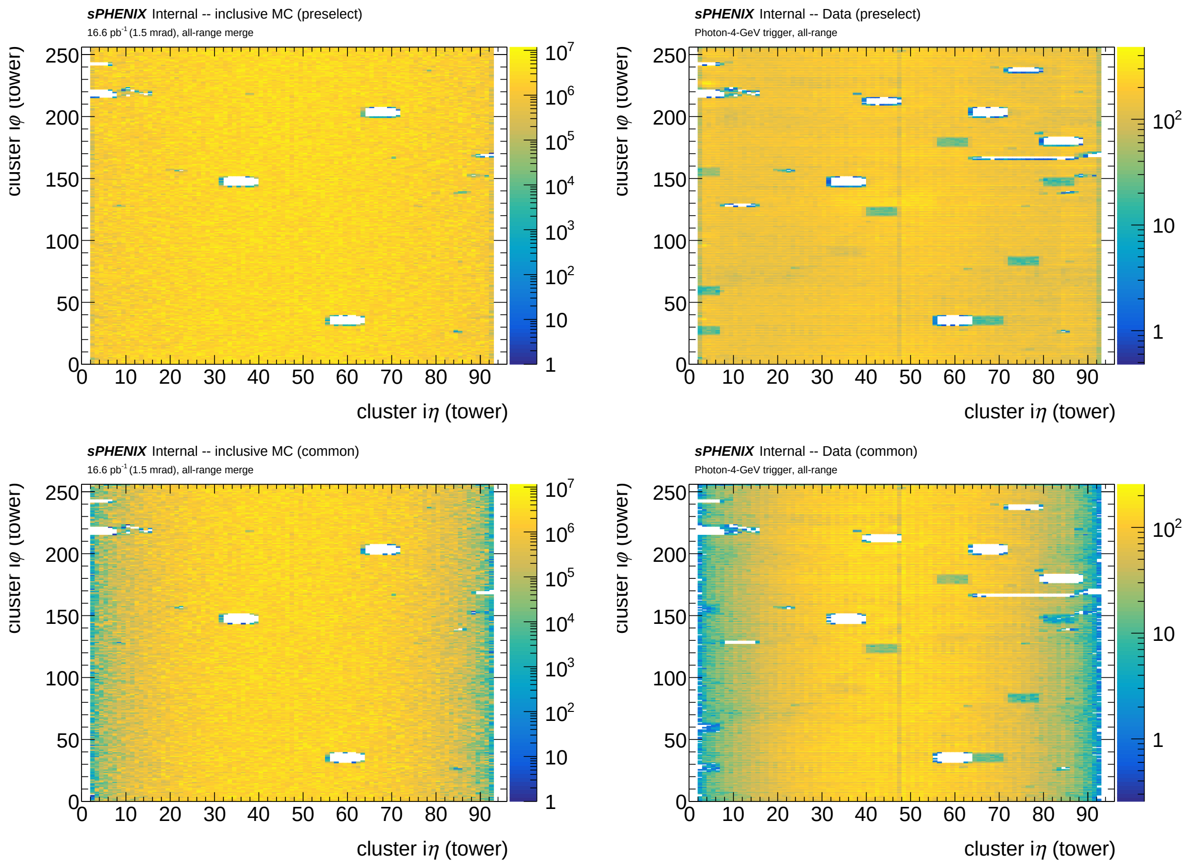


Figure 65: Tower occupancy maps (96×256 in $(i\eta, i\phi)$) for the lumi-weighted inclusive MC (left column) and data (right column) at the `preselect` (top row, $E_T^{\gamma} \geq 8$ GeV with no-saturation and no shape cuts) and `common` (bottom row, preselect plus the common shape-variable pre-selection) selection levels. Dark stripes in the data maps mark dead towers not reproduced in the MC acceptance.

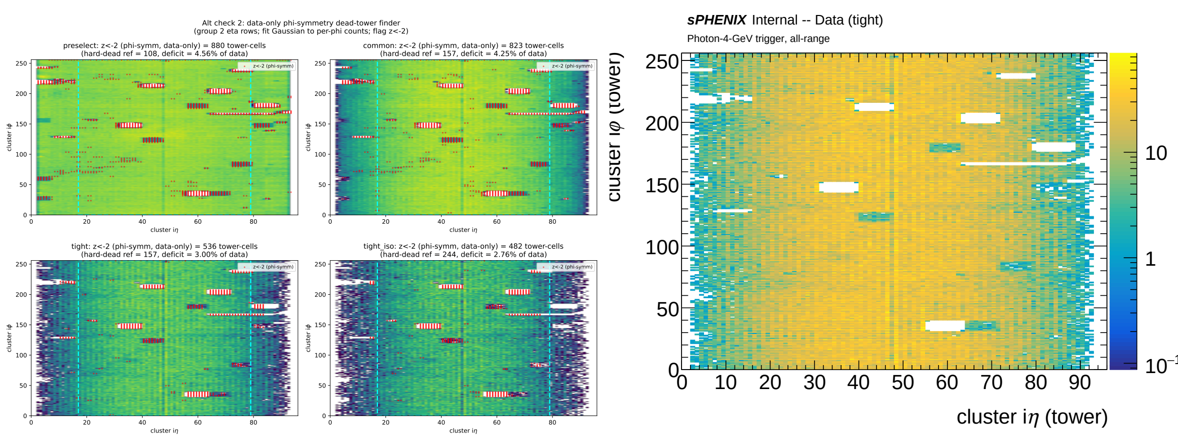


Figure 66: Left: ϕ-symmetry-flagged cells ($z < -2$ in the per-η-band Gaussian fit) overlaid on the data tower map at the three selection levels. Right: data tower map at the `tight` selection level, which carries the lowest cluster density and therefore the broadest per-band fit width; the resulting mask contains 646 towers.

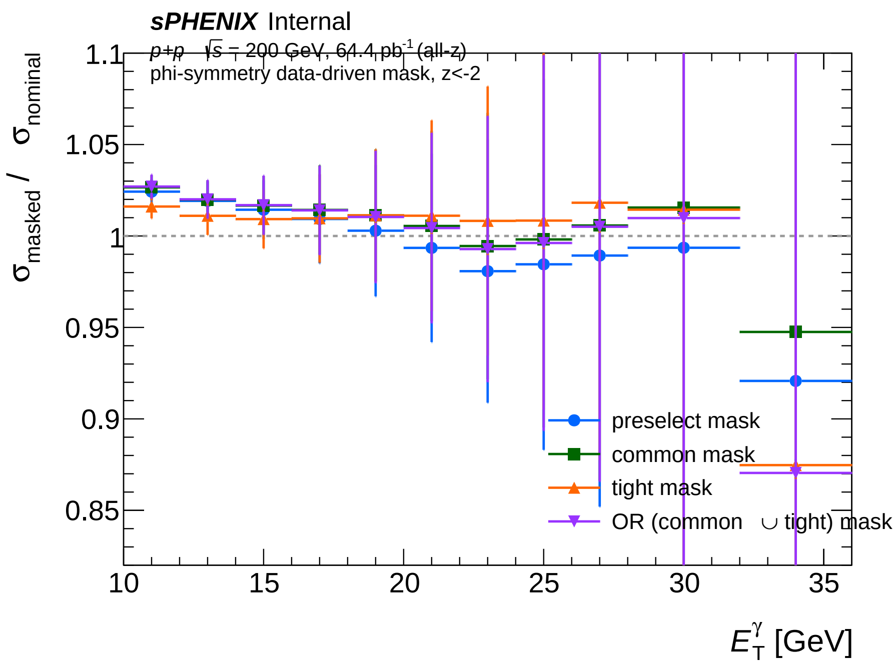


Figure 67: Per-E_{T}^{γ}-bin ratio $\sigma_{\mathrm{masked}}/\sigma_{\mathrm{nom}}$ for the four ϕ-symmetry mask variants. All four sit within $\pm 3\%$ of unity across $E_{\mathrm{T}}^{\gamma} \in [10, 30]$ GeV. The larger fluctuation in the $[32, 36)$ GeV bin is driven by low-statistics cells in the masked region and does not affect the integrated shift.

13 JETPHOX PDF Comparison: CT14nlo vs CT14lo vs NNPDF3.1

The NLO theory reference used for comparison with the measured cross section is JETPHOX with the CT14nlo parton distribution functions. To quantify the PDF-driven theory spread, the JETPHOX production has been rerun with two alternative PDF sets — the LO CT14lo set and the NLO NNPDF3.1 set ($\alpha_s(M_Z) = 0.118$) — using identical kinematic and fragmentation settings. Pairing the LO CT14lo set with the NLO matrix element is a known LO-in-NLO inconsistency (the missing NLO gluon evolution is absorbed into an effective LO fit that is not strictly valid at all x when convolved with the NLO hard cross section), so it is shown here only as a methodological cross-check of the LO-vs-NLO PDF effect.

13.1 Method

All three JETPHOX productions share the same configuration ($\sqrt{s} = 200$ GeV p+p, photon $|y| < 0.7$, direct + fragmentation components with the BFG II fragmentation functions, fixed-cone isolation with $R = 0.3$ and $E_{\mathrm{T}}^{\mathrm{iso,truth}} < 4$ GeV, and renormalisation/factorisation/fragmentation scales $\mu_R = \mu_F = \mu_f = c_m \cdot p_{\mathrm{T}}$ with $c_m \in \{0.5, 1.0, 2.0\}$). Per scale, 100 segments of 10^6 events each are generated, totalling 10^8 events per scale point. The full PDF error set is computed for every point (56 CT14 Hessian eigenvector pairs for CT14lo/CT14nlo and 100 MC replicas for NNPDF3.1). The three productions are independent: only the PDF LHAPDF identifier (`CT14lo`, `CT14nlo`, `NNPDF31_nlo_as_0118`) differs, and the MC random-number-generator seeds are offset so no event streams are shared between productions.

13.2 Results

The central-scale ($\mu = 1.0 \cdot p_T$) per-bin cross sections for the three PDFs are summarised in Table 9, together with the NLO/LO PDF ratios. Moving from the (LO) CT14lo to a (NLO) PDF shifts the prediction monotonically in E_T^γ: at low p_T (8–14 GeV) the NLO PDFs lie 3–11% *above* CT14lo, at mid p_T (14–26 GeV) they lie 4–25% *below*, and at the highest bins (26–36 GeV) they lie 27–50% below. Integrated over the analysis range $E_T^\gamma \in [8, 36]$ GeV, the total direct + fragmentation cross section increases from 4942 pb (CT14lo) to 5294 pb for CT14nlo (+7.1%) and 5330 pb for NNPDF3.1 (+7.9%).

The crossover near $E_T^\gamma \approx 14$ GeV reflects the well-known LO-vs-NLO gluon difference: at the moderate $x \sim 2p_T/\sqrt{s} \sim 0.1$ accessed at low p_T, the LO CT14lo gluon under-shoots because NLO K-factors are absorbed into the LO fit; at higher $x \gtrsim 0.2$ accessed at high p_T, the LO fit over-compensates to match jet data at the LO matrix element, and the NLO matrix element $\times$ NLO PDF product properly suppresses this excess.

The two NLO PDFs (CT14nlo and NNPDF3.1) agree with each other to $\pm 3\%$ at low p_T and grow apart to $\pm 20\%$ in the highest $[32, 36)$ GeV bin, where the PDF-family spread becomes comparable to the renormalisation-scale envelope ($\pm 40\%$ from $\mu = 0.5/1.0/2.0$). This spread is much smaller than the CT14lo vs. CT14nlo shift, confirming that the LO-vs-NLO PDF effect — rather than the specific fit family — is the dominant PDF-driven theory uncertainty. Figure 68 shows the cross sections and the ratios to CT14lo as a function of E_T^γ.

Table 9: Central-scale ($\mu = 1.0 \cdot p_T$) JETPHOX cross section ($d^2\sigma/dp_T\,dy$ in pb/GeV, $|y| < 0.7$) for CT14lo, CT14nlo and NNPDF3.1, with ratios to CT14lo in the last two columns. Integrated 8–36 GeV: CT14lo 4942 pb, CT14nlo 5294 pb (+7.1%), NNPDF3.1 5330 pb (+7.9%).

E_T^γ bin [GeV]	CT14lo	CT14nlo	NNPDF3.1	CT14nlo/CT14lo	NNPDF/CT14lo
8–10	1519	1673	1690	1.101	1.113
10–12	539.9	571.2	575.4	1.058	1.066
12–14	218.5	222.2	225.1	1.017	1.030
14–16	98.4	96.2	94.4	0.978	0.959
16–18	47.0	44.0	41.9	0.937	0.891
18–20	23.5	20.8	19.1	0.887	0.815
20–22	9.82	8.60	8.01	0.876	0.816
22–24	6.54	5.53	4.95	0.846	0.756
24–26	3.66	2.93	2.73	0.799	0.745
26–28	2.10	1.41	1.37	0.669	0.651
28–32	0.940	0.608	0.579	0.647	0.616
32–36	0.323	0.236	0.165	0.731	0.510

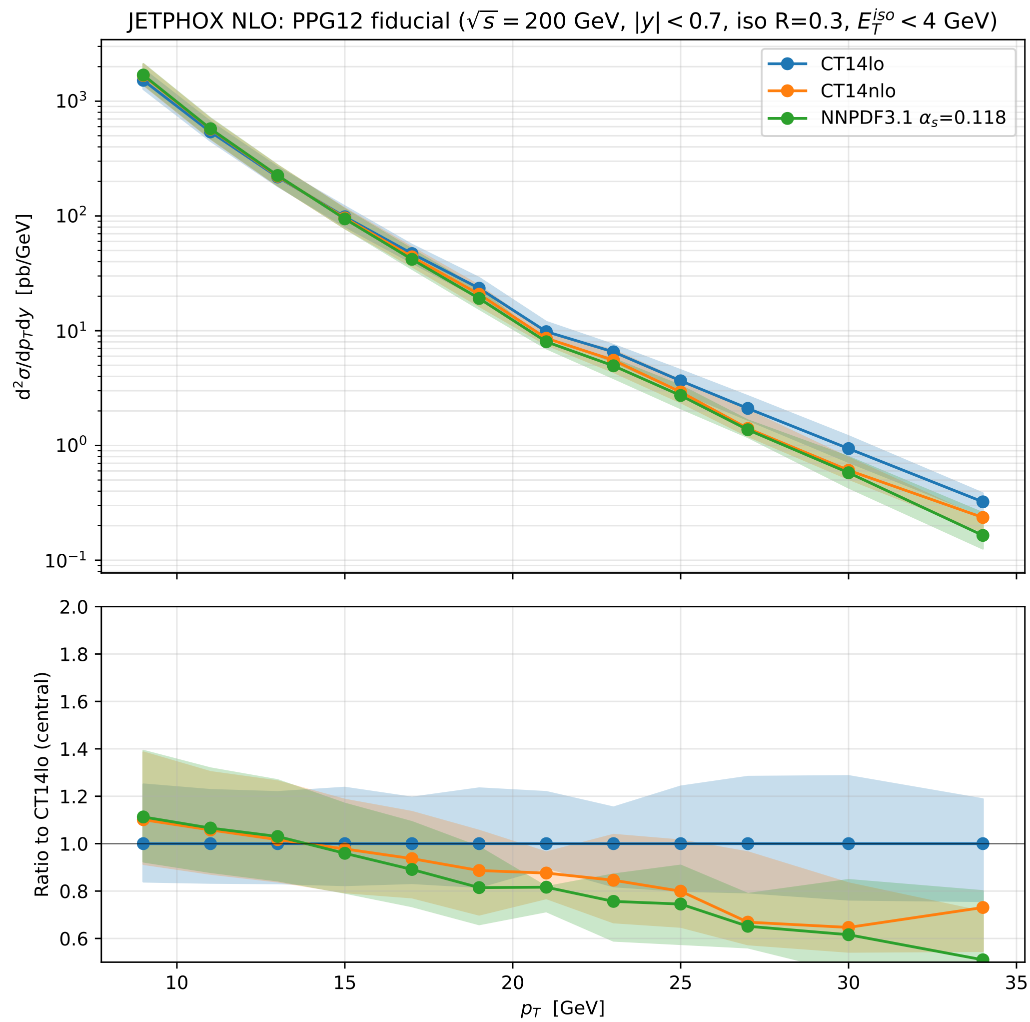


Figure 68: Top panel: JETPHOX central-scale ($\mu = 1.0 \cdot p_\mathrm{T}$) cross section vs. E_T^{γ} for CT14lo (reference), CT14nlo and NNPDF3.1 NLO. Bottom panel: per-bin ratio to CT14lo. The LO CT14lo curve lies below the two NLO predictions at low p_T and above them at high p_T; the two NLO PDFs agree to within $\pm 3\%$ at low p_T and up to $\pm 20\%$ at the highest bin.

14 Back-to-Back Jet Cross-Check of the NPB BDT

The non-physics-background (NPB) BDT (Section 3.4.2) is trained against beam-induced streak clusters that are not associated with a real p+p collision. Such clusters carry no recoil jet, while signal direct/fragmentation photons and high-p_{T} π^0 candidates are accompanied by a balancing parton from the $2 \to 2$ hard scattering. Requiring a back-to-back recoil jet (b2b jet) above some p_{T} threshold therefore yields a sample that is physically depleted of NPB clusters *independently* of any shower-shape-based veto. Comparing the unfolded cross section in this b2b-enriched sample with and without the NPB BDT cut is thus a closure test: if the BDT is unbiased on real photons, the two should agree to within statistics. Cross-checking against the full-sample nominal selection in addition tells us whether the b2b enrichment itself disturbs the measurement.

14.1 Method

Four systematic variants are produced on top of the analysis pipeline:

- `b2bjet_pt5`: nominal selection plus a recoil-jet requirement with $p_{\mathrm{T}}^{\mathrm{jet}} \geq 5$ GeV back-to-back in ϕ ($|\Delta\phi| > 3\pi/4$). NPB BDT cut kept at the nominal NPB > 0.5.
- `b2bjet_pt5_npb0`: as above but with the NPB BDT threshold lowered to 0 (cut off).
- `b2bjet_pt7`: same as `b2bjet_pt5` but with $p_{\mathrm{T}}^{\mathrm{jet}} \geq 7$ GeV.
- `b2bjet_pt7_npb0`: same as `b2bjet_pt7` but with NPB cut off.

Each variant runs the full data + MC pipeline (purity, unfolding, efficiency correction, lumi normalization) with the same $\mathscr{L} = 64.37$ pb^{-1} as the nominal. The cross section $d^2\sigma/\mathrm{d}\eta\,\mathrm{d}E_{\mathrm{T}}^{\gamma}$ is then computed bin-by-bin and compared to the nominal selection.

14.2 Results

Figure 69 overlays the four variant cross sections on top of the nominal in the upper panel and shows the variant-to-nominal ratio in the lower panel. Two observations follow:

NPB BDT closure. The filled-marker (NPB on) and open-marker (NPB off) curves of the same color overlay tightly across the analysis E_{T}^{γ} range. Bin-by-bin, $|\sigma_{\mathrm{pt5}}/\sigma_{\mathrm{nom}} - \sigma_{\mathrm{pt5\_npb0}}/\sigma_{\mathrm{nom}}| \leq 1.0\%$ and $|\sigma_{\mathrm{pt7}}/\sigma_{\mathrm{nom}} - \sigma_{\mathrm{pt7\_npb0}}/\sigma_{\mathrm{nom}}| \leq 0.5\%$, with the largest values in the lowest E_{T}^{γ} bin. The NPB cut therefore does not change the answer when a genuine b2b recoil jet is required, which is the expected behaviour if the BDT is unbiased on real photons.

b2b vs nominal. The b2b-enriched cross section sits 6–10% *below* the nominal in the lowest E_{T}^{γ} bin (12–14 GeV) and converges to within $\pm 10\%$ of the nominal for $E_{\mathrm{T}}^{\gamma} \geq 14$ GeV. The deficit is consistently larger for the $p_{\mathrm{T}}^{\mathrm{jet}} \geq 7$ GeV variant (10.0% at 12–14 GeV) than for the $p_{\mathrm{T}}^{\mathrm{jet}} \geq 5$ GeV variant (6.9% at the same bin), which is the expected ordering if the recoil-jet acceptance at low photon E_{T}^{γ} is sensitive to the modelling of the soft fragmentation tail in the underlying PYTHIA-8 jet sample: at low E_T^γ the recoil parton is soft and hadronizes into a low-p_T jet that the b2b cut sometimes rejects, with stronger sensitivity at the higher jet-p_T threshold. The agreement at $E_T^\gamma \geq 14$ GeV, where the recoil jet is well above either threshold, is what validates the NPB BDT: the cross section recovered from a b2b-enriched, NPB-physically-depleted sample matches the nominal NPB-BDT-vetoed sample, so the BDT is not introducing a measurable bias on the recovered cross section.

The low-E_T^γ deficit is therefore interpreted as a recoil-jet-acceptance effect inherited from the PYTHIA-8 fragmentation modelling rather than as an NPB-cut bias, and is not propagated as a systematic on the nominal cross section.

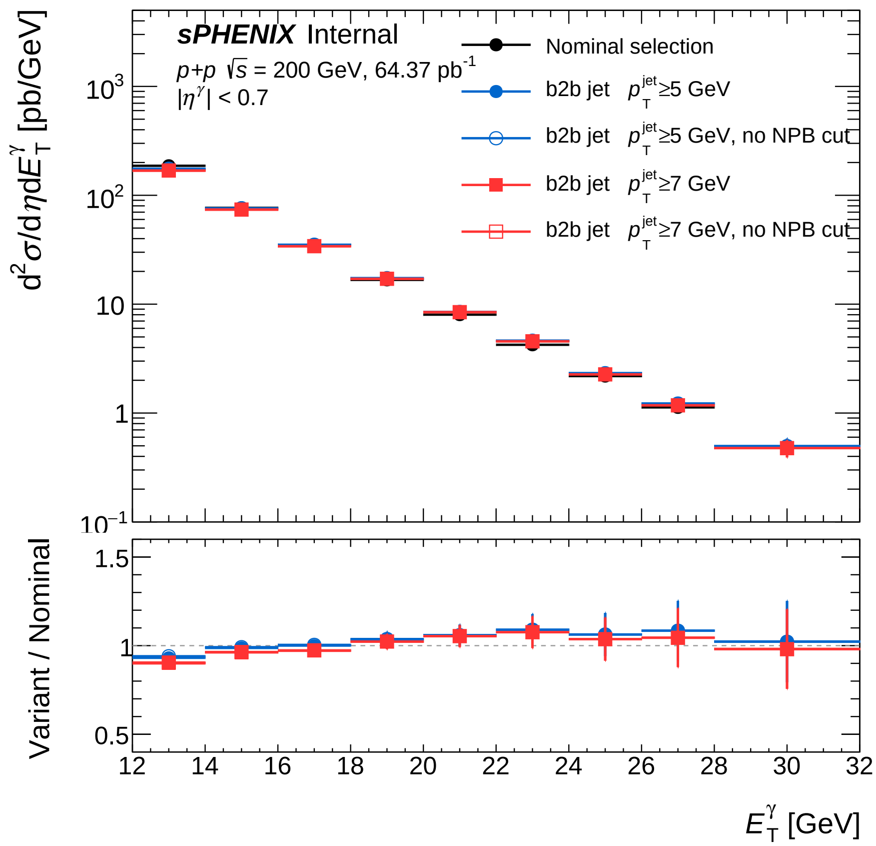


Figure 69: Cross-check of the NPB BDT using a back-to-back recoil-jet requirement. Top: differential cross section $d^2\sigma/\mathrm{d}\eta\,\mathrm{d}E_T^\gamma$ vs. photon E_T^γ for the nominal selection (black) and four b2b-enriched variants: $p_T^{\mathrm{jet}} \geq 5$ GeV with (filled blue) and without (open blue) the NPB BDT cut, and $p_T^{\mathrm{jet}} \geq 7$ GeV with (filled red) and without (open red). Bottom: variant-to-nominal ratio. The NPB-on and NPB-off variants agree to within 1% across the analysis E_T^γ range, validating the NPB BDT. The b2b-enriched cross sections sit a few % below the nominal at the lowest E_T^γ (with a larger deficit for the $p_T^{\mathrm{jet}} \geq 7$ GeV threshold) and recover to within $\pm 10\%$ for $E_T^\gamma \geq 14$ GeV, consistent with a recoil-jet-acceptance effect from the PYTHIA-8 fragmentation modelling at low recoil-parton p_T.

Figure 70 quantifies the underlying recoil-jet acceptance directly. For truth-matched prompt photons in nominal PYTHIA-8 inclusive MC, the fraction with a reconstructed back-to-back jet ($|\eta^{\mathrm{jet}}| < 0.6$, $|\Delta\phi| > 3\pi/4$) above a given p_T^{jet} threshold rises strongly with photon E_T^γ and is well below unity at the lowest E_T^γ. The threshold-dependent gap tracks the larger b2b/nominal cross-section deficit seen for $p_T^{\text{jet}} \geq 7$ GeV relative to $p_T^{\text{jet}} \geq 5$ GeV in Fig. 69, supporting the recoil-jet-acceptance interpretation rather than an NPB-BDT-induced bias.

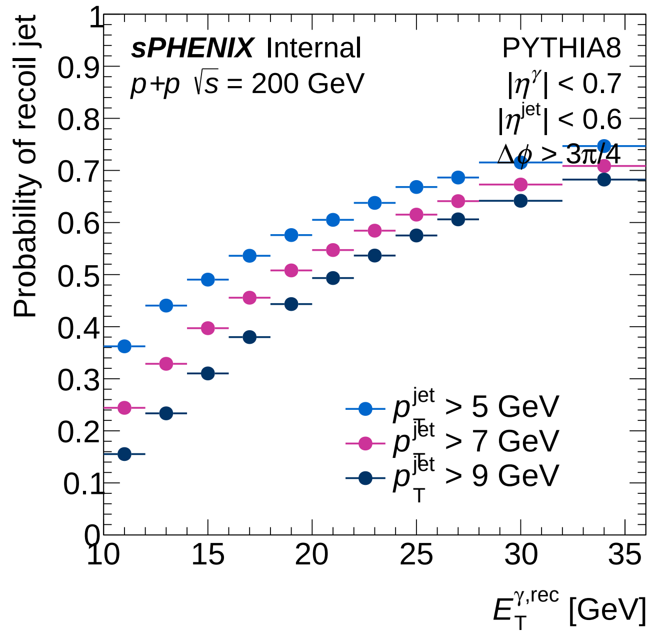


Figure 70: Probability of finding a reconstructed back-to-back recoil jet ($|\eta^{\text{jet}}| < 0.6$, $|\Delta\phi| > 3\pi/4$) above $p_T^{\text{jet}} = 5$, 7, and 9 GeV as a function of photon E_T^γ, evaluated on truth-matched prompt photons in nominal PYTHIA-8 inclusive MC. The probability is $\sim 36\%$ (24%) at $E_T^\gamma = 10$–12 GeV for $p_T^{\text{jet}} \geq 5$ (7) GeV and rises to $\sim 75\%$ (71%) at $E_T^\gamma = 32$–36 GeV, confirming that recoil-jet acceptance is the limiting factor at low photon E_T^γ.

15 PHENIX comparison: bin-width and acceptance corrections

The PHENIX measurement of inclusive direct photons in $|\eta| < 0.25$ [1] is reported as a representative cross section per E_T^γ bin, derived from the invariant cross section $E\, d^3\sigma/dp^3$. To overlay it on the present isolated-photon measurement at $|\eta| < 0.7$ on equal footing, two corrections are applied to the published points before display in Figure 55: a bin-width recovery from a smooth fit of the PHENIX spectrum, and a pseudorapidity-density rescaling derived from prompt-photon truth simulation.

15.1 Bin-width recovery

The PHENIX invariant cross section is converted to the differential form $d^2\sigma/(d\eta\, dE_T^\gamma)$ through the Jacobian $2\pi\, E_T^\gamma$ and fit with a smooth modified power law

$$f(E_T^\gamma) = A\left[1 + (E_T^\gamma)^2/b\right]^c \tag{9}$$

over the published 5–26 GeV range. The fit is shown in the top panel of Figure 71; the bottom panel shows the per-bin pull

$$\mathrm{pull}_i = \frac{y_i^{\mathrm{data}} - \langle f\rangle_i}{\sigma_i^{\mathrm{stat}}},$$

which is bounded by $|2.5|$ across the entire fit range and yields $\chi^2/\mathrm{ndf} = 28.07/15$. The PHENIX value in each bin is reported at the bin centre after PHENIX's own bin-shift correction, so converting it to a strict bin average requires only a smooth multiplicative shape factor read off the fit:

$$y_i^{\text{bin-avg}} = y_i^{\mathrm{pub}} \cdot \frac{\langle f\rangle_i}{f((E_T^\gamma)_i^c)}, \qquad \langle f\rangle_i = \frac{1}{(\Delta E_T^\gamma)_i}\int_{\mathrm{bin}\, i} f(E_T^\gamma)\, dE_T^\gamma, \tag{10}$$

where $(E_T^\gamma)_i^c$ is the bin centre. The published value (and its statistical scatter) is preserved; only the few-percent bin-shape distortion of a steeply falling spectrum within finite-width bins is removed.

15.2 Pseudorapidity-density rescaling

The pseudorapidity-density ratio

$$R(E_T^\gamma) = \frac{(dN/d\eta)_{|\eta|<0.25}}{(dN/d\eta)_{|\eta|<0.7}} \tag{11}$$

is evaluated from prompt-photon truth simulation (PYTHIA-8 8, direct+fragmentation, no isolation requirement) on the present analysis truth-p_T grid. The ratio is shown in Figure 72: it is consistent with unity to within 0.5% below $E_T^\gamma \approx 14$ GeV and rises monotonically to $R \approx 1.22$ at $E_T^\gamma = 36$–45 GeV, reflecting the rapidity narrowing of $2 \to 2$ prompt-photon production at higher Bjorken-x. Each bin-averaged PHENIX value $y_i^{\text{bin-avg}}$ from Equation 10 is divided by $R((E_T^\gamma)_i^c)$ to produce the $|\eta| < 0.7$-equivalent spectrum displayed alongside the present measurement in Figure 55. Imposing the truth-isolation condition $E_T^{\mathrm{iso,truth}} < 4$ GeV on the prompt-photon sample shifts $R(E_T^\gamma)$ by less than 0.1% in every bin, so the same factor applies to PHENIX's inclusive fiducial within the precision of either measurement. The fractional uncertainty of each PHENIX point is preserved through the rescaling.

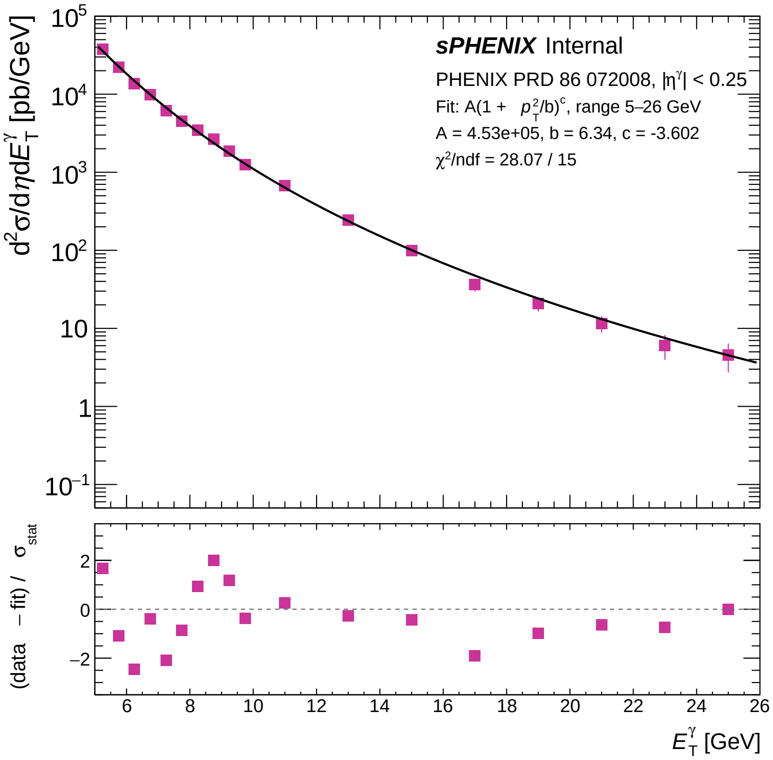


Figure 71: Top: the published PHENIX inclusive direct-photon cross section $d^2\sigma/(d\eta\, dE_T^\gamma)$ (markers) and the modified-power-law fit of Equation 9 (curve) over the 5–26 GeV range. Bottom: per-bin pull = (data − bin-averaged fit) / σ_{stat} for each PHENIX bin.

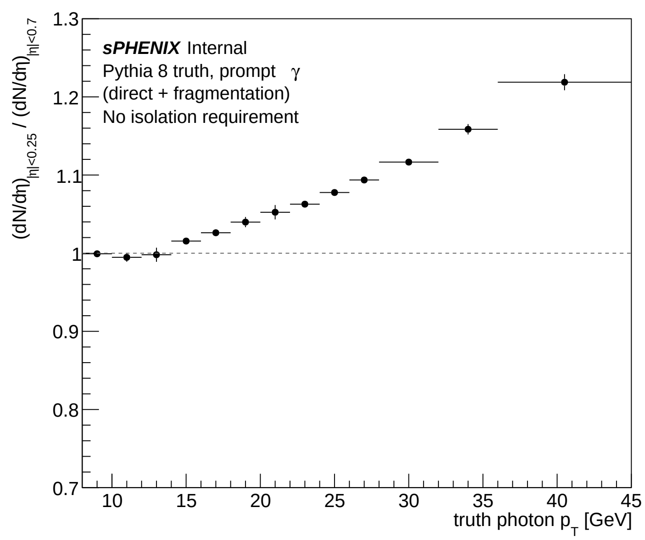


Figure 72: Pseudorapidity-density ratio $R(E_T^\gamma) = (dN/d\eta)_{|\eta|<0.25}/(dN/d\eta)_{|\eta|<0.7}$ for prompt photons evaluated from PYTHIA-8 8 truth simulation (direct+fragmentation, no isolation requirement). The ratio is consistent with unity below $E_T^\gamma \approx 14$ GeV and rises monotonically with E_T^γ as the rapidity distribution narrows at higher x.

16 MBD Vertex Efficiency: Photon vs Jet Cross-Check

The MBD-vertex-found fraction (`vz_reco` $\neq -9999$) is computed in PYTHIA-8 8 versus leading truth p_T, separately for the photon channel (`photon5/10/20`, leading truth photon p_T) and the jet channel (`jet5/8/12/20/30/40`, leading R=0.4 truth-jet p_T). Per-sample truth-p_T windows prevent sample-boundary double-counting; per-event weights are σ_i / N^i_{gen}. Figure 73 overlays the two channels: they agree to within 1% across the full reported range and fall monotonically from $\sim$0.80 at p_T^{lead}=5 GeV to $\sim$0.66 at 35 GeV, consistent with reduced forward activity at higher Bjorken-x. This channel-level closure motivates the systematic transfer described in Section 5.5.

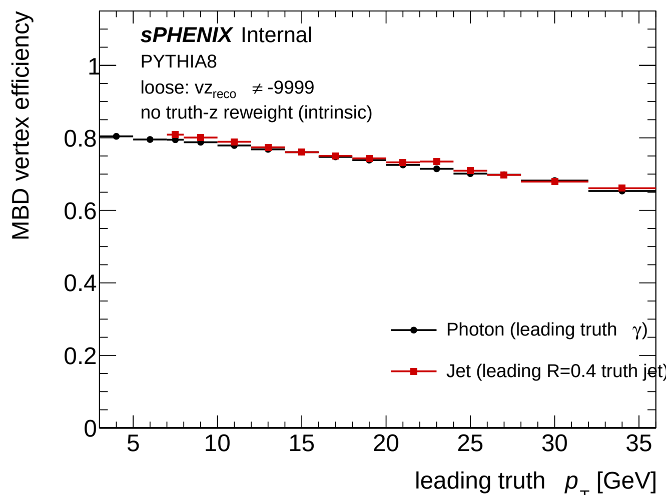


Figure 73: MBD-vertex-found fraction (`vz_reco` $\neq -9999$) versus leading truth p_T, for photon (black) and jet (red) PYTHIA-8 8 MC.

17 Double-Interaction (Pileup) Effects

17.1 Introduction

The analysis uses a full-GEANT DI MC blend. Dedicated `photon{5,10,20}_double` and `jet{8,12,20,30,40}_double` samples are produced by full detector simulation of two overlapping p+p collisions: one hard-scatter event drawn from the same PYTHIA-8 generator as the corresponding single-interaction sample (e.g. PhotonJet10 for `photon10_double`), and one minimum-bias PYTHIA-8 (Detroit-tune) event overlaid as pile-up. The two collisions are placed at independently-drawn primary vertices from the generator beam profile (§17.3). These DI samples are mixed with the nominal single-interaction samples at the cluster-weighted physics fractions of Table 10. The blend provides both the DI efficiency study (§17.4) and the shower-shape data–MC validation (§17.5).

17.2 Pileup Fractions

The mean number of interactions per bunch crossing, μ, is derived from the Poisson-corrected MBD coincidence rate:

$$\mu = \frac{\sum_{\text{runs}} N_{\text{MBD,corr}}}{r_B \cdot \sum_{\text{runs}} \Delta t}, \tag{12}$$

where $r_B = 111 \times 78{,}200 = 8.68 \times 10^6$ Hz is the RHIC bunch-crossing rate. The event-level interaction fractions under the zero-truncated Poisson are

$$f_1 = \frac{\mu\, e^{-\mu}}{1 - e^{-\mu}}, \qquad f_2 = \frac{(\mu^2/2)\, e^{-\mu}}{1 - e^{-\mu}}, \qquad f_{\geq 3} = 1 - f_1 - f_2. \tag{13}$$

Fractions are computed per run and luminosity-weighted. Cluster-weighted fractions account for the higher cluster multiplicity of multi-interaction events:

$$w_{\text{single}} = \frac{f_1}{\sum_k k\, f_k}, \qquad w_{\text{double}} = \frac{2\, f_2 + 3\, f_{\geq 3}}{\sum_k k\, f_k}, \tag{14}$$

with triple-and-higher interactions folded into the double bin for the purposes of the MC blend (the `photon10_double` / `jet12_double` productions model the two-collision case well; extra collisions are rare enough to be approximated by DIs).

Table 10: Luminosity-weighted pileup fractions and cluster-level mixing weights. Triple+ folded into f_2 for blending. μ_{corr} is the MBD-corrected mean interactions per bunch crossing.

Crossing angle	Run range	μ_{corr}	f_1	f_2	w_{single}	w_{double}
0 mrad	47289–51274	0.24	0.879	0.111	0.776	**0.224**
1.5 mrad	51274–54000	0.08	0.960	0.039	0.921	**0.079**

The cluster-weighted fractions w_{double} (last column of Table 10) are the numbers used throughout this analysis. They multiply the per-event cross section weight of each sample and set the double-interaction mixing fraction f_d in the vertex reweighting.

17.3 Vertex-Shift Mechanism

In a double-interaction event the MBD reconstructs a single vertex whose best estimator is the average of the two collision points:

$$z_{\mathrm{vtx}}^{\mathrm{reco}} \approx \frac{z_{\mathrm{vtx},1} + z_{\mathrm{vtx},2}}{2}. \tag{15}$$

The RHIC luminous region has $\sigma_{\mathrm{lum}} \approx 22\,\mathrm{cm}$ at 1.5 mrad and $\sigma_{\mathrm{lum}} \approx 54\,\mathrm{cm}$ at 0 mrad, while the MC generator was produced with a wider $\sigma_{\mathrm{gen}} \approx 65\,\mathrm{cm}$ common to both periods. The two vertices in a DI event are drawn from the generator distribution (MC) or from the luminous region (data), so the displacement between the primary truth vertex and the reconstructed vertex, $\Delta z \equiv z_{\mathrm{vtx}}^{\mathrm{reco}} - z_{\mathrm{vtx}}^{\mathrm{truth}}$, can reach several tens of cm in either case. The cluster pseudorapidity shift induced at the EMCal-barrel radius $R_{\mathrm{EMCal}} = 93.5\,\mathrm{cm}$ follows directly from the cluster geometry:

$$\Delta\eta \approx -\frac{\Delta z}{R_{\mathrm{EMCal}}\cosh\eta}. \tag{16}$$

At $\eta = 0$ this reduces to $|\Delta\eta| \approx |\Delta z|/93.5\,\mathrm{cm}$.

Table 11: Vertex-shift distributions in single- and double-interaction photon10 MC.

Quantity	Single MC	Double MC
RMS of Δz [cm]	7.6	34.5

17.4 Efficiency Impact

17.4.1 Truth–Reco Matching

Truth-to-reco association uses the GEANT track-ID from the CaloRawClusterEval module, taken as the unambiguous ground truth. No additional geometric requirement is imposed because the vertex-shift described in Section 17.3 can produce large reconstructed-cluster η shifts in DI events even though the underlying cluster is fully reconstructed and carries the correct energy. The residual genuine cluster loss in DI MC (no track-ID-matched cluster found at all) is at the $\sim 12\%$ level and is retained as a genuine pileup degradation in the reconstruction-stage efficiency. A per-event check confirms that the two interactions do not introduce additional prompt-photon truth content: mean truth photons per event are 0.493 (single) vs. 0.495 (double), and mean truth $E_{\mathrm{T}}^{\mathrm{iso,truth}}$ is 0.17 vs. 0.26 GeV— well below the 4 GeV fiducial threshold.

17.4.2 Per-Stage Efficiency

The factorized efficiency chain $\varepsilon_{\mathrm{reco}} \to \varepsilon_{\mathrm{iso|reco}} \to \varepsilon_{\mathrm{id|reco+iso}} \to \varepsilon_{\mathrm{all}}$ reveals the genuine pileup effects stage by stage. Table 12 presents the integrated efficiency for single, double, and mixed (0 mrad) photon10 MC. Figure 74 shows the same information as a function of truth p_{T}.

Table 12: Per-component efficiency with `trkID` truth-reco matching, photon10 samples integrated over $E_T^\gamma = 14$–$30\,\text{GeV}$. Mixed = $0.776 \times$ single $+ 0.224 \times$ double (0 mrad cluster-weighted fractions from Table 10).

Efficiency component	Single MC	Double MC	Mixed 0 mrad	Effect (single → double)
$\varepsilon_{\text{reco}}$ (reconstruction)	93.5%	73.1%	88.4%	−20 pp (cluster loss + shifted kinematics)
ε_{iso} (isolation | reco)	74.2%	71.0%	73.5%	−3 pp (extra cone energy)
ε_{id} (BDT | reco+iso)	82.7%	69.4%	80.0%	−**13 pp** (vertex-shifted kinematics)
ε_{all} (total)	57.4%	36.0%	52.0%	−21 pp (reco + BDT dominate)

Three structural observations:

- **Reconstruction** ($\varepsilon_{\text{reco}}$): the 20 pp gap between single (93.5%) and double (73.1%) MC splits into genuine cluster loss (no-`trkID`-match rate: 7.9% double vs. 2.4% single, i.e. $\sim$5.5 pp of the drop) and common-cut failures from shifted kinematics ($\sim$15 pp).
- **Isolation** (ε_{iso}): essentially unchanged (3 pp degradation). Extra cone energy from the second collision is present but small compared with the 4 GeV isolation threshold.
- **Photon ID** (ε_{id}): a **13 pp degradation** in double MC (69.4% vs. 82.7%), entirely driven by vertex-displaced shower-shape inputs to the parametric E_T-dependent BDT threshold. The effect is small in single MC because only a small fraction of single-interaction clusters experience large vertex-induced kinematic shifts. In double MC the vertex-shifted population has a BDT pass rate of only $\sim 60\%$ compared with 82.7% for the unshifted population, and this population dominates the degradation.

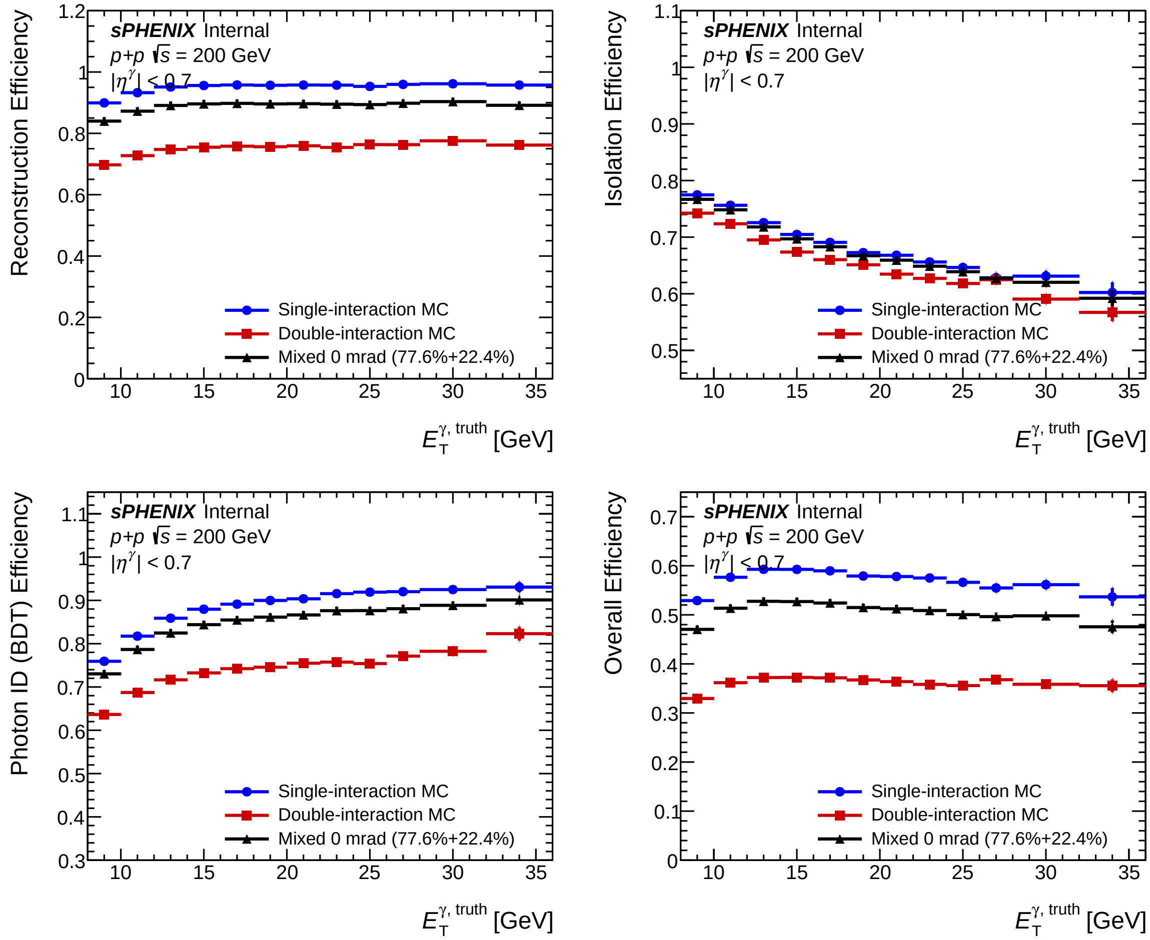


Figure 74: Per-stage efficiency vs. truth p_T for single (blue), double (red), and mixed 0 mrad (black) photon10 MC. Upper left: reconstruction. Upper right: isolation. Lower left: photon ID (BDT). Lower right: total. The BDT stage shows the cleanest 13 pp gap between single and double MC, driven by vertex-shifted kinematics feeding an E_T-dependent BDT threshold.

17.4.3 Implications for the Nominal Analysis

The integrated DI-induced efficiency reduction is $\sim 9\%$ at 0 mrad and $\sim 3\%$ at 1.5 mrad, scaling with the cluster-weighted pileup fraction. Reconstruction loss is the dominant channel. BDT degradation and isolation contribute at the percent level. The nominal cross-section pipeline absorbs these effects directly: the efficiency calculation, the response matrix, and the inclusive-jet background template are all built from the SI+DI mixed MC ensemble, with each event carrying a per-event weight $w = \sigma_\mathrm{eff} \cdot f_\mathrm{mix} \cdot \mathrm{TVR}(z_h, z_\mathrm{mb})$, where $f_\mathrm{mix} = 1 - f_\mathrm{DI}$ for SI samples (`*_nom`), $f_\mathrm{mix} = f_\mathrm{DI}$ for DI samples (`*_double`), and TVR is the truth-vertex reweight described in Section 18. There is no separate factorized DI-loss correction applied outside the response matrix. The f_DI uncertainty is propagated as a single-parameter envelope (Section 5.8).

17.5 Shower-Shape Data-vs-Mixed-MC Validation

17.5.1 0 mrad Period (22.4% Double Interaction)

At 0 mrad the pileup fraction is high. A pure single-interaction MC misrepresents the data for pileup-sensitive shower-shape observables. The most sensitive variable is w_η^{CoG}: vertex shifts along the beam axis broaden the reconstructed η profile via Eq. (16), with the discrepancy growing with E_T as the spectrum-driven statistical uncertainty shrinks. The photon-ID BDT score amplifies these residuals through its nonlinear discriminant. Mixing in 22.4% of DI MC closes the data–MC gap substantially for both at $E_\mathrm{T} \geq 14$ GeV. Energy-spread-driven observables (f_1, $E_{1\times1}/E_{3\times3}$) are less DI-sensitive than w_η^{CoG}.

17.5.2 1.5 mrad Period (7.9% Double Interaction)

At 1.5 mrad the DI fraction is $\sim 3\times$ smaller than at 0 mrad, so the DI contribution to the mixed MC is correspondingly smaller. Single-only baseline agreement at the preselection cut is already good for most variables ($\chi^2/\mathrm{ndf} \lesssim 10$). Where mixed MC helps, the improvement is concentrated at the highest E_T bins.

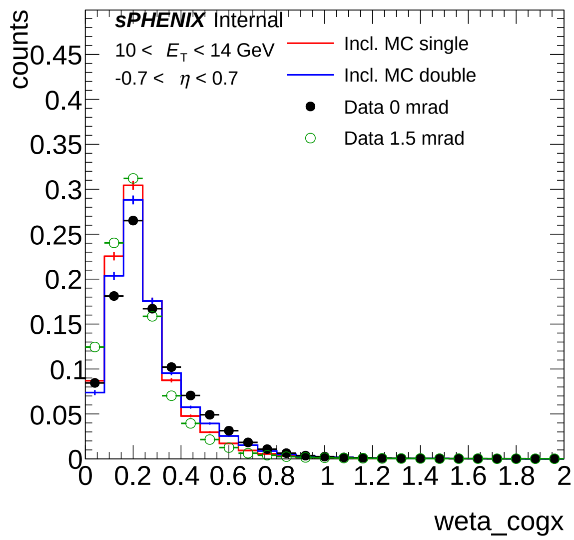
Figures 75 and 76 overlay pure single-interaction MC (red) with pure double-interaction MC (blue) and data from both periods: 0 mrad (filled black circles) and 1.5 mrad (open green circles), all area-normalized at the preselection cut. Data at 0 mrad systematically tracks the double-MC tail while data at 1.5 mrad tracks closer to single MC, consistent with the 22.4% vs. 7.9% DI fractions and providing a direct sample-level cross-check on the blending pipeline. The 8-panel overlay is split into two 4-panel figures for readability: the vertex-shift-sensitive observables (w_η^{CoG} and the composite BDT score) in Fig. 75, and the ring-energy observables (f_1 and $E_{1\times1}/E_{3\times3}$) in Fig. 76.

17.5.3 χ^2 Summary

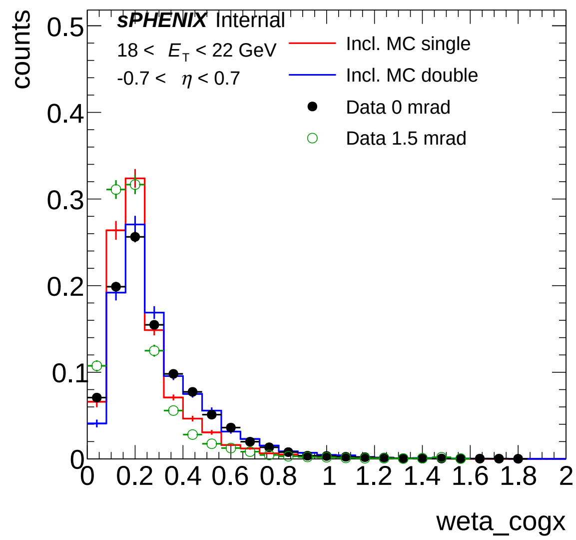
Table 13 reports representative Data − Inclusive-MC χ^2/ndf for the two leading discriminants (w_η^{CoG} and the photon-ID BDT score) at the lowest and highest E_T bins, comparing the single-only baseline against the mixed (SI + DI blended) MC at both crossing angles.

Table 13: Representative data − inclusive-MC χ^2/ndf at the preselection cut for the single-only baseline and the mixed (SI + DI blended) MC at the two crossing angles. Mid- and high-E_T bins benefit from the DI blending. The lowest-E_T bin is sensitive to the residual DI–MC mismodeling discussed in the text.

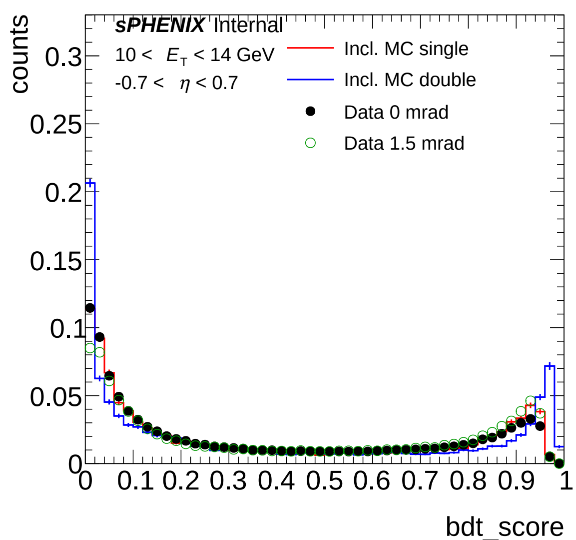
		0 mrad		**1.5 mrad**	
Variable	E_T [GeV]	SI-only χ^2/ndf	Mixed χ^2/ndf	SI-only χ^2/ndf	Mixed χ^2/ndf
w_η^{CoG}	10–14	47.40	28.96	9.72	42.31
w_η^{CoG}	22–28	15.58	0.75	21.10	1.31
BDT score	10–14	6.85	20.30	5.98	33.08
BDT score	22–28	9.62	1.75	1.89	1.13



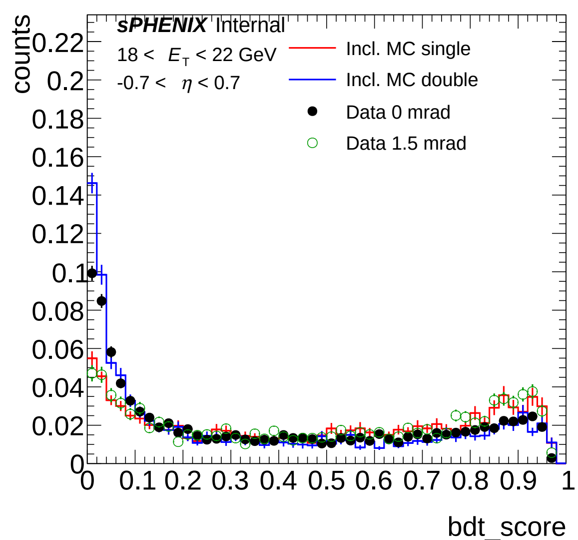
(a) w_η^{CoG}, 10–14 GeV



(b) w_η^{CoG}, 18–22 GeV

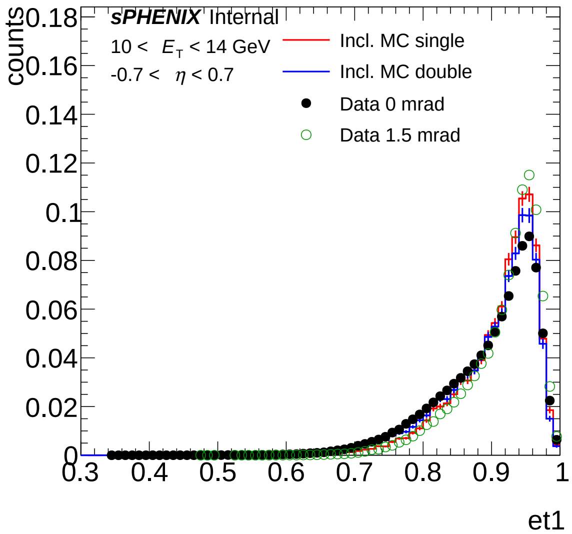


(c) BDT score, 10–14 GeV

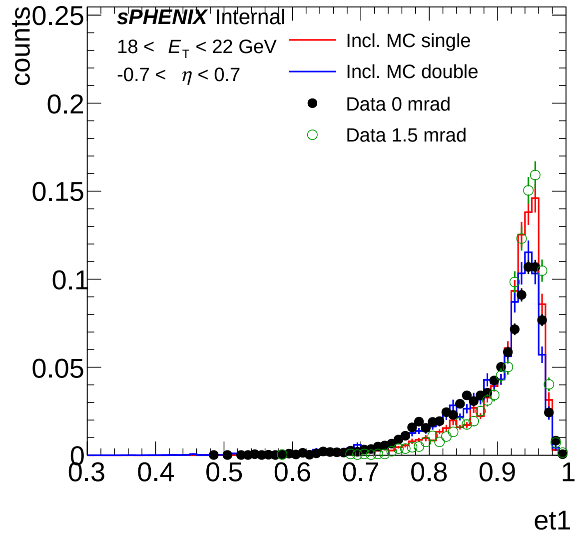


(d) BDT score, 18–22 GeV

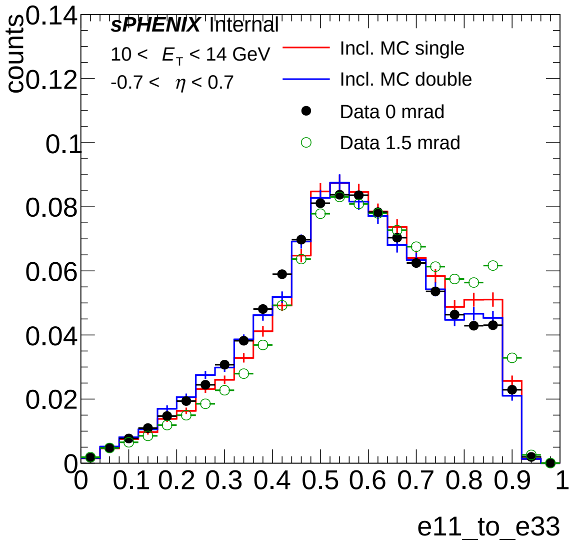
Figure 75: Sample-level shower-shape overlay, vertex-shift-sensitive observables: single-interaction MC (red), double-interaction MC (blue), data 0 mrad (filled black circles), data 1.5 mrad (open green circles), all area-normalized. Left column: $10 < E_T < 14$ GeV. Right column: $18 < E_T < 22$ GeV. Top row: w_η^{CoG}. Bottom row: BDT score. 0 mrad data systematically tracks the DI tail; 1.5 mrad data tracks closer to single MC.



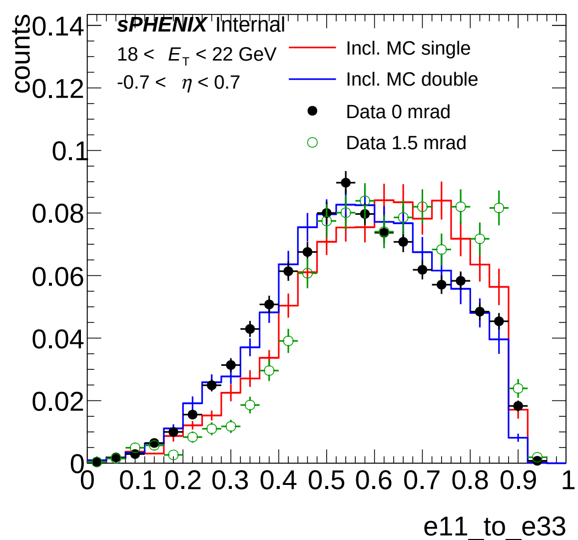
(a) f_1, 10–14 GeV



(b) f_1, 18–22 GeV



(c) $E_{1\times1}/E_{3\times3}$, 10–14 GeV



(d) $E_{1\times1}/E_{3\times3}$, 18–22 GeV

Figure 76: Sample-level shower-shape overlay, ring-energy observables: colour coding as in Fig. 75. Left column: $10 < E_T < 14$ GeV. Right column: $18 < E_T < 22$ GeV. Top row: f_1. Bottom row: $E_{1\times1}/E_{3\times3}$. Both observables show smaller DI-induced single/double separation than w_η^{CoG}/BDT, and the DI-sensitivity ordering between periods still holds.

17.6 Data-Driven Scan of the Blending Fraction

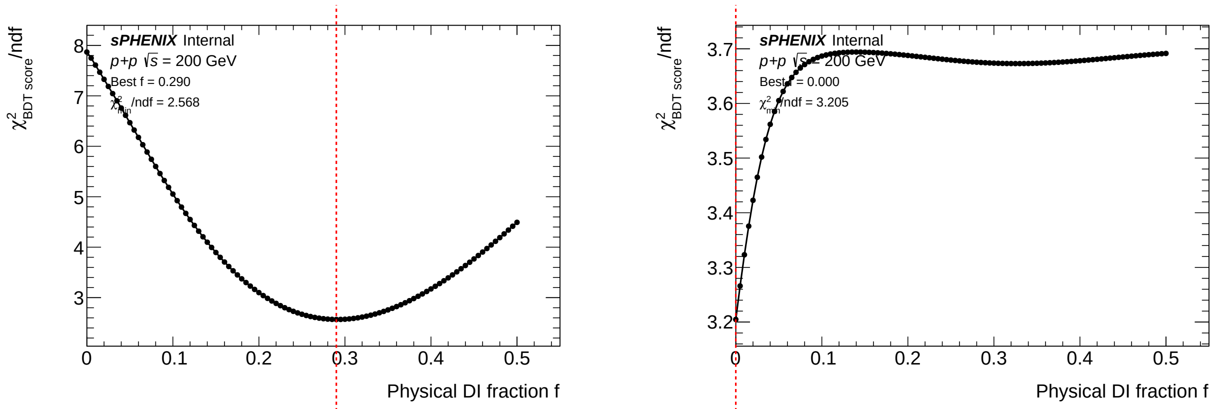
The nominal blending uses the analytic cluster-weighted DI fractions $f_{\rm DI}$ =0.224 (0 mrad) and $f_{\rm DI}$ =0.079 (1.5 mrad) computed in Sec. 17.2 from the single/double Poisson decomposition of the per-bunch interaction multiplicity. The systematic uncertainty on $f_{\rm DI}$ is set by treating the blending fraction as a free parameter and minimizing the χ^2 between data and the inclusive MC built from de-baked single- and double-interaction templates: the resulting $f_{\rm best}$ at each crossing angle defines a one-sided variation about the analytic anchor, and the envelope is symmetrized by mirroring that shift to the opposite side. The fixed-anchor data–mixed-MC χ^2/ndf summarized in Table 13 provides the reference baseline that the scan extends.

Method. Per-event MC weights in `ShowerShapeCheck.C` carry a constant multiplicative factor `mix_weight` equal to $1 - f_{\rm DI}^{\rm anal}$ for SI samples (`*_nom`) and $f_{\rm DI}^{\rm anal}$ for DI samples (`*_double`); the cross-section weight, the period luminosity ratio, and the per-event truth-vertex reweight are independent multiplicative factors. The eight per-period SI samples (three signal photon + five inclusive-jet $p_{\rm T}$ bins) are hadded into a single SI-total file, and likewise for the eight DI samples. The truth-shape templates are recovered by dividing the SI-total and DI-total per-bin contents by their respective baked-in `mix_weight`: $T_{\rm SI} = {\rm SI}_{\rm total}/(1 - f_{\rm DI}^{\rm anal})$ and $T_{\rm DI} = {\rm DI}_{\rm total}/f_{\rm DI}^{\rm anal}$. The closed-form round-trip $(1 - f_{\rm DI}^{\rm anal})T_{\rm SI} + f_{\rm DI}^{\rm anal}T_{\rm DI} = {\rm SI}_{\rm total} + {\rm DI}_{\rm total}$ is verified numerically against the existing `signal_combined` + `jet_inclusive_combined` hadds to TH2F float32 precision. The inclusive MC at any candidate physical cluster fraction f is then built by direct shape interpolation,

$$\mathrm{MC}_{\rm incl}(f) = (1 - f)\, T_{\rm SI} + f\, T_{\rm DI}. \tag{17}$$

The fit is a one-parameter χ^2 scan against the data BDT-score distribution at the preselection cut (NPB-clean, jet-dominated). The BDT score is the cleanest single observable for this purpose: it aggregates the eleven shower-shape inputs through a discriminator already trained on the sample-level SI/DI separation, so the bin-by-bin fluctuations track the relevant physics rather than per-variable modeling defects. The χ^2 is shape-only (data and MC unit-area-normalized in each $E_{\rm T}$ bin before χ^2) so the absolute jet cross-section uncertainty does not leak into $f_{\rm DI}$. Each crossing angle is scanned independently over $f \in [0.00, 0.50]$ in steps of $\Delta f = 0.005$. The total chi2 sums the per-bin residuals across all five $E_{\rm T}$ cells at the preselection cut ($5 \times 50 = 250$ BDT bins, ndof $= 250$).

Results. Figure 77 shows the BDT-score χ^2/ndf as a function of f for both crossing angles. The 0 mrad scan exhibits a clean parabolic minimum at $f_{\rm best} = 0.290 \pm 0.008$ (parabolic fit at $\chi^2_{\rm min} + 1$, statistical scaling only; see caveat below), *above* the analytic anchor 0.224. The 1.5 mrad scan minimum sits at the lower edge of the physical scan range ($f_{\rm best} = 0$), *below* the analytic anchor 0.079 (an unconstrained parabolic fit extrapolates to $f \approx 0.067$, also below analytic). Numerical results are summarized in Table 14.



(a) 0 mrad. Best-fit $f_{\text{best}} = 0.290$ (analytic 0.224).

(b) 1.5 mrad. Best-fit $f_{\text{best}} = 0.000$ (scan-edge; analytic 0.079).

Figure 77: BDT-score χ^2/ndf as a function of the physical double-interaction cluster fraction f used in Eq. (17), summed over all E_T bins at the preselection cut. The inclusive MC is rebuilt as $(1-f)\,T_{\text{SI}} + f\,T_{\text{DI}}$ using the same de-baking procedure described in the text. The red dashed line marks f_{best} from the grid; the analytic anchors are 0.224 (0 mrad) and 0.079 (1.5 mrad).

Table 14: Data-driven χ^2/ndf scan summary. χ^2/ndf is from the shape-only BDT-score chi2 at the preselection cut, all E_T bins combined, ndof = 250. $\Delta\chi^2(f_{\text{DI}}^{\text{anal}})$ is the increase in chi2 when fixing f to the analytic anchor instead of f_{best}.

Crossing	$f_{\text{DI}}^{\text{anal}}$	f_{best}	χ^2/ndf at f_{best}	χ^2/ndf at $f_{\text{DI}}^{\text{anal}}$	$\Delta\chi^2$
0 mrad	0.224	0.290	2.57	2.84	68.4
1.5 mrad	0.079	0.000 (edge)	3.20	3.67	116.6

Caveats. The residual χ^2/ndf at f_{best} is far from unity ($\chi^2/\text{ndf}_{\text{min}} = 2.57$ and 3.20), so the $\Delta\chi^2 = 1$ prescription does not yield a statistically meaningful confidence interval on f in the strict frequentist sense; the parabolic ± 0.008 uncertainty quoted above is the curvature of the chi2 surface and should be inflated to $\sim\sqrt{\chi^2/\text{ndf}_{\text{min}}}$ times that as a pragmatic Birge-ratio-style uncertainty estimate. The two crossings also pull f_{best} in *opposite directions* relative to the analytic anchor: 0 mrad above by +30% relative, 1.5 mrad to the lower edge. Because the per-cluster DI fraction at 0 mrad is $\sim 2.8\times$ that at 1.5 mrad, a coherent DI-shape mismodeling at fixed sign would shift f_{best} in the same fractional direction at both crossings; observing opposite-sign pulls instead points to a non-DI tension — the chi2 surface in this scan is dominated by residual data–MC discrepancies unrelated to f_{DI} (TVR closure, jet-fragmentation modeling, BDT training-vs-data calibration). A second cross-check using the broader nine-shower-shape combination (w_η^{CoG}, `wphi_cogx`, f_1–f_4, $E_{1\times1}/E_{3\times3}$, $E_{1\times7}/E_{7\times7}$, $E_{3\times2}/E_{3\times5}$, excluding BDT and NPB to avoid a circular discriminator) yields $f_{\text{best}} = 0.34$ and 0.00 at the two crossings respectively, with even larger $\chi^2/\text{ndf}_{\ \text{min}}$ ($\sim$4.36 and 5.04); the qualitative picture (opposite-sign pulls, larger residual than the analytic envelope) is robust.

Systematic envelope. The central values are kept at the analytic anchors $f_{\text{DI}} = 0.224$ (0 mrad) and 0.079 (1.5 mrad) — the chi2 fit is used to size the envelope, not to relocate the central prediction, because the caveats above show the chi2 surface is dominated by data–MC tension unrelated to f_{DI} rather than a coherent DI-fraction error. A single f_{DI} variation per crossing angle is run at the data-driven f_{best}:

- **0 mrad**: variation $f_{\mathrm{DI}} = 0.290$, shift $|f_{\mathrm{best}} - f_{\mathrm{DI}}^{\mathrm{anal}}| = 0.066$ (+30% of analytic).
- **1.5 mrad**: variation $f_{\mathrm{DI}} = 0.000$, shift $|f_{\mathrm{best}} - f_{\mathrm{DI}}^{\mathrm{anal}}| = 0.079$ (100% of analytic).

For each crossing angle the absolute per-bin cross-section deviation from the nominal-blending run, $|\Delta\sigma|$, is taken as the symmetric two-sided systematic $\pm|\Delta\sigma|$ on the cross section. The two single-variation runs are propagated through the full cross-section pipeline and the resulting per-bin systematic is reported in Sec. 5.8.

17.7 Cluster Size Under DI

Cluster size (EMCal tower count n_{owned} plus the bounding-box widths Δi_η, Δi_ϕ) is the basic shape property underlying every shower-shape-based discriminator used in this analysis. The DI-induced shifts are mild for photons ($\Delta\langle n_{\mathrm{owned}}\rangle \sim 0.2$ towers across the analysis range) but several times larger for jets, because soft neighbouring towers from the second collision can be pushed above the 70 MeV clustering threshold in a jet environment. After the tight photon-ID selection, data, jet MC, and photon MC all converge to a signal-like cluster size within $\pm5\%$, and the residual DI effect on $\langle n_{\mathrm{owned}}\rangle$ is at the sub-tower level. The cluster-size shift on the ABCD background template is therefore most visible at the preselection stage and is absorbed to below $\sim 1\%$ after the tight cut.

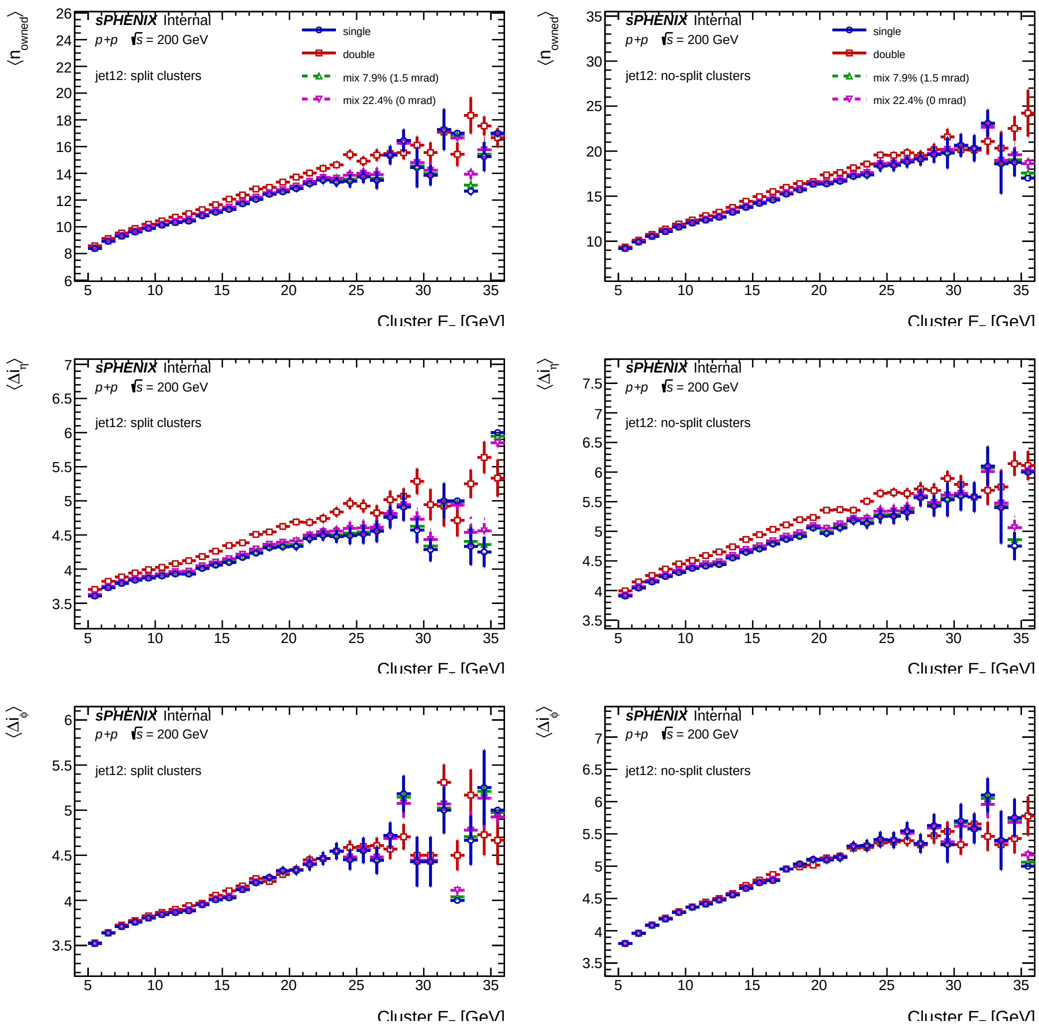
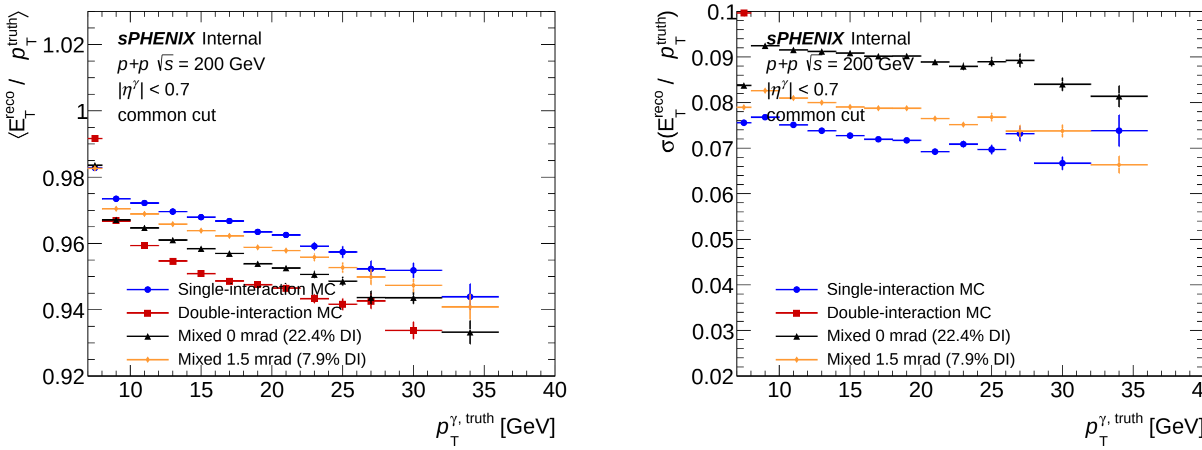


Figure 78: DI cluster-size reference for jet12: single-interaction MC vs. double-interaction MC, with mixed curves at the 1.5 mrad ($f = 7.9\%$) and 0 mrad ($f = 22.4\%$) physics fractions.

17.8 Energy Response Under DI

The photon energy response $R = E_T^{\mathrm{reco}} / p_T^{\mathrm{truth}}$ is measured bin-by-bin in truth p_T across the single, double, and mixed MC, both with and without the truth-vertex reweight. The extraction is driven by `efficiencytool/compare_energy_response.C`, which builds six sample sets per p_T bin (single, double, mixed_0 mrad, mixed_0 mrad_vtxrw, mixed_1p5mrad, mixed_1p5mrad_vtxrw) and fits each to a double-sided Crystal-Ball (DSCB) function to extract the peak μ and resolution σ. The blended vertex scan used for the MC vertex reweight is built internally from the `photon10_nom` and `photon10_double` vertex distributions.



(a) Response mean μ, DI comparison

(b) Response width σ, DI comparison

Figure 79: Photon energy response $R = E_{\mathrm{T}}^{\mathrm{reco}} / p_{\mathrm{T}}^{\mathrm{truth}}$ at the common-cut level, across single, double, and mixed (0 mrad / 1.5 mrad) MC without the truth-vertex reweight. Left: DSCB peak $\mu(p_{\mathrm{T}})$; right: DSCB width $\sigma(p_{\mathrm{T}})$. Double MC shifts the peak by $\lesssim 0.5\%$ and broadens the width by $\lesssim 10\%$ at $p_{\mathrm{T}} \sim 20\,\mathrm{GeV}$; the mixed curves sit between single and double at the appropriate fractions.

The DI-induced peak shift of $\lesssim 0.5\%$ and width broadening of $\lesssim 10\%$ at the preselection level sits well within the energy-scale systematic budget, and is further reduced after the iterative truth-vertex reweight. The leading DI sensitivity therefore remains on the efficiency (§17.4) and the ABCD background template (§17.7).

17.9 DI Reco–Truth Kernel

In jet12_double MC at 1.5 mrad, the conditional reco kernel $P(z_r | z_h = 0)$ has an RMS of $\sim 34\,\mathrm{cm}$ (with z_{mb} marginalized), wider than the $\sim 22\,\mathrm{cm}$ data target. Even forcing the primary truth vertex to $z_h = 0$ cannot narrow the reco distribution below this kernel width. This is the structural reason a truth-only reweight on z_h alone is insufficient for DI events, and motivates the $w(z_h)\, w(z_{\mathrm{mb}})$ factorization used in the iterative method of §18.

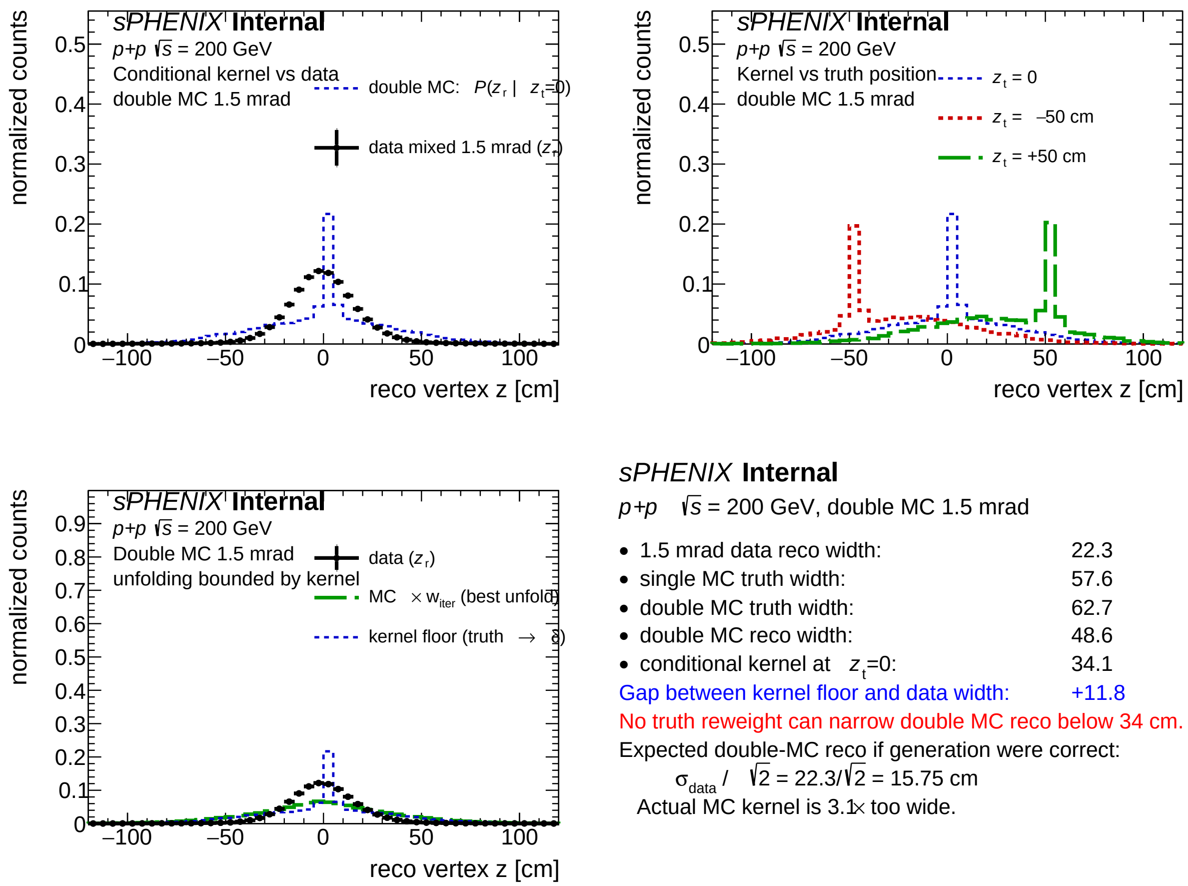


Figure 80: Conditional reco kernel $P(z_r|z_h)$ in jet12_double MC at 1.5 mrad (with z_{mb} marginalized), with the 1.5 mrad data reco z_r shape overlaid. The kernel's $\sim$34 cm RMS exceeds the $\sim$22 cm data peak, which is the geometric lower bound on truth-only reweighting.

17.10 Cross-Check: Truth-Vertex Reweight Closure

The truth-vertex reweight (§18) is a data-driven correction whose closure on the z_{reco} distribution in the analysis window is measured per crossing angle. The cross-section impact, estimated by comparing the nominal to the maximum-deviation iteration of the reweight, is $O(1\text{–}2\%)$, concentrated at low E_{T}^{γ} where the vertex-cut acceptance is most sensitive. This is treated as a cross-check rather than as a quadrature term in the main systematic budget; the residual mismatch is absorbed into the photon-ID response-driven group through the BDT-trained shower-shape response chain.

18 Truth-Vertex Reweighting

The truth-vertex reweight is derived from data using an iterative method, factorized as $w(z_h)$ for single-interaction MC and $w(z_h)\,w(z_{\mathrm{mb}})$ for double-interaction MC. The factorization is necessary because a reco-only weight $f(z_r)$ depends only on the symmetric coordinate $\bar{z} = (z_h + z_{\text{mb}})/2$ and is flat in the antisymmetric coordinate $\varepsilon = (z_h - z_{\text{mb}})/2$; the antisymmetric component is therefore unconstrained by reco-vertex closure alone, and the primary truth vertex $z_h = \bar{z} + \varepsilon$ retains a half-narrow, half-wide Gaussian width $\sigma(z_h) = \sqrt{\frac{1}{2}\sigma^2_{\text{lum}} + \frac{1}{2}\sigma^2_{\text{MC}}} \simeq 47$ cm at 1.5 mrad. Without a separate constraint on ε, the double-MC truth fiducial acceptance is biased low by about 19 percentage points relative to the data target, propagating to a $+1.4\%$ bias on the extracted cross section at 1.5 mrad and $\sim 0.3\%$ at 0 mrad.

18.1 Method: Data-Driven Iterative Truth-Vertex Reweight

The method derives a truth-vertex weight function $w(z)$ directly from the data reco distribution, with no closed-form Gaussian assumption on the luminous-region shape.

Forward model. Let f_d be the cluster-weighted double-interaction fraction (0.079 at 1.5 mrad, 0.224 at 0 mrad) and $f_s = 1 - f_d$. Under a candidate weight $w(z)$, the mixed-MC reco distribution is

$$M(z_r) = f_s \sum_{e \in T_s} w(z_h^e)\, \delta(z_r - z_r^e) + f_d \sum_{e \in T_d} w(z_h^e)\, w(z_{\text{mb}}^e)\, \delta(z_r - z_r^e), \tag{18}$$

evaluated event-by-event over the single-interaction sample T_s and the double-interaction sample T_d. For double-interaction events the weight enters *twice*, once per truth vertex. This factorization gives the iteration access to the antisymmetric coordinate ε that a reco-only weight $f(z_r)$ cannot constrain.

Binning and closure window. Both periods use 80 bins over $[-200, 200]$ cm (bin width 5 cm), with identical bin edges for $w(z)$ and for the reco-vertex histograms. The closure window is auto-derived per period as the 99-th percentile of $|z_r^{\text{data}}|$: 1.5 mrad uses $z_{\text{cut}} = 83$ cm (34 bins), 0 mrad uses $z_{\text{cut}} = 144$ cm (58 bins). Outside the window, data statistics are too low for the iteration to constrain $w(z)$.

Seed. $w_0(z) = \mathcal{N}(0, \sigma^2_{\text{seed}})/\mathcal{N}(0, \sigma^2_{\text{gen}})$, with $\sigma_{\text{gen}} = 64.997 \pm 0.015$ cm measured from the photon10 primary truth vertex (identical between periods, since the same MC is used for both), and $\sigma_{\text{seed}} = \text{std}(z_r^{\text{data}})$ measured from data — 21.48 cm at 1.5 mrad, 54.32 cm at 0 mrad. σ_{seed} requires *no* physics input: no crossing-angle correction, no luminous-region calculation. It is just the empirical width of the reco distribution we are trying to match.

Iterative update. At iteration i the residual is $r_i(z_r) = D(z_r)/M_i(z_r)$. The update is Lucy–Richardson-like, multiplicative, and damped:

$$w_{i+1}(z) = w_i(z) \cdot \left[\text{smooth}(r_i(z))\right]^{\alpha}, \qquad \alpha = 0.5, \tag{19}$$

where the residual at reco bin z_r is applied to the weight at the matching truth bin $z = z_r$. After each update w is renormalized so $\langle w(z_h) \rangle = 1$ averaged over the single-MC truth distribution (preserving the physics DI fraction). Damping prevents over-correction; 1-bin Gaussian smoothing on the residual prevents bin-by-bin oscillation; the update factor is clipped to $[0.5, 2]$ per iteration; boundary bins are edge-clamped. The final $w(z)$ is data-driven everywhere. Convergence is $\max|r_i(z) - 1| < 0.01$ inside the closure window, with a hard cap of 30 iterations.

18.2 Per-Period Results

Both periods are fit on the photon10/photon10_double mix. Table 15 collects the headline numbers.

quantity	1.5 mrad	0 mrad
run range	51274–54000	47289–51274
luminosity [pb^{-1}]	17.1642	47.2076
f_{double} (cluster-weighted)	0.079	0.224
data $\|z_r\|$ 99% percentile [cm]	82.0	143.9
closure window z_{cut} [cm]	83	144
bins inside closure window	34	58
$\sigma_{\text{seed}} = \text{std}(z_r^{\text{data}})$ [cm]	21.48	54.32
σ_{gen} from MC z_h (generator) [cm]	64.997 ± 0.015	
initial χ^2 (in window)	1,419,454	26,589
final χ^2 (in window)	453	158
final χ^2/ndf (in window)	13.7 (ndf 33)	2.78 (ndf 57)
final max $\|r-1\|$ (in window)	0.111	0.008

Table 15: Headline closure of the iterative truth-vertex reweight. Inside the analysis fiducial $|z| < 60$ cm the per-bin closure is at the 1–2% level for both periods.

Hero figure. Figure 81 is the single-figure summary of the iterative method across both periods: data reco vs. reweighted MC reco, residuals, the recovered $w(z)$, and the headline numbers.

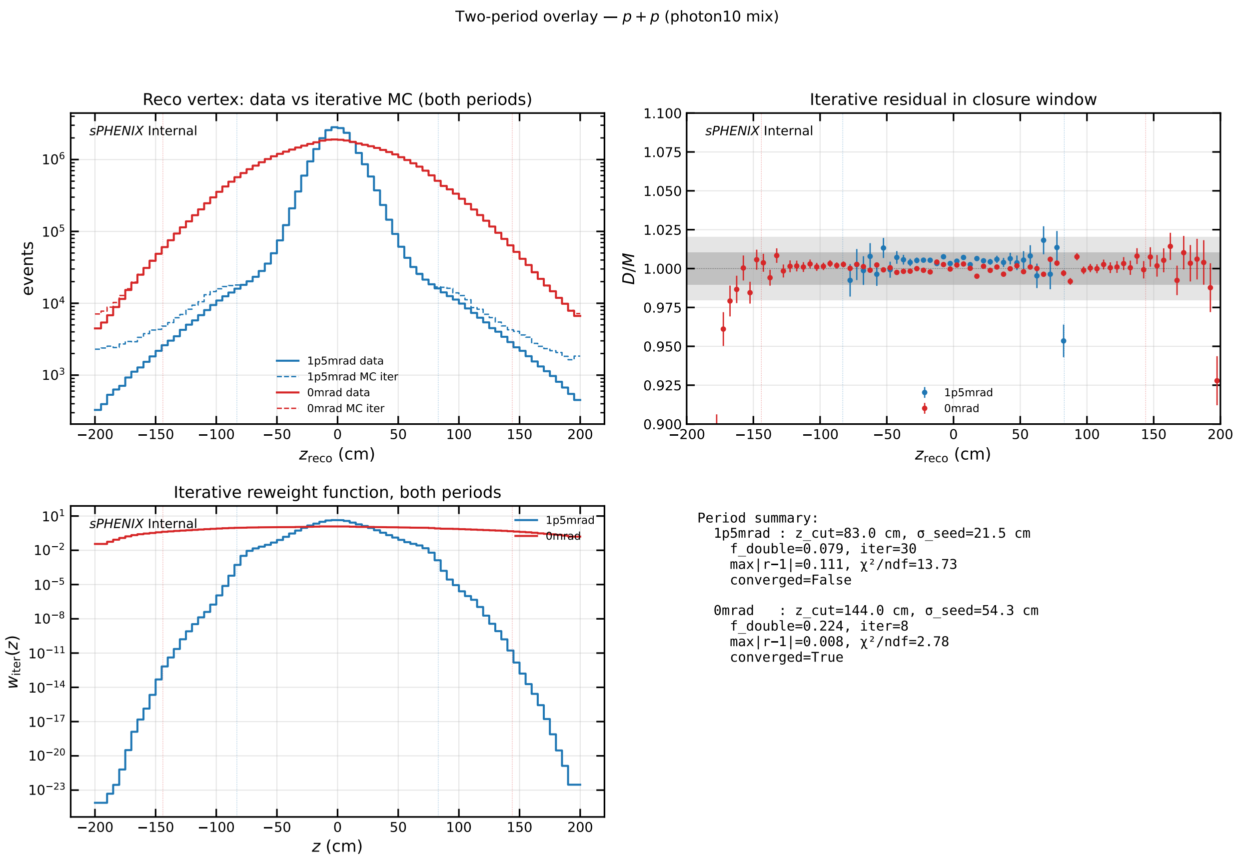


Figure 81: Iterative data-driven truth-vertex reweight summary, both crossing angles. Top-left: data reco z_r (solid) vs. mixed MC reco z_r after iterative reweight (dashed), both periods overlaid (log scale). Top-right: residual $r(z_r) = D(z_r)/M_{\mathrm{iter}}(z_r)$ inside each period's closure window (vertical dashed lines). Bottom-left: recovered truth-vertex weight $w(z)$ for both periods (log scale). Bottom-right: summary statistics. The 0 mrad case is essentially closed; the 1.5 mrad case is closed in the bulk and stalls at the closure-window edge.

Truth vertex comparison across methods. Figure 82 overlays the primary truth vertex z_h distribution of the mixed MC under three weighting methods: no reweight (raw MC, $\sigma \simeq 64$ cm), the production reco-level weight $f(z_r)$, and the new iterative truth-level weight $w(z_h)$. At 1.5 mrad the production method leaves z_h at $\sigma \simeq 27$ cm — narrower than the raw MC but still far wider than the true luminous-region width ($\sigma_{\mathrm{lum}} \simeq 22$ cm). The iterative method narrows z_h to $\sigma \simeq 16$ cm, which is narrower than the data reco width $\sigma_{\mathrm{seed}} = 21.5$ cm precisely as expected (the MBD vertex finder adds resolution smearing, so each truth vertex must be narrower than the reco width). At 0 mrad both methods converge at $\sigma \simeq 54$–57 cm.

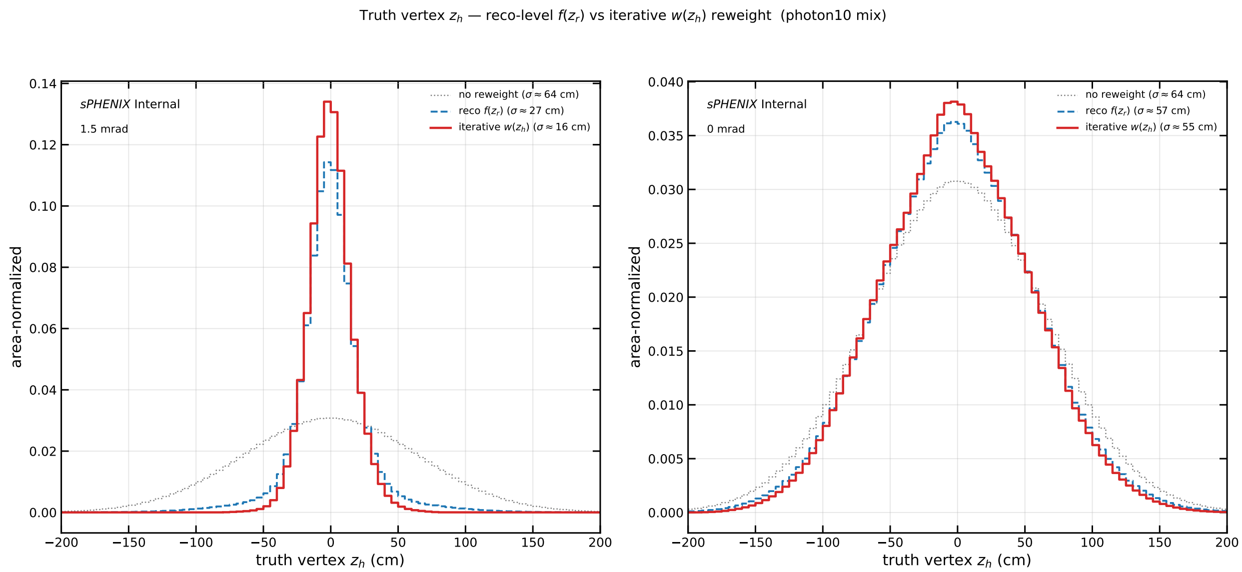


Figure 82: Truth vertex z_h distribution of the mixed photon10 MC under three reweighting methods, per period. Left: 1.5 mrad, where $f(z_r)$ (blue dashed) fails to narrow the truth distribution sufficiently while the iterative $w(z_h)$ (red) narrows it past the data reco width. Right: 0 mrad, where both methods agree because the MC beam profile is already close to the data luminous region.

0 mrad: clean convergence. Figure 83 shows the 6-panel closure diagnostic for 0 mrad. The seed Gaussian with $\sigma_{\text{seed}} \simeq 54$ cm already captures most of the shape (initial max $|r-1| = 8\%$, initial $\chi^2/\text{ndf} \simeq 467$); the parametric best-fit Gaussian at $\sigma_{\text{tgt}} \simeq 54.6$ cm is essentially identical ($\chi^2/\text{ndf} \simeq 454$) because the data is nearly Gaussian. The iteration cleans up the residual over 8 updates and drops below the 1%-per-bin tolerance, with final max $|r-1| \simeq 0.8\%$ and $\chi^2/\text{ndf} \simeq 2.78$. The recovered $w(z)$ is smooth and broadly Gaussian with a small central enhancement.

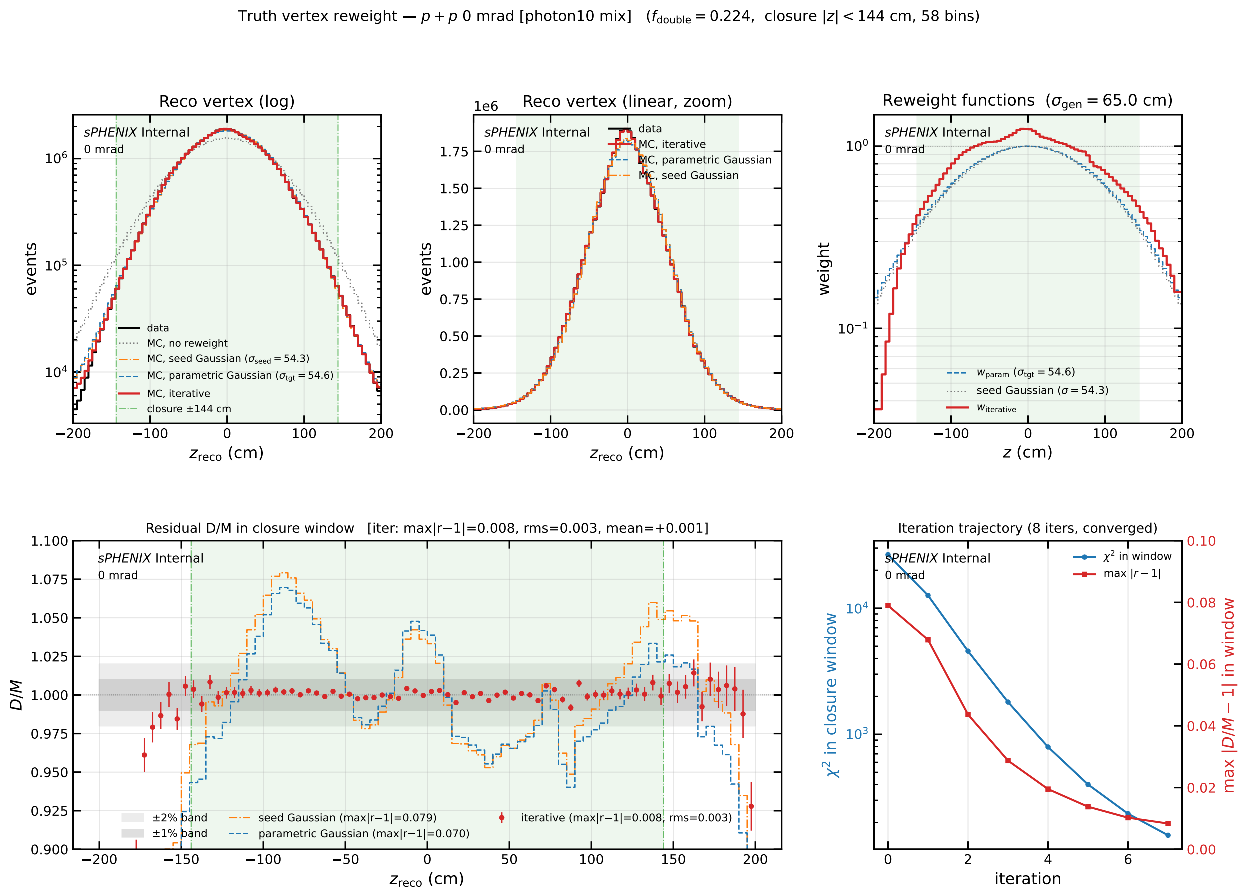


Figure 83: 0 mrad iterative closure (6 panels). Top-left/centre: data reco vs. mixed MC reco under no reweight (gray dotted), the seed Gaussian $w(z) \propto \mathcal{N}(0, \sigma^2_{\text{seed}})/\mathcal{N}(0, \sigma^2_{\text{gen}})$ (orange dash-dot), the parametric best-fit Gaussian (blue dashed), and the iterative reweight (red). Top-right: the corresponding $w(z)$. Bottom-left: residual D/M in the closure window with Poisson errors and $\pm 1\%/\pm 2\%$ reference bands. Bottom-right: per-iteration χ^2 and max $|r-1|$ trajectory. At 0 mrad the seed and the parametric Gaussian are nearly identical because the data reco is already close to a single Gaussian; the iterative method improves on both.

1.5 mrad: bulk closure, edge stall. Figure 84 shows the 6-panel diagnostic for 1.5 mrad. The seed Gaussian with $\sigma_{\text{seed}} \simeq 21$ cm starts from max $|r-1| = 64\%$ and an in-window χ^2 of 1.42×10^6. The first ~ 5 iterations make dramatic progress; the iteration then enters a slow plateau driven by bins at the closure-window edge $|z_r| \simeq 83$ cm. After 30 iterations χ^2 in the window has dropped by $\sim 3100\times$ to $\simeq 453$, and max $|r-1|$ sits at $\simeq 0.11$, set by those stuck edge bins. Inside the bulk $|z_r| < 30$ cm the closure is ~ 1–2% per bin; between 30 and 60 cm the residual climbs to ~ 5–10%. The recovered $w(z)$ shows a strong central core boost and a clear non-Gaussian shape in $|z| \simeq 40$–60 cm that no symmetric Gaussian ansatz can represent.

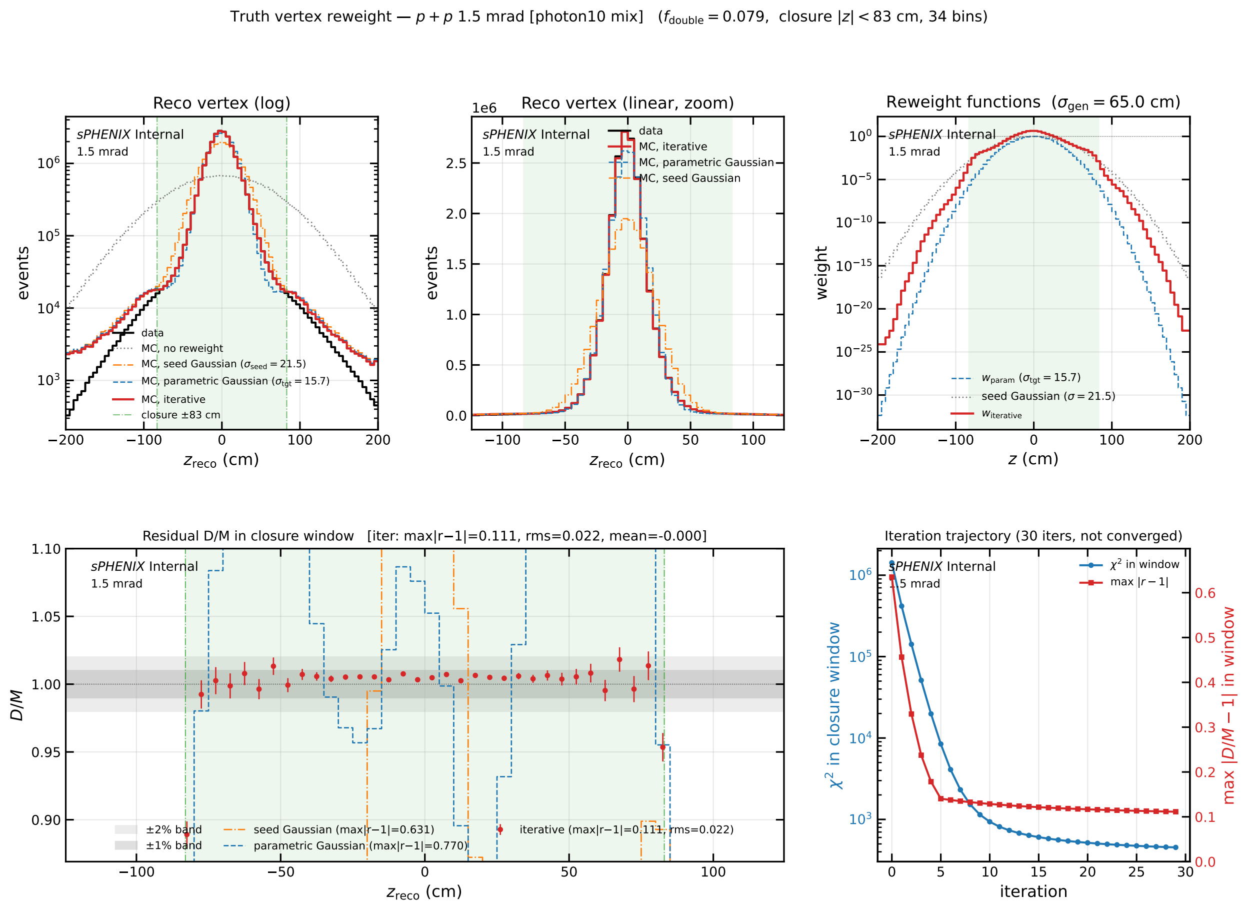


Figure 84: 1.5 mrad iterative closure (6 panels), same layout as Fig. 83. At 1.5 mrad the seed Gaussian ($\sigma_{\mathrm{seed}} = 21.5$ cm, orange dash-dot) and the parametric best-fit Gaussian ($\sigma_{\mathrm{tgt}} = 15.7$ cm, blue dashed) are distinct; both mis-predict the reco-vertex shape by 30–60% at the closure-window edge. The iterative method (red points) closes the bulk to 1–2% inside $|z_r| < 30$ cm and stalls at max $|r-1| \simeq 0.11$ at $|z_r| \simeq 83$ cm. The recovered $w(z)$ in the top-right panel shows a strong central core boost and a clear non-Gaussian shape in $|z| \simeq 40$–60 cm.

Method comparison. Figure 85 overlays the iterative reweight against three reference baselines at 1.5 mrad: no reweight, the seed Gaussian (initial point of the iteration), and the parametric best-fit Gaussian (1-parameter σ_{tgt} scan with minimum at 15.7 cm). The no-reweight MC overshoots data by a factor of several; both Gaussian baselines match the width on average but mismatch the shape at 30–70%; only the iterative method follows the narrow data core *and* tracks the non-Gaussian mid-tail.

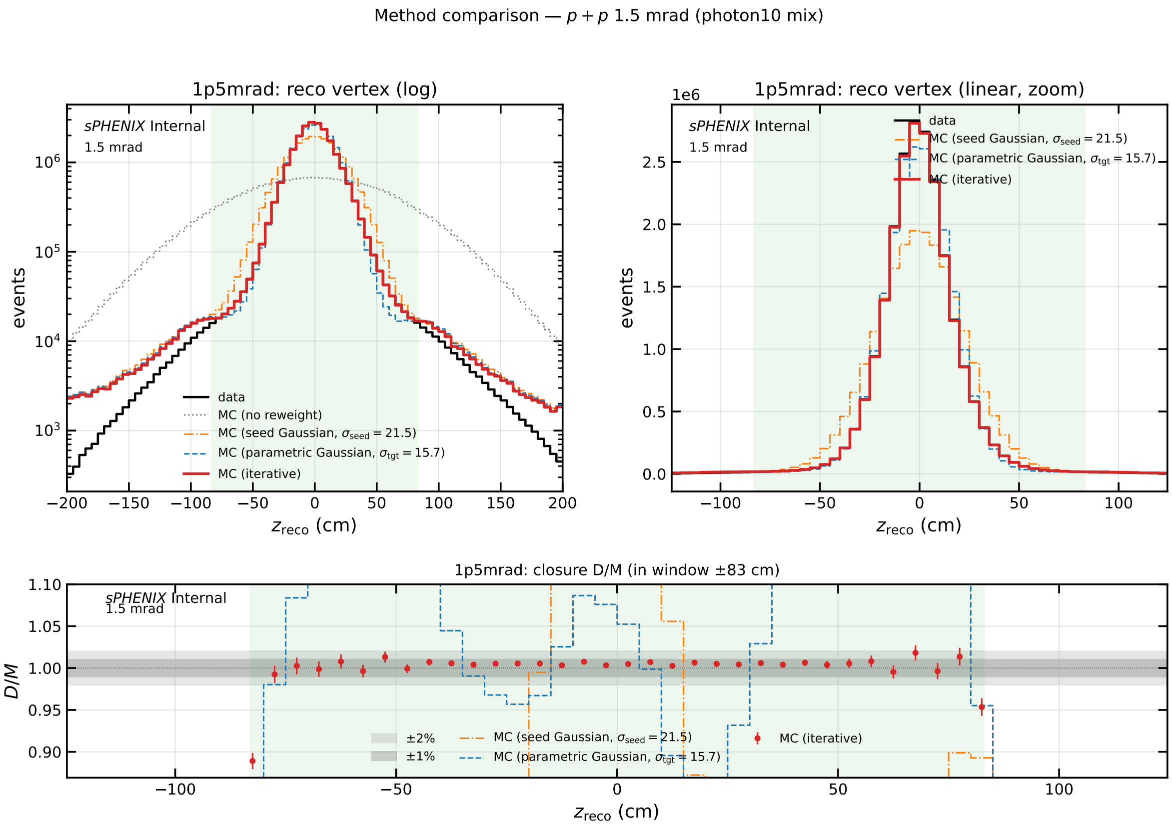


Figure 85: Method comparison at 1.5 mrad: data reco vs. mixed MC reco under no reweight (gray dotted), seed Gaussian $\sigma_{\text{seed}} = 21.5\,\text{cm}$ (orange dash-dot), parametric best-fit Gaussian $\sigma_{\text{tgt}} = 15.7\,\text{cm}$ (blue dashed), and the iterative data-driven reweight (red), with ratio panel below. The iterative method is the only one that follows the narrow data core while also tracking the non-Gaussian mid-tail structure.

Parametric σ_{tgt} scan. Figure 86 shows the stage-1 parametric χ^2 scan that precedes the iterative method. A single Gaussian truth-vertex weight $w(z) \propto \mathcal{N}(0, \sigma^2_{\text{tgt}})/\mathcal{N}(0, \sigma^2_{\text{gen}})$ is evaluated at each scan point. At 1.5 mrad the minimum is at $\sigma_{\text{tgt}} \simeq 15.7\,\text{cm}$ with $\chi^2/\text{ndf} \simeq 3237$; the iterative method reaches 13.7, a $\sim 236\times$ improvement. At 0 mrad the minimum is at $\sigma_{\text{tgt}} \simeq 54.6\,\text{cm}$ with $\chi^2/\text{ndf} \simeq 454$; the iterative method reaches 2.78, a $\sim 163\times$ improvement.

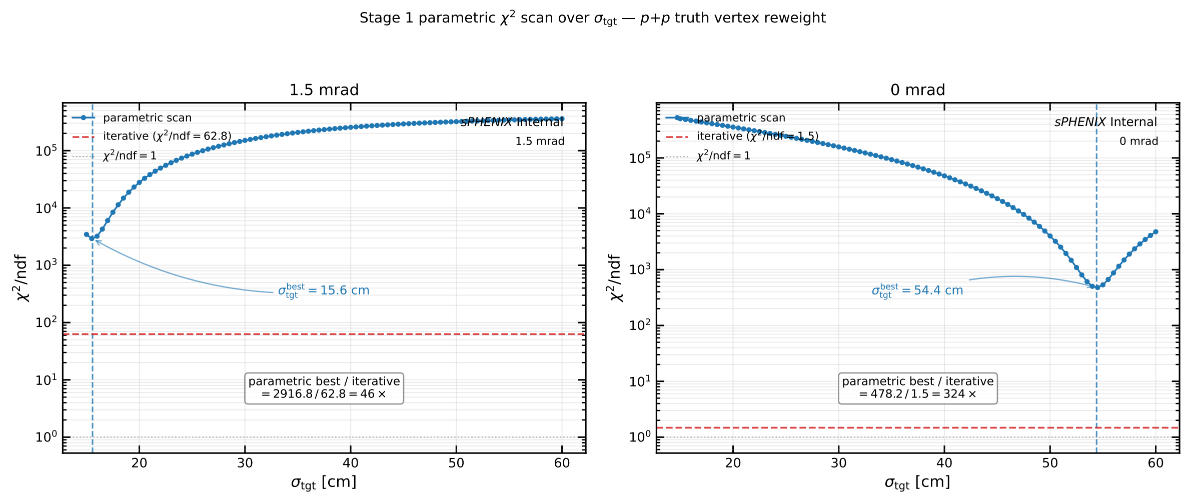


Figure 86: Stage-1 parametric χ^2/ndf as a function of the target truth-vertex Gaussian width σ_{tgt}, for the 1.5 mrad (left) and 0 mrad (right) periods. Vertical dashed line: parabolic-refined minimum. Horizontal dashed line (red): final χ^2/ndf achieved by the iterative method. Horizontal dotted line: χ^2/ndf = 1. The gap between the parametric minimum and the iterative floor quantifies the non-Gaussian shape mismatch that no single-Gaussian ansatz can capture.

18.3 Physics Interpretation

Why 0 mrad is easy. The 0 mrad luminous region ($\sigma_{\text{lum}} \simeq 54$ cm) is close to the generator beam profile ($\sigma_{\text{gen}} \simeq 65$ cm), and the data reco shape is nearly Gaussian. The seed is already a good first approximation and the iteration just cleans up small departures (slightly enhanced central core).

Why 1.5 mrad is hard. Three things conspire:

- The geometric overlap reduction at the 1.5 mrad crossing angle compresses the longitudinal luminous region to $\sigma_{\text{lum}} \simeq 22$ cm, nearly a factor of three narrower than the generator profile. The reweight must extract a narrow target from a wide source; the dynamic range of $w(z)$ at the central core is correspondingly large ($w(0) \simeq 8$).
- The data reco distribution is *not* a clean Gaussian. It has a narrow core plus a mid-tail shoulder near $|z_r| \simeq 45$–60 cm. The iterative method discovers this structure directly — the recovered $w(z)$ is non-Gaussian in the same z range — whereas a single-parameter Gaussian cannot represent it.
- The MBD resolution tail leak from core-truth events fills the bins near the closure-window edge ($|z_r| \simeq 83$ cm). The iteration becomes insensitive to $w(z)$ at that z because the contributions to those reco bins come dominantly from truth vertices at *other* z values via the wide double-interaction reco-truth kernel (Appendix 17.9). This sets the residual floor at $\max |r - 1| \simeq 11\%$.

What the recovered $w(z)$ looks like. At 1.5 mrad the recovered $w(z)$ has three distinguishing features (top-right panel of Fig. 84): (a) a strong central peak with $w(0) \simeq 8$ that pulls doubles whose truth vertices both lie near the IP into the data-like core; (b) a mid-z dip that suppresses the generator's natural $|z| \sim 30$–$50\,\mathrm{cm}$ population; and (c) a clear $\pm$ asymmetry in $|z| \simeq 40$–$60\,\mathrm{cm}$ tracking the asymmetric mid-tail bump in the data. None of these features is representable by a symmetric single-parameter Gaussian. At 0 mrad, $w(z)$ is essentially the seed Gaussian with a small central enhancement.

The reweight is fit on the photon10 (and photon10_double) sample with a cluster-weighted blend at the period-specific DI fraction; extension to the full sample blend (including jet12) is anticipated to shift the closure metrics by $O(1\%)$.

19 Generator Cross-Check: PYTHIA-8 vs. HERWIG

The full 1.5 mrad analysis pipeline is rerun replacing the nominal PYTHIA-8 8 Detroit signal MC with HERWIG 7 photon-jet samples (`photon10_herwig`, `photon20_herwig`) while keeping the data input and all background MC unchanged. The truth-p_{T} prior is rederived in the same fashion as the nominal: an unweighted iter=1 unfolding of the data with the HERWIG response matrix, refit to a rational form. To isolate the irreducible response-matrix difference from the prior choice, the unfolded comparison plot uses a common reweight target (PYTHIA-8's iter=1 unfolded data) so that any residual difference reflects only the response matrix.

The four panels of Figures 87–90 compare the two pipelines on (i) isolation efficiency ϵ_{iso}, (ii) photon-ID efficiency ϵ_{ID}, (iii) MBD-vertex efficiency ϵ_{MBD}, and (iv) the unfolded yield before efficiency division. The differences are concentrated in ϵ_{iso} and ϵ_{MBD}: HERWIG photons sit in less-active mid-rapidity environments at fixed truth E_{T}^{γ}, raising both efficiencies by a few percent. ϵ_{ID} agrees within $\pm 1.5\%$ across the analysis range, and the unfolded woeff ratio collapses to within $\pm 4\%$ once a common reweight target is used.

The ϵ_{iso} difference is propagated as a generator-driven systematic in the efficiency group (Section 5.4). The ϵ_{MBD} difference is *not* propagated from this comparison, because the data-driven MBD-coincidence efficiency from PPG10 (Section 5.5) already covers the generator-modelling uncertainty on the MBD-trigger response. The unfolded-yield comparison is included as a closure cross-check only.

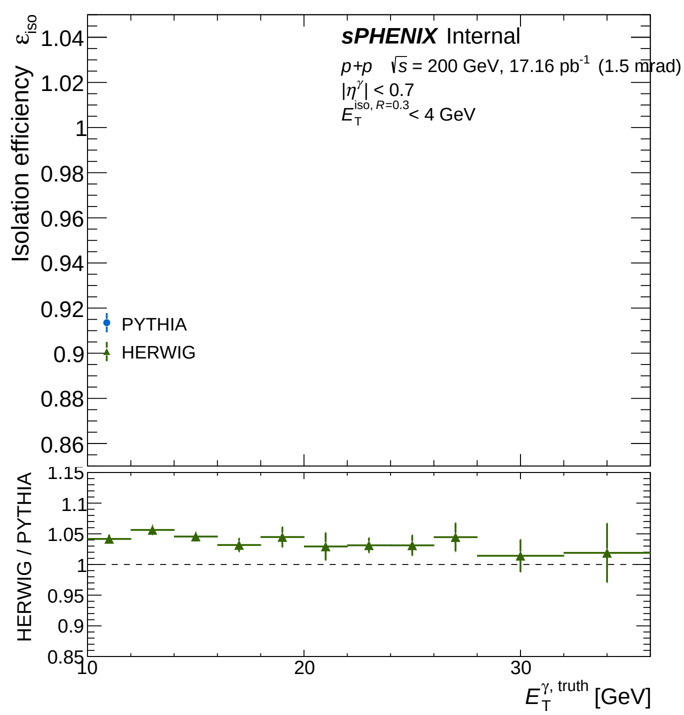


Figure 87: Isolation efficiency ϵ_{iso} vs. truth E_{T}^{γ} for the nominal PYTHIA-8 8 pipeline (blue) and the HERWIG 7 cross-check (green), with the HERWIG/PYTHIA-8 ratio in the lower panel. HERWIG sits 3–6% above PYTHIA-8 across the reported range.

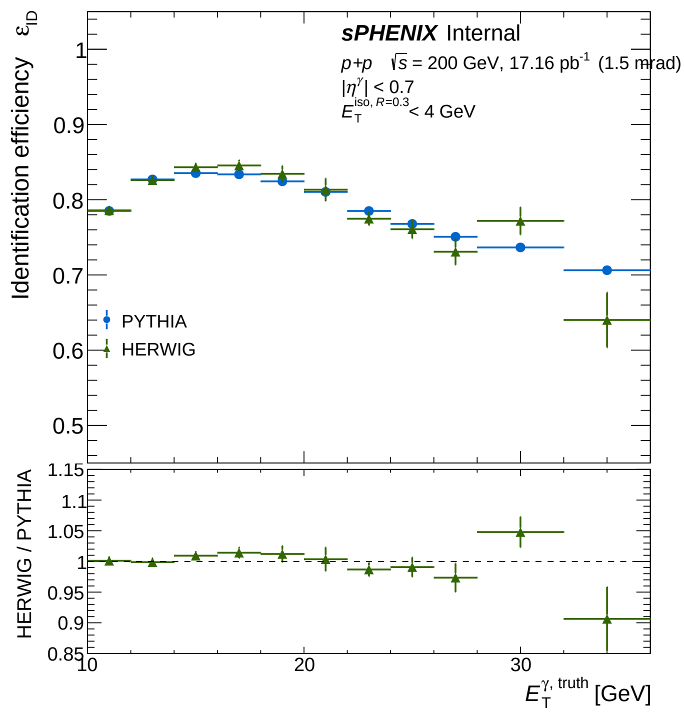


Figure 88: Photon-ID efficiency ϵ_{ID} vs. truth E_T^{γ} for the nominal PYTHIA-8 8 pipeline and the HERWIG 7 cross-check. The two agree within $\pm 1.5\%$ across the analysis range.

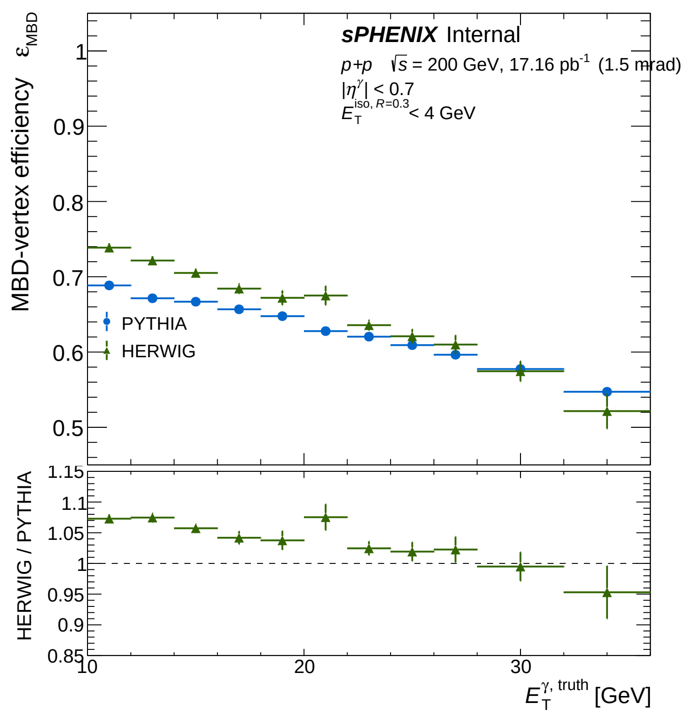


Figure 89: MBD-vertex efficiency ϵ_{MBD} vs. truth E_{T}^{γ} for the nominal PYTHIA-8 8 pipeline and the HERWIG 7 cross-check. HERWIG sits 4–7% above PYTHIA-8, reflecting different forward-UE modelling at fixed leading-photon E_{T}^{γ}. This generator difference is covered by the data-driven MBD-coincidence systematic of Section 5.5 and not double-counted from this comparison.

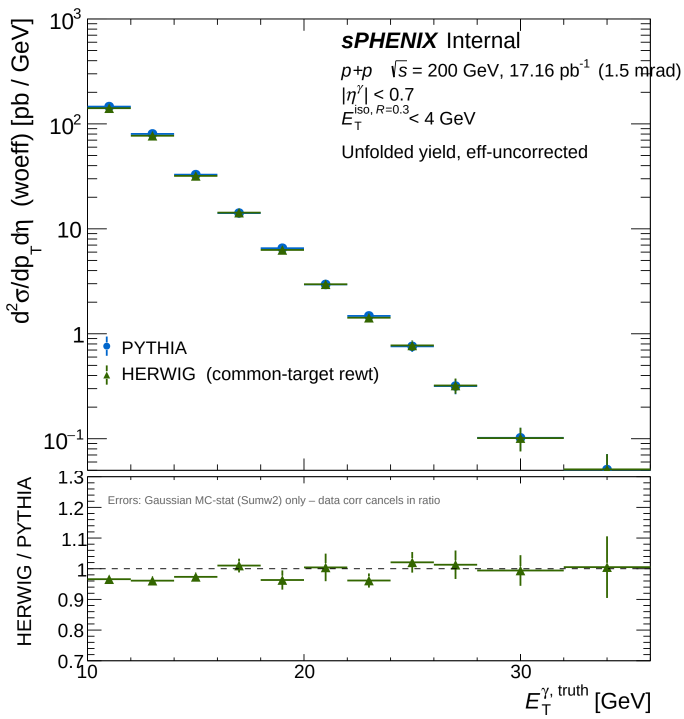


Figure 90: Unfolded BG-subtracted yield before efficiency division ($h^{\mathrm{woeff}}_{\mathrm{unfold\,sub}}$) vs. truth E_T^{γ} for the nominal PYTHIA-8 8 pipeline (blue) and the HERWIG 7 cross-check with a common reweight target (green). The data input is bit-identical between the two pipelines, so the data-statistical fluctuation cancels in the ratio. Errors on the ratio are the Gaussian MC-stat (Sumw2) of the response-matrix truth marginal only. The residual is within $\pm 4\%$ across the analysis range, comfortably within the unfolding-iteration systematic already assigned in Section 5.7.